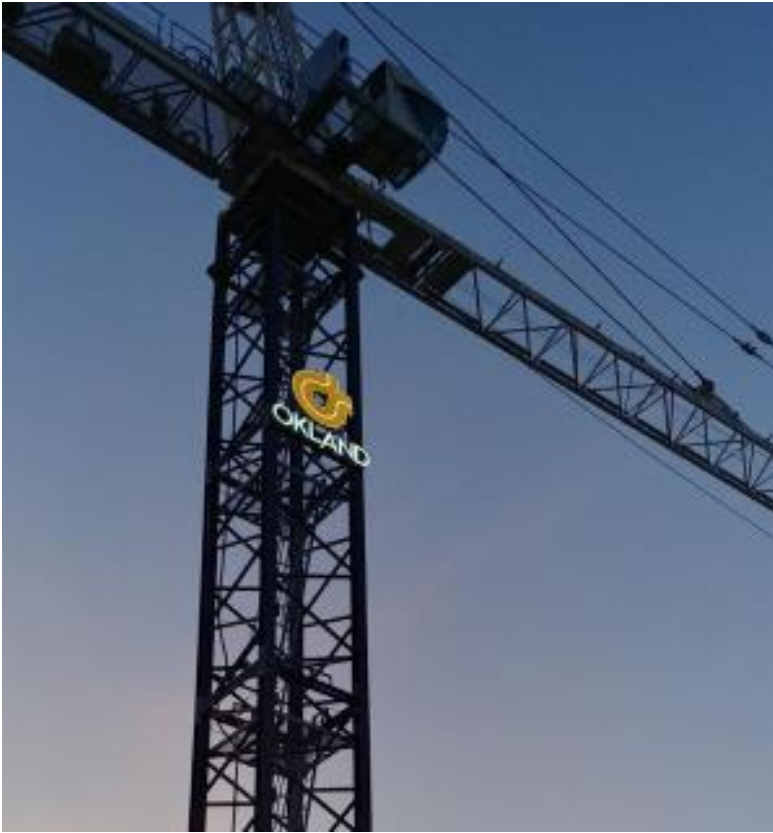




OKLAND SAFETY, HEALTH, AND ENVIRONMENTAL MANUAL

Subcontractor Specific

OKLAND SAFETY, HEALTH, AND ENVIRONMENTAL POLICY

WE BELIEVE:

Nothing we do is more important to Okland than the safety, health, and well-being of our employees and their families, our subcontractor workers and their families, our clients and their families, and the general public that may be impacted by our construction activities.

All injuries, occupational illnesses, and incidents are preventable; we can achieve and sustain a zero-incident culture.

At Okland, we are all safety leaders who act with a sense of urgency to eliminate or effectively control safety and health hazards.

WE EXPECT:

Each worker to have an individual commitment to work safely and comply with the requirements of our safety and health program.

Each worker to show their safety leadership by not only looking after the well-being of themselves but also of each other.

Every activity we do to be adequately planned out and to be in compliance with all applicable rules and regulations.

WE COMMIT:

To own the overall safety and health program for our company; to convey, with clarity, our expectation of the effective implementation of the Okland Safety and Health program; and to provide the resources and policies that are needed to establish and maintain a positive, proactive, and corporate-wide safe and healthy culture.



Bill Okland, CEO



Brett Okland, President

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LINKS TO EXTERNAL DOCUMENTS

L1-1	Visitor Orientation (Non-Blue Vest)	http://docs.okland.com/msa/safety/L1-1.pdf
L1-2	Visitor Orientation (Blue Vest)	http://docs.okland.com/msa/safety/L1-2.pdf
L2-1	Pre-Task Plan (PTP) Form	http://docs.okland.com/msa/safety/L2-1.pdf
L2-2	Pre-Task Plan (PTP) Guide Sheet	http://docs.okland.com/msa/safety/L2-2.pdf
L3	Job Hazard Analysis (JHA)	http://docs.okland.com/msa/safety/L3.pdf
L4	Rigging & Electrical Color-Code	http://docs.okland.com/msa/safety/L4.pdf
L5	Hot Work Permit	http://docs.okland.com/msa/safety/L5.pdf
L6 - 1	Soil Type A – Over 4’ Excavation & Trench Access Permit	http://docs.okland.com/msa/safety/L6-1.pdf
L6 - 2	Soil Type B – Over 4’ Excavation & Trench Access Permit	http://docs.okland.com/msa/safety/L6-2.pdf
L6 - 3	Soil Type C – Over 4’ Excavation & Trench Access Permit	http://docs.okland.com/msa/safety/L6-3.pdf
L7	Site Specific Steel Erection Plan and Checklist	http://docs.okland.com/msa/safety/L7.pdf
L9	Lift Planning Worksheet – Hydraulic/Lattice Crane	http://docs.okland.com/msa/safety/L9.pdf
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L13-2	CAUTION Yellow Tape Sign	http://docs.okland.com/msa/safety/L13-2.pdf
L13-3	CARE Tape Sign	http://docs.okland.com/msa/safety/L13-3.pdf
L20	Release Waiver and Indemnity Agreement for Equipment Use	http://docs.okland.com/msa/safety/L20.pdf
L28	MEWP Operators Inspection Form	http://docs.okland.com/msa/safety/L28.pdf
L30	Baker Scaffold Quick Reference Sheet	http://docs.okland.com/msa/safety/L30.pdf
L31	Fabricated Frame Scaffold Quick Reference Sheet	http://docs.okland.com/msa/safety/L31.pdf
L33-1	Soil Disturbance Permit – EXAMPLE ONLY	http://docs.okland.com/msa/safety/L33-1.pdf
L33-2	Soil Disturbance Permit Process Map	http://docs.okland.com/msa/safety/L33-2.pdf

L33-3	Utility Backfill Standards	http://docs.okland.com/msa/safety/L33-3.pdf
L33-4	Survey Layout Standards	http://docs.okland.com/msa/safety/L33-4.pdf
L35	Floor Opening Hole Cover Standard	http://docs.okland.com/msa/safety/L35.pdf
L36	Barricade Tape Guidelines	http://docs.okland.com/msa/safety/L36.pdf
L40	Stationary Supported Scaffold Permit	http://docs.okland.com/msa/safety/L40.pdf
L41	Mudsill Reference Sheet	http://docs.okland.com/msa/safety/L41.pdf
L42	Permit Sleeve/Magnet Tag Tracking Policy	http://docs.okland.com/msa/safety/L42.pdf
L43	Okland Required PPE Variance Policy	http://docs.okland.com/msa/safety/L43.pdf
L44	Water Crash Cart Policy	http://docs.okland.com/msa/safety/L44.pdf
L49	Drilling Piles/Caissons Excavation Permit	http://docs.okland.com/msa/safety/L49.pdf
L50	Temporary Heater Permit	http://docs.okland.com/msa/safety/L50.pdf
L51	Demolition Permit – Selective Demolition	http://docs.okland.com/msa/safety/L51.pdf
L52	Demolition Permit – Total Demolition	http://docs.okland.com/msa/safety/L52.pdf
L54	Guardrail Removal and Modification Permit	http://docs.okland.com/msa/safety/L54.pdf
L55	Danger – Overhead Work Sign	http://docs.okland.com/msa/safety/L55.pdf
L56	Danger – Energized Electrical Room Sign	http://docs.okland.com/msa/safety/L56.pdf
L57	Scaffold Stair Tower Permit	http://docs.okland.com/msa/safety/L57.pdf
L58	Trash Chute Permit	http://docs.okland.com/msa/safety/L58.pdf
L59	Overhead Protection at Building Perimeter Policy	https://docs.okland.com/msa/safety/L59.pdf
L60	Non-Crane Lift Planning Worksheet	http://docs.okland.com/msa/safety/L60.pdf
L61	Tower Crane Assembly Checklist	http://docs.okland.com/msa/safety/L61.pdf
L62	Tower Crane Disassembly Checklist	http://docs.okland.com/msa/safety/L62.pdf
L63	Tower Crane Assembly/Disassembly Authorization Form	http://docs.okland.com/msa/safety/L63.pdf
L66	Man & Material Hoist Assembly Checklist	http://docs.okland.com/msa/safety/L66.pdf
L67	Okland’s Personnel/Material Hoist Authorization Form	http://docs.okland.com/msa/safety/L67.pdf
L68	Critical Lift Criteria & Permit	http://docs.okland.com/msa/safety/L68.pdf
L69	Mobile Equipment in Public Right of Way Permit	http://docs.okland.com/msa/safety/L69.pdf

EXPECTATIONS

Subcontractors shall establish Policies, Standards, and Procedures to ensure the safety & health of workers and others exposed to Subcontractor's activities.

At a minimum, Subcontractor's Policies, Standards, Procedures and Performance shall meet and be in strict compliance with:

- All local, state, and federal governmental regulations (i.e., legal obligations)
- Contractual obligations including the requirements set forth in the contractual agreement between Okland and Owner
- This Safety & Health Manual and those documents incorporated by reference within it
- Project established requirements
- The manufacturer's recommendations for all tools, materials, and equipment used by Okland and/or Subcontractor

If there is a conflict as to which requirement is to be adhered to between local, state, and/or federal governmental regulations, contractual obligations, this S&H Manual and those documents incorporated by reference within it, Project requirements, and/or manufacturer recommendations, the most stringent components of each shall apply while always maintaining compliance with legal obligations. In many cases this will result in Subcontractor having to comply with legal obligations as well as additional items required in this S&H Manual. This is intentionally the design, and in part, the intent of this S&H Manual.

Management/supervision shall plan safety into each work task. Although the ultimate success of our Safety, Health and Environmental Program depends on the full commitment and cooperation of each individual worker, management/supervision is responsible to ensure that applicable rules and procedures are established and enforced and that effective training programs are employed.

Safety, occupational health, and environmental protection must never be sacrificed for production. These elements are integral parts of quality control, cost reduction and job efficiency. Each supervisor must be personally concerned with the performance demonstrated by workers under his/her supervision as their performance relates to these elements.

SCOPE

The requirements set forth in this manual shall apply to all work managed by Okland Construction Co., Inc.

The requirements in this manual have been placed in their respective sections only to facilitate the cross-referencing of the requirements with specific OSHA regulations.

However, unless otherwise stipulated in this manual, all requirements in this manual shall apply to all scopes of work irrespective of the requirement's heading or location it has been placed within the manual.

ACRONYMS & DEFINITIONS

For the sole purpose of this manual, the following definitions shall apply.

ADA	Americans with Disabilities Act
ANSI	American National Standards Institute
AWG	American Wire Gauge
CFR	Code of Federal Regulations
Competent Person	One who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them
EPA	Environmental Protection Agency
GFCI	Ground Fault Circuit Interrupter
Infeasible	It is impossible to perform the work in compliance with the Okland specified requirement(s). Subcontractor/Supervisor has the burden of establishing infeasibility, regardless of cost or immediately available manpower, tools, equipment and/or materials.
JHA	Job Hazard Analysis
LOTO	Lock-Out Tag-Out
MSA	Master Subcontract Agreement
MUTCD	Manual on Uniform Traffic Control Devices
Narrow Frame Scaffold	A scaffold that is 46" or less in its least dimension
NFPA	National Fire Protection Association
OEM	Original Equipment Manufacturer
Okland	Okland Construction Co., Inc., and all affiliated companies respective to the contract documents
OSHA	Occupational Safety and Health Administration
Owner	The entity Okland Construction Co., Inc. has established a Prime Contract with
PEL	Permissible Exposure Limit
PFAS	Personal Fall Arrest System
PPE	Personal Protective Equipment
PPM	Parts Per Million
Project	Those areas indicated in the contract documents where construction work is performed for Okland. This includes, but is not necessarily limited to, accessible areas of the Project, staging areas, fenced construction areas, jobsite trailers, warehouses, company-provided parking areas, vehicles and equipment on the Project, driveways, lockers, toolboxes, or other related storage areas used by employers and/or employees.

Project Management Team	The Okland Project Director, Project Manager, Project Superintendent(s), Project Engineer(s), and any <u>full time</u> Project Safety Manager(s) assigned to the Project and/or multiple Projects managed by the same team.
PTP	Pre-Task Plan
QR	Qualified Rigger
QSP	Qualified Signal Person
Qualified Person	One who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training, and experience, has successfully demonstrated his/her ability to solve or resolve problems related to the subject matter, the work, or the project.
ROPS	Roll Over Protective Structure
SDS	Safety Data Sheets (formerly MSDS)
Subcontractor	The entity that Okland is contracted with to provide work on the Project and their respective lower-tiered Subcontractors.
Supervisor	An employee designated by his/her employer to supervise activities relating to work on the Project and who meets Okland's requirements to be a Supervisor on an Okland Project
Supported Scaffold	Any scaffolding other than suspended scaffolding (e.g., scaffolds supported by outrigger beams, brackets, poles, legs, uprights, posts, frames, or similar rigid supports)
SWPPP	Storm Water Pollution Prevention Plan
Zero Tolerance	Not allowing work to begin or continue if it is not safe or not in compliance with the established requirements

SECTION 1

GENERAL REQUIREMENTS

GENERAL REQUIREMENTS

THE WILLIAMS-STEIGER OCCUPATIONAL SAFETY AND HEALTH ACT OF 1970

Section 5. Duties

- a. Each employer
 - i. shall furnish to each of his employees' employment and a place of employment which are free from recognized hazards that are causing or are likely to cause death or serious physical harm to his employees;
 - ii. shall comply with occupational safety and health standards promulgated under this Act.
- b. Each employee shall comply with occupational safety and health standards and all rules, regulations, and orders issued pursuant to this Act which are applicable to his own actions and conduct.

“THE GENERAL DUTY CLAUSE”

Section 5(a)(1) of the William-Steiger Occupational Safety and Health act of 1970 has become known as “The General Duty Clause”. It is a catch-all clause for citations if OSHA identifies unsafe conditions for which no specific clause exists.

In practice, OSHA court precedent and the review commission have established that if the following elements are present, a “general duty clause” citation may be issued.

- a. The employer failed to keep the workplace free of a hazard to which employees of that employer were exposed.
- b. The hazard was recognized (Examples might include: through safety personnel, employee(s), organization(s), trade organization(s) or industry customs).
- c. The hazard was causing or was likely to cause death or serious physical harm.
- d. There was a feasible and useful method to correct the hazard.

GENERAL PROJECT REQUIREMENTS

1. Stop Work Authority:

- a. Every worker on the project shall have Stop Work Authority if they believe the activity is not safe or not in compliance with the established requirements of this manual.
 - i. If work is stopped, the supervisor over the work shall be immediately notified.
 - ii. In addition, the supervisor for the worker initiating the Stop Work action shall also be notified immediately.
 - iii. Work shall not proceed until the concern noted has been effectively addressed or found to be in compliance with the requirements of this manual.

2. Supervision:

- a. Subcontractor shall have a competent, English proficient supervisor on site at all times while work is being performed under the Subcontractor's contracted scope of work.
- b. Each supervisor shall implement and enforce the safety & health requirements.
- c. Subcontractor's designated on-site safety representative shall have authority within the employer's organization to stop, correct and/ or change work activities performed by their employees and their subtier's employees.
- d. Prior to beginning work, Subcontractor shall identify in writing, to Okland, the name of its onsite safety representative, competent person(s) (as defined and where required by OSHA) and the individual who oversees safety-related issues from their home office.
- e. Subcontractors shall not be on Okland projects without an Okland supervisor (i.e., Okland Project Responsible Person) being on site.

3. Site-Specific Safety Plan:

- a. If Subcontractor's safety performance is found to not be consistent with the requirements set forth in this manual, at the discretion of Okland, Subcontractor shall develop and submit to Okland, a Site-Specific Safety Plan that details the specific activities, hazards, and controls for Subcontractor's scope of work.
 - i. At no time shall Subcontractor's Site-Specific Safety Plan stipulate requirements less than those established in this manual.

4. Lone Worker:

- a. Workers conducting the following activities shall only do so when they can be visually seen and heard by at least one other worker. If the other worker(s) is/are not on the lone-worker's crew, the pre-task plan must identify who will be within sight and earshot of the lone worker. The identified person must be conducting a joint activity with the lone worker.
 - i. Activities where a Personal Fall Arrest System (PFAS) is required to be used.
 - ii. Activities where work is being performed on elevated decks/floors where:
 1. A fall hazard exists on a roof that is being controlled with a warning line system; or
 2. One or more ladders of any type are the sole means of access/egress.
 - iii. Activities in a trench 4' or greater in depth.
 - iv. Entering a permit or a non-permit required confined space.
 - v. Conducting live electrical work of any voltage.
 - vi. Use of a respirator for any reason.

- vii. Conducting demolition work.
- viii. Work on or around the project outside of normal hours.
- ix. Operating earth or material handling equipment (e.g., forklifts, skid-steers, back-hoes, scrapers, graders) where at least one of these conditions exist:
 - 1. Operation is in remote areas on the project.
 - 2. Operation is in a remote yard.
 - 3. Operation poses a risk of other workers or the public being struck by the equipment.
- x. Activities involving the maintenance or repair of mobile or fixed equipment (e.g., MEWPs, man/material hoists, forklifts, skid-steers, back-hoes, scrapers, graders) where there is a potential for the released of stored energy (e.g., electrical, hydraulic, gravity).

5. Meetings:

- a. Subcontractor shall participate in preconstruction meetings. Subcontractor's Project Manager, site Superintendent, foremen and designated safety representative shall attend this meeting.
- b. Okland will conduct a weekly Subcontractor coordination meeting. Subcontractor's site Superintendent shall attend this meeting.
- c. Okland may conduct additional meetings to address safety-related issues as necessary. Subcontractor's Project Manager, site Superintendent, foremen, designated safety representative and employees shall attend these meetings as directed by Okland.

6. Documentation:

- a. **Okland shall have the right to but not the duty to review** any and all of Subcontractor's documentation required by OSHA regulations and/or this manual. This right does not extend to a worker's **confidential medical** records. This review shall not constitute approval in any form.

7. Permits & Permit Sleeves:

- a. Subcontractor shall comply with the Okland [Permit Sleeve / Magnet Tag Tracking Policy](#) for all Okland issued permits (e.g., hot work, soil disturbance, over 4' excavation and trench access, piling drilling, over 24' scaffold erection).
- b. Additional Okland and/or project specific training may be required for supervisors and workers prior to issuing them permits.

8. Required Age:

- a. No person under the age of **eighteen (18)** shall be allowed on the Project, which includes access through Project gates, irrespective of the minor staying in a vehicle without the explicit authorization from Okland's Superintendent and Okland's Project Manager.
- b. No person under the age of **eighteen (18)** shall be allowed to work on the Project.

9. Tobacco:

- a. Use of tobacco products, including e-cigarettes, are prohibited on the project except in areas designated by Okland.

10. Animals:

- a. No animals shall be allowed on the Project including pets, irrespective of the animal staying in the cab of a vehicle, in an office, in a lock-up or similar work/break area.

11. Audio Devices:

- a. Audio devices such as radios, speakers, and similar that are audible in the general work area are prohibited.

12. Communication Devices:

- a. The use of personal cellular/mobile telephones, on the Project are prohibited unless used in designated areas (e.g., fire extinguisher areas, water stations, break areas, etc.).
- b. The use of cellular/mobile telephones, on the Project while in a safety sensitive position (e.g., operating equipment, exposed to a fall hazard, climbing ladders, walking around mobile equipment) is prohibited.
- c. Okland's onsite project management team may, at their discretion, institute additional restrictions on the use of electronic devices including cellular/mobile telephones.

13. Unsafe Conditions/Acts:

- a. Workers are to report unsafe acts/conditions to their supervisor immediately. If the worker feels as though his/her safety concern is not being addressed, they are encouraged and expected to report their concern to the Okland Superintendent and/or the Okland Project Manager.
- b. Workers shall not work under unsafe conditions.
- c. Workers shall not conduct unsafe acts.
- d. Horseplay shall not be tolerated.
- e. Fighting shall not be tolerated (this includes physically touching any other person with the intent to do harm, threaten, or intimidate).

14. Disciplinary Action:

- a. Subcontractors shall establish and enforce an effective disciplinary program to address their employees whom Subcontractor observes not in compliance or found to have been not in compliance with the requirements stipulated in this manual.
- b. In addition, any person Okland observes not in compliance or found to have been not in compliance with the requirements stipulated in this manual is subject to being removed from the Project at the sole discretion and for a duration stipulated by Okland's Project management. Such removal may extend beyond the specific project in which the non-compliance action occurred, and the person may be denied access to all Okland projects at the discretion and for a duration stipulated by Okland's Corporate Risk Management and Operations Team.
- c. Disciplinary action will be consistent with Okland's Zero Tolerance approach to noncompliance with Okland's safety, health, and environmental program.

VISITORS

A visitor is any individual who visits a project site and who does not have a contractual relationship with Okland Construction. Individuals who are performing or supervising work on the project are not considered visitors. Each visitor must have a signed Release of Liability-Waiver agreement with Okland Construction and complete the required safety orientation course.

1. Before accessing the project, all prospective visitors to the project shall report to the Okland project office to check in with the Okland Superintendent or other authorized Okland supervisor.
2. All visitors will be designated as either “Visitors” (see #5) or “Blue Vest Visitors” (see #6)
 - a. See “Orientation and Waiver Requirements” matrix below.
3. **Visitors** are individuals who (i) are familiar with construction sites, (ii) have experience being on construction sites, and (iii) are wearing all of the project’s required PPE.
 - a. Okland’s Superintendent or other authorized Okland supervisor must do the following before Visitors are granted access to the project:
 - i. Ensure each Visitor has proper authorization from an Okland project management team member before each Visitor is allowed on the project site;
 - ii. Ensure each visitor reads and signs a [Visitor Orientation and Release of Liability-Waiver Form](#);
 - iii. Ensure each visitor is instructed that photos and videos are not permitted without approval from the Project Director;
 - iv. Question the Visitor on the specific area(s) that he or she would like to visit;
 - v. Advise the visitor(s) on time limits of the visit and describe any restricted areas, including any area(s) the visitor has requested to visit;
 - vi. Ensure the visitor will be wearing the project’s required PPE at all times while on the project site;
 - vii. Verify each visitor parked their vehicle in the proper location.
4. **Blue Vest Visitors** are individuals who (i) are not familiar with construction sites; (ii) do not have sufficient experience visiting construction sites; or (iii) are not wearing all of the project’s required PPE. In addition to the Visitor rules listed above, Blue Vest Visitors must:
 - a. Read and sign the [Okland Blue Vest Visitor Orientation and Release of Liability-Waiver Form](#);
 - b. Wear an Okland-issued blue visitor safety vest;
 - c. Meet the following minimum requirements for apparel and PPE while visiting the project:
 - i. Closed toed shoes;
 - ii. No shorts (although not recommended, skirts or dresses are allowable for Blue Vest Visitors);
 - iii. Hard hat; and
 - iv. Glasses (safety glasses are preferred, but non-safety prescription eyeglasses are also acceptable).
 - d. Be accompanied by an Okland-approved escort at all times while on the project site.

VISITOR ORIENTATION & WAIVER REQUIREMENTS

INDIVIDUAL	BLUE VEST ORIENTATION FORM/WAIVER	VISITOR ORIENTATION FORM/WAIVER	FULL ORIENTATION
OKLAND			
Active Management	N/A	N/A	YES
Guests of Okland	YES	N/A	N/A
Child	YES	N/A	N/A
Employees not Involved in Project	N/A	N/A	YES
OWNERS			
Active Management (Owner Reps Included)	N/A	YES	N/A
Guests of Owners	YES	N/A	N/A
Owner's Child	YES	N/A	N/A
Non-related Parties	YES	N/A	N/A
ARCHITECTS AND ENGINEERS (NON-DESIGN/BUILD)			
Active Management	N/A	YES	N/A
Guests of AE	YES	N/A	N/A
Owner's Child	YES	N/A	N/A
Non-related Parties	YES	N/A	N/A
TRADE PARTNERS & SUPPLIERS			
Trade Partners - Active on Project	N/A	N/A	YES
Trade Partners Office Personnel	N/A	YES	N/A
Manufactures/Materials Rep	N/A	YES	N/A
Guests or Trade Partner	YES	N/A	N/A
VISITORS			
Student Visitors	YES	N/A	N/A
Curious Community Members	YES	N/A	N/A
VIP's	YES	N/A	N/A

GENERAL LIABILITY

1. As the general public does not have the knowledge and awareness of construction-related hazards, all work in and/or adjacent to the public right-of-way shall be conducted with added precaution. In addition, since there are no specific regulations addressing what added controls need to be implemented to prevent injury to the general public, each exposure shall be evaluated on a case-by-case basis. **Subcontractors and workers shall plan for added control measures anytime the general public might be exposed to construction-related hazards.** However, Okland reserves the right to require Subcontractor to increase its general liability control measures if Okland determines that existing and/or planned control measures are inadequate. Such increased measures shall be at Subcontractor's cost unless otherwise agreed to by Okland.
2. Appropriate control measures shall be implemented to address/abate all attractive nuisances. For the purpose of this manual, an attractive nuisance is a hazardous condition that meets the following criteria:
 - a. The place where the condition exists is one upon which the Subcontractor knows or **has reason to know** that the general public is likely to trespass, and;
 - b. The condition is one of which the Subcontractor knows or **has reason to know** and realizes or **should realize** will involve an unreasonable risk of injury, death or serious bodily harm to the general public;
 - c. The general public, because of their lack of knowledge and understanding to construction related hazards, may not discover the condition or realize the risk involved in inter-meddling with it or in coming within the area made dangerous by it;
 - d. The utility to the Subcontractor of maintaining the condition and the burden of eliminating the danger are slight as compared with the risk to the general public involved, and;
 - e. The Subcontractor fails to exercise reasonable care to eliminate the danger or otherwise to protect the general public.
3. Subcontractor shall not perform any work in, modify, or close any public sidewalk or road without a City permit and State permit (if it is under the authority of the State) and without the explicit authorization of Okland's Project Manager and Okland's Superintendent.
4. An Okland Safety Manager shall be consulted prior to any work in the public right-of-way.

SECURITY

Subcontractor is responsible for the security of their property (which includes but is not limited to their tools, their employee's tools, their trailers, their conexes, their building materials, and similar) while on the Project.

1. Okland will evaluate the general security needs for the Project. When feasible and practical Okland will erect a fence-line around the main Project boundaries. Regular access/egress gates will be established and open during construction hours.
2. If a Subcontractor requests to open a gate or portion of the fence-line in addition to the regular Project access/egress gates **and** Okland authorizes the request, the requesting Subcontractor shall "man" the gate with a trained, competent and qualified worker at the requesting Subcontractor's cost. If for any reason Okland determines that the worker manning the gate is not properly trained, competent and/or qualified to conduct his/her duties as a gate guard, Subcontractor shall immediately relieve the worker of his/her duties and have the position re-assigned.
3. Okland has the authority to prohibit any unauthorized personnel from accessing the Project or any portion thereof.
4. Subcontractor, on behalf of its employees, grants to Okland the right to periodically conduct random searches of vehicles on site, lunch boxes, tool boxes, etc. for controlled or prohibited substances and/or stolen tools, materials, etc. Okland has the right but not the duty to perform such searches.
5. Subcontractor and its employees shall not remove non-personal tools and/or equipment from the Project without permission from their supervisor and Okland.
6. Subcontractor and its employees shall not give away and/or remove building materials from the Project without explicit written permission from Okland's Superintendent.

SECTION 2

SPECIFIC SAFETY REQUIREMENTS

SPECIFIC SAFETY REQUIREMENTS

Subpart A – General

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Subcontractor's request to OSHA (including State Plan States) for and/or the award of variances from OSHA safety regulations shall not override Subcontractor's obligation to comply with Subcontractor's contract documents which includes the requirements set forth in this manual.
2. Subcontractor shall immediately inform Okland's Project Manager and Superintendent of all OSHA and/or other regulatory agency inspections and/or notices of complaint.

Subpart B – General Interpretations

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Subcontractors shall ensure that the requirements set forth in this manual are included in their contract documents, purchase orders and/or service agreements, with any/all lower-tiered Subcontractors.
2. Subcontractor shall be responsible for compliance with all safety regulations and requirements associated with their respective scope of work irrespective of Subcontractor subcontracting portions of its scope of work to lower-tiered Subcontractors.
3. Subcontractor shall immediately abate all recognized and/or identified hazards in their work environment.
4. If Subcontractor believes that the action needed to abate a hazard is not their responsibility, Subcontractor shall either correct the hazard or remove any exposed workers from exposure to the hazard and immediately notify Okland's Superintendent.

Subpart C – General Safety and Health Provisions

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Safety Inspections:
 - a. Subcontractor's designated Project Superintendent, general foreman, foremen, and lead personnel shall conduct general daily safety inspections of Subcontractor's work areas, materials, tools, equipment, and personnel.
 - b. Subcontractor's designated Project Superintendent shall conduct a weekly **documented** safety inspection of their work area, materials, and equipment. This inspection shall be inclusive of all of Subcontractor's work including all work scopes that Subcontractor has

subcontracted to lower-tiered Subcontractors. Okland has the right to review but not the duty to collect this documentation. These inspections shall be documented in a manner acceptable to Okland.

- c. In Okland's Arizona market, Subcontractor shall conduct weekly documented Heat Safety focused inspections between June 15th and September 30th of each year. Inspections are to be recorded in the Project's designated Management System Platform (B3F, Procore, Autodesk, etc.).
- d. In Okland's Arizona market, Subcontractor shall conduct weekly documented Monsoon Preparedness focused inspections between June 15th and September 30th of each year. Inspections are to be recorded in the Project's designated Management System Platform (B3F, Procore, Autodesk, etc.).

2. Work Activity Planning - Written:

- a. Prior to the start of each work shift, and as needed throughout the day as tasks change, Subcontractor's individual crew Foreman shall plan for and review the necessary safety measures for that work shift with their employees. This work plan shall include an inspection of their work area in an effort to identify and correct unsafe conditions prior to the start of work. Documentation of this process shall be kept on file at the jobsite. Okland's [Pre-Task Plan form \(PTP\)](#) shall be utilized for this purpose. Subcontractors can utilize their own forms for this purpose as long as their form/process provides for an equal process.
 - i. PTPs shall be kept in the work area so adjustments can be made when warranted and/or so additional team members can review the PTP with their supervisor if they join the work task after the PTP has been completed.
- b. For work activities that are unusual or pose a higher degree of risk related to Subcontractor's scope of work, or when requested by Okland, Subcontractor shall complete and submit to Okland a written hazard assessment prior to the beginning of such activity. This assessment is commonly referred to as a [Job Hazard Analysis \(JHA\)](#). The JHA shall be specific to the Project. The JHA shall detail the controls that will be implemented and enforced to achieve compliance with all applicable requirements. Modifications to the Subcontractor's initial JHA will be required as conditions and/or means/methods change. Okland shall have the right to review, but not the duty to approve, the JHA. Subcontractor shall implement the controls identified in the JHA.
- c. While conducting the PTP and JHA process the [Hierarchy of Controls](#) must be considered to ensure that the most feasibly effective controls are implement.

3. Safety Education:

- a. Project Orientation:
 - i. Prior to conducting work on an Okland Project, all workers shall:
 - 1. Complete Oklands Annual Safety Orientation.
 - a. This orientation shall be completed virtually, via Okland's Trade Partner Orientation Platform
 - b. This orientation shall be completed **PRIOR** to attending Okland's Site Specific Orientation.
 - c. The duration of the Annual Safety Orientation is approximately:
 - i. English: 2 hours and 10 minutes
 - ii. Spanish: 2 hours and 30 minutes

- d. The Annual Safety Orientation shall be completed, at least annually by each worker.
 - 2. Attend a Site-Specific safety orientation conducted by Okland at their project office.
 - a. The duration of the Site-Specific Orientation is typically 30-minutes. However, additional time may be required to address client, facility, regulatory, or other Site-Specific hazards/requirements.
 - ii. These orientations are not a training course.
 - iii. All workers attending the orientation shall be properly trained and authorized **by their employer** to conduct their respective duties.
 - iv. Site-Specific Orientations conducted by Okland will be in English.
 - v. If Subcontractor wants individuals who do not speak and/or understand English to work on the Project, Subcontractor shall schedule a separate non-English site-specific orientation with Okland's Superintendent to accommodate these individuals. Subcontractor shall give a 48-hour notice to Okland for the separate non-English site-specific orientation. **It is the responsibility of Subcontractor to provide an interpreter for the site-specific orientation.**
 - b. Okland + Orientations:
 - i. Additional orientations will be required for Okland work authorized by "Permit" (e.g., Hot Work, Soil Disturbance, etc.) as well as other specific high risk work activities (e.g., crane operation, rigging, etc.)
 - c. Continued Safety Trainings:
 - i. Subcontractor shall conduct continued safety trainings with their employees addressing hazards their employees are exposed to directly by their scope of work, and indirectly from work being conducted by others. The training topics and verification of all workers in attendance shall be documented. A copy of this documentation shall be made available to Okland upon request.
 - d. Other Conditions, Equipment & Material:
 - i. As the need arises and/or as direct by Okland, Subcontractor shall conduct additional safety education with its employees when unusually hazardous conditions are present or are likely to become present, the use of specialized equipment is necessary and/or when the use of hazardous materials will be required.
- 4. Reporting and Investigating of Injuries and Incidents:
 - a. Subcontractor shall immediately inform the Okland Superintendent of all work-related injuries/illnesses to Subcontractor's employees (including lower-tier Subcontractor's employees) and of all safety-related incidents including "near incidents/near misses/close-calls."
 - b. Subcontractor shall submit to Okland's Superintendent, within 24-hours of the injury or incident, written investigation reports pertaining to the incident/injury. Reports shall be completed by the worker(s) involved in the incident/injury and supervisor(s) of the worker(s) involved in the incident/injury. Additionally, if any witnesses have information pertaining to the incident/injury the witnesses also shall complete a report.
 - c. Subcontractor shall immediately report all known actual and/or potential losses to the Okland Superintendent and Okland Project Manager.
 - d. Okland shall have the right but not the duty to suspend operations and personnel from the site until satisfied that all information pertaining to safety related incidents is examined and investigated either by Okland and/or by Subcontractor.

- e. Subcontractor shall participate and cooperate in any injury or incident investigation it, its employees, or its tiered subcontractor is involved in and shall complete and submit to Okland any report forms requested by Okland within legal compliance.
- f. Subcontractor shall be required to attend and participate in any investigative/Root Cause Analysis (RCA), follow up, and /or disciplinary review meetings as a result of injury or incidents.
- g. Subcontractor shall participate in Okland's Root Cause Analysis process as requested by Okland for incidents in which Subcontractor is directly or indirectly involved in.

5. Housekeeping:

- a. It shall be the responsibility of the Subcontractor to keep the premises clear and clean of their debris throughout the day on an "as-you-go" basis and to provide equipment and labor necessary to remove all debris and surplus material from the site. If, in the opinion of Okland, this requirement is not being met, Okland may provide labor and/or equipment necessary to perform the required clean-up. All associated costs shall be deducted from the Subcontractor's contract.
- b. Subcontractor shall provide an adequate number of trash receptacles to facilitate the collection of debris generated from their respective scope of work. Once full, the trash receptacle(s) shall be emptied immediately.
- c. Stairways and walkways shall be kept clear of debris, materials, tools, and other obstructions.
- d. Loose material that has the potential to be picked up and carried by the wind shall be adequately secured to prevent displacement of the material by the wind.
- e. Dumpsters shall be placed in locations free from ignition sources.
- f. Dumpsters shall be placed at least 15' away from any portable chemical toilet.

6. Emergency Action Plan:

- a. Subcontractor shall comply with the any and all Project specific Emergency Action Plans.
 - i. To assist with the notification of an evacuation, ALL Okland projects are equipped with hand-held air horns. These air horns are kept in the Okland project office.
 - 1. To differentiate between an air-horn used for crane hoisting purposes and an evacuation notification, evacuation notifications shall be conducted by sounding 3 quick air horn blasts.
 - 2. Qualified Riggers/Signal Persons shall NEVER use 3 quick air horn blasts for crane hoisting purposes.
- b. These plans will be periodically evaluated and revised as necessary.
- c. Subcontractors will be advised of changes to the plans during weekly coordination meetings and monthly safety meetings. Subcontractor shall comply with the revisions.
 - i. Communication:
 - 1. **Emergency Telephone Numbers** – A list of emergency telephone numbers shall be posted in Subcontractors project office locations.
 - 2. **Mobile Telephone** – Subcontractor shall provide all of its on-site supervisors with a mobile telephone to allow for general and emergency communication.
 - 3. **Interpreters** – Subcontractor shall provide an interpreter for all required site-specific orientations, trainings and meetings for their employees who do not understand and/or speak English. In addition, Subcontractor shall provide for an interpreter to be on-site at all times when its employees who do not understand and/or speak English are on site.

- d. Subcontractor shall ensure that training on the requirements of Emergency Action Plans is conducted with all personnel under Subcontractors control on at least a quarterly basis.

Subpart D – Occupational Health and Environmental Controls

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. First Aid:
 - a. First Aid kits and supplies shall be a standard industrial kit and be appropriate for Subcontractors scope, location, and number of employees.
 - b. Where the eye(s) of any person may be exposed to any injurious chemical, Subcontractor shall provide and ensure that a suitable eye wash station is immediately available in the immediate work area for emergency use. Eye wash stations shall be of sufficient size to adequately flush the chemical from the eye(s).
2. Sanitation:
 - a. **Toilets** – Unless otherwise stipulated in contract documents, each subcontractor is required to provide toilet facilities for their employees on the project.
 - i. Workers found relieving themselves in areas other than the Project established toilet facilities shall be **immediately** and **permanently** removed from all Okland Projects.
 - ii. Hand cleansing or sanitizing agents shall be provided for each toilet unit.
 - iii. Toilets shall be placed in an area not exposed to overhead hazards (e.g., on the perimeter of buildings where overhead work is taking place) and shall be at least 15' away from any dumpster.
 - b. **Food and Drink** – Food and drink shall only be permitted and allowed to be consumed in areas designated by Okland.
 - c. **Smoking and Tobacco** – Smoking and use of tobacco products shall only be permitted in areas designated by Okland and in compliance with legal obligations.
 - d. **Potable Water** – Unless otherwise stipulated in contract documents, each subcontractor is required to provide potable water for their employees on the project.
 - i. When reusable water containers are utilized for potable water, they are to be cleaned at least every two days with a sanitizing agent to prevent the spread of disease.
3. Illumination: (see [Subpart K](#) of this document).
4. Hazard Substances and Hazard Communication
 - a. Subcontractor shall develop and implement an effective written hazard communication program specific to their scope of work.
 - b. Subcontractor shall submit to Okland's Superintendent a Safety Data Sheet (SDS) and estimated quantities for each hazardous chemical, substance, material, product, or similar that the Subcontractor will be using on the Project. This information shall be provided prior to the product's arrival on the Project and shall be in both electronic and paper form.
 - c. All chemical containers including small daily use and one-time use containers must be labeled.

- d. Under no circumstance shall food or beverage containers be utilized to hold hazardous chemicals, substances, materials, products, or similar of any nature.
- e. All chemical containers shall be covered with a lid when not in use.
- f. Unless otherwise stipulated in the contract documents each subcontractor is responsible for the waste they generate from the use of hazardous chemicals, substances, materials, products, or similar as well as the associated costs.
- g. All waste generated from the use of hazardous chemicals, substances, materials, products, or similar shall be disposed of in accordance with manufacturer's recommendations as well as all local, State and Federal regulations. Hazardous waste and/or hazardous chemicals, substances, materials, products, or similar shall not be disposed of in general waste receptacles/dumpsters nor shall they be disposed of in drains, manholes, vaults, or tanks.
- h. Unless otherwise stipulated in the contract documents, each subcontractor is responsible to remove all unused hazardous chemicals, substances, materials, products, or similar materials from the project immediately after the scope of work they were used for is completed.
- i. Each employer shall train its employees who are assigned duties which may cause exposure to hazardous chemicals, substances, materials, products, or similar or who may be exposed to hazardous chemicals, substances, materials, products, or similar on the appropriate safety & health requirements for working with and/or around the hazardous substances.
- j. Subcontractor shall provide a spill kit that is of adequate size and type to address spills of chemicals involved in subcontractor's scope of work.
- k. When conducting activities that will disturb on coatings on surfaces (e.g., welding, cutting, burning, grinding, chipping, abrasive blasting, rivet busting, etc.), a pre-assessment shall be conducted to determine if the coating is hazardous or will become hazardous once it is disturbed. If a hazard is identified, the hazard must be effectively controlled.

5. Mold:

- a. Subcontractor shall immediately report any water damage, leaks, or water intrusion to the Okland Superintendent.
- b. Subcontractor shall immediately report the presence of mold that Subcontractor observes on any portion of the project to the Okland Superintendent.
- c. Unless otherwise directed by Okland, Subcontractor shall not remove or disturb visible mold contaminated materials. If such direction is given it shall be conducted in compliance with industry standard safety requirements and under the guidance of Certified Industrial Hygienist.
- d. Subcontractor shall be responsible for any mold remediation that is necessary as the result of subcontractor's actions or inactions.

6. Lasers:

- a. Areas in which lasers are used shall be posted with standard laser warning placards.

7. Wind:

- a. The Okland project superintendent The Okland project superintendent, or the Okland Project Responsible Person (PRP), and the supervisor for each Subcontractor shall monitor daily and weekly forecasts and ensure that all materials, building components, temporary fences, and equipment are adequately stored and secured for the anticipated wind forces forecasted.
- b. Okland and Subcontractor shall comply with the following actions when wind speeds, sustained or frequent gusts, meet or exceed the below listed Miles Per Hour (MPH):

- i. **20 MPH** = The Okland project superintendent, or the Okland Project Responsible Person (PRP), and the supervisor for each Subcontractor shall evaluate their exterior elevated work and determine if the work can safely proceed or needs to be suspended.
 - 1. Examples of work activities that should be suspended are:
 - (a) Critical crane lifts of any type (unless OEM allows for a higher, or requires a lower wind speed for the configuration the crane is in)
 - (b) Hoisting of personnel with a crane man-basket
 - (c) Use of luffing type cranes (unless OEM allows for a higher wind speed for the configuration the crane is in)
 - (d) Hoisting of large surface area loads (e.g., gang forms, tilt panels, window systems, sheet piles) with cranes, forklifts and/or other mobile equipment (e.g., boom-trucks, excavators, front-end loaders)
 - (e) Wrapped scaffolds, regardless of height (unless engineered for a higher wind speed)
- ii. **30 MPH** = In addition to complying with the requirements for 20 MPH winds, all elevated work exposed to the wind shall be suspended.
 - 1. Examples of elevated work activities that shall be suspended are:
 - (a) All crane activities
 - (b) All hoisting activities with mobile equipment where the load is hoisted in a suspended fashion (e.g., loads below the tines of a forklift, loads below the bucket of an excavator, loads below the bucket of a front-end loader)
 - (c) Aerial lifts
 - (d) Bucket trucks
 - (e) Scissor lifts
 - (f) Scaffolds of any type (unless engineered for a higher wind speed)
 - (g) Work from ladders
 - (h) Forklift man-baskets
 - (i) Activities on roofs
 - (j) Elevated activities near the perimeter of elevated floors
 - (k) Working from the sides of walls, formwork, rebar assemblies, or similar
 - (l) Manual handling of material with large surface areas (e.g., sheetrock, plywood, glazing)
- iii. **40 MPH** = In addition to complying with the requirements for 20 MPH and 30 MPH winds, all outdoor activities shall be suspended.
 - 1. Examples of outdoor work activities that shall be suspended are:
 - (a) Any activity where the worker is not protected by an enclosed cab of equipment or vehicle.
 - (b) Any demolition activity.
 - (c) Scaffold stair towers (unless engineered for a higher wind speed).
 - (d) Use of exterior Man/Material hoists (unless the OEM or AHJ allows for/requires a higher/lower wind speed).
 - (e) Placement of concrete with a concrete pump truck when the boom is elevated.
 - (f) Placement of concrete with a fixed concrete placing boom exposed to the wind.

- c. Examples of the wind speed visual clues can be found at: <https://www.weather.gov/pgr/wind>
- d. Many factors can affect wind speed on a project (e.g., elevation, terrain, proximity to other structures). General wind speeds for your closest large airport can be found at: <https://www.weather.gov/>
- e. Alternately, there are many mobile apps. on the market that you can download to monitor the windspeed for your project's location.
 - i. Examples are:
 - 1. Windy: <https://www.windy.com>
 - 2. UAV Forecast: <https://www.uavforecast.com/>
- f. Following wind events that are greater than 40 MPH, and prior to production work resuming, the Okland project superintendent, or the Okland Project Responsible Person (PRP), or their designee, and the supervisor for each Subcontractor shall inspect the work areas to determine if safety hazards were created during the wind event (e.g., unstable scaffolding, unstable stair towers, damage to cranes, displacement of materials, building components, temporary fences, and equipment, etc.). Hazardous conditions shall be immediately rectified or properly barricaded until rectified.

8. Lightning:

- a. Okland and Subcontractor shall comply with the following actions when lightning is detected within the miles-to-project proximity listed below:
 - i. **Within 10-miles** = All exterior elevated work shall be suspended until 30-minutes following the last lightning strike. Note: If additional strikes are detected, the 30-minute wait period shall be reset.
 - 1. Examples of elevated work activities that shall be suspended are:
 - (a) Placement of concrete with a concrete pump truck when the boom is elevated.
 - (b) Placement of concrete with a concrete placing boom.
 - (c) All crane activities.
 - (d) All hoisting activities with mobile equipment where the load is hoisted in a suspended fashion (e.g. loads below the tines of a forklift, loads below the bucket of an excavator, loads below the bucket of a front-end loader).
 - (e) Aerial lifts.
 - (f) Bucket trucks.
 - (g) Scissor lifts.
 - (h) Scaffolds of any type including stair towers.
 - (i) Work from ladders.
 - (j) Forklift man-baskets.
 - (k) Activities on roofs.
 - (l) Working from the sides of walls, formwork, rebar assemblies, or similar.
- b. **Within 5-miles** = All outdoor activities shall be suspended until no lightning is detected for a period of 30-minutes.
 - i. Examples of work activities that shall be suspended are:
 - 1. Any outdoor activity where the worker is not protected by an enclosed cab of equipment or vehicle.
- c. There are many mobile apps. on the market that you can download to monitor lightning for your project's location.

i. Examples are:

1. WeatherBug: <https://www.weatherbug.com/appdownload/>
2. My Lightning Tracker & Alerts: <https://apps.apple.com/us/app/my-lightning-tracker-alerts/id1175031987>

9. Heat Illness Prevention:

- a. Subcontractor shall establish an effective Heat Illness Prevention Policy with definable actions that are consistent with Federal, State & Local regulations.
- b. Subcontractor shall provide all resources necessary to comply with the project's, and their internal, Heat Illness Prevention Plan.
- c. Subcontractor shall conduct weekly documented Heat Safety focused inspections between June 15th and September 30th of each year. Inspections are to be recorded in the Project's designated Management System Platform (B3F, Procore, Autodesk, etc.).

Subpart E – Personal Protective Equipment (PPE)

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Subcontractor shall maintain on site an adequate supply of safety equipment and a PPE inventory appropriate for their respective scope of work.
2. Minimum work attire and Personal Protective Equipment:
 - a. The following work attire and personal protective equipment is required to be worn 100% of the time while conducting work on the Project:
 - i. **Hard hat meeting ANSI Z89.1 2009.** The worker's name and that of his/her employer shall be visibly displayed on the front part of the hard hat. Bump caps and cowboy style hard hats are prohibited.
 - ii. **Eye protection meeting ANSI Z87.1 with "side-shield" protection.** In addition, if prescription glasses are required to be worn by a worker, the prescription glasses shall meet the ANSI Z87.1 requirements with side-shields or the worker shall utilize the appropriate ANSI Z87.1 eye protection over the prescription glasses.
 1. Dark lens safety glasses shall not be worn indoors.
 2. Dark lens safety glasses shall not be worn outdoors when lighting is below 5-foot candles.
 - iii. **Boots.**
 1. Subcontractor workers shall wear, at a minimum, over the ankle work boots. Sport shoes (including ANSI approved) are prohibited.
 2. Safety-toed boots and metatarsal protection are required to be worn when operating any equipment that poses a risk of injury to the top of a worker's foot (e.g., walk-behind or jumping-type compactor, jackhammers)
 - iv. **Shirt.** Shirts, at a minimum, shall fully cover the worker's torso to the waist and have sleeves that cover the upper arms.

- v. **Pants.** Pants at a minimum shall fully cover the worker from the worker's waist to the worker's ankles. Sweat-pants do not meet the intent of this requirement and are prohibited.

- vi. **High visibility shirt, vest, or jacket.**

- 1. For daytime work, the worker's most outer shirt, vest, or jacket shall be a bright and vibrant color (e.g., orange, fluorescent yellow, fluorescent green, or similar). This is required to be worn anywhere on the project site inside or outside.
- 2. For nighttime work, areas with limited lighting, and work around equipment, the worker's most outer shirt, vest, or jacket shall be a bright and vibrant color (e.g., orange, fluorescent yellow, fluorescent green, or similar) **and** shall also be retro-reflective or fitted with retro-reflective striping. The retro-reflective material shall be visible at a minimum distance of 1,000'.
- 3. Anytime workers are exposed to public vehicular traffic, the requirements for High-Visibility Safety Apparel listed in the Manual on Uniform Traffic Control Devices (MUTCD) shall apply to ALL workers with the exposure NOT just the flaggers.
- 4. If "hot work" is being conducted, the worker's high visibility shirt, vest, or jacket shall also be constructed from flame-resistant materials.

- 3. Gloves must be worn, as warranted, to mitigate exposure to hand & finger injuries.

- a. At a minimum, the glove type and/or cut rating shall be sufficient to protect the worker from the hazard(s) they are exposed to (e.g., ANSI rated cut resistant gloves when handling sharp objects, impact resistant gloves when there is an elevated potential of being struck on the top of the hand/finger by objects, chemical rated gloves specifically selected to adequately protect from the specific chemical hazard, etc.)

- 4. Okland issued hard hats & vests:

- a. Okland branded and issued hard hats and vests are provided for Okland employees and visitors of Okland. These items are issued to personnel for use while performing Okland authorized activities. They are expected to be returned to Okland when the employee ceases to be employed by Okland and when visitors complete their visit to the Okland project.

- 5. Green hard hats are not allowed on any Okland Project unless the worker wearing it is a 'Qualified Rigger' as defined by OSHA and is actively engaged in rigging activities for the project.

- a. Unless otherwise agreed to in contract documents, Qualified Riggers who are actively rigging and/or signaling hoisted loads, of any type, shall be wearing a green hardhat, a high visibility hard hat sock/cover, or an alternative means of visibility identifying the rigger with highly visible hard hat decals that can be observed by the crane operator from a distance.
 - i. This includes rigging/signaling activities that do not involve a crane (e.g., rigging with chain falls, come-alongs, electric hoists, coffering hoists, forklifts, earthmoving equipment)

- b. Oakland employed Qualified Riggers/Signal Persons (QR/SP), that are NOT full-time QRs/SPs, shall have their Oakland issued helmets demarcated with green decals pursuant to the location stipulated by Oakland.
6. Hard hats shall be replaced as per the manufacturer's recommendation.
 7. Face-shields shall be worn at all times while operating a grinder and/or chop-saw, irrespective of what type of blade or disk is being utilized with the tool or what type of material the tool is being utilized to cut/grind.
 8. Face-shields shall be worn at all times concrete chipping tools pose a hazard to the face.
 9. When workers are required to conduct work while kneeling, knee pads shall be worn.
 10. At all times that hand-operated/controlled/held power tools that generate substantial vibration are being utilized, vibration-dampening gloves shall be worn by the operator of such tool.
 11. Safety-toed boots and metatarsal protection is required to be worn when operating any walk-behind or jumping-type compactor.
 12. Where respiratory protection is necessary, the use of respirators shall be a last resort. Alternate controls shall first be evaluated and where feasible, effectively implemented.
 13. Workers using a respirator, for any reason, shall only do so when they can be visually seen and heard by other workers within their company.
 14. Subcontractors who have to utilize respiratory protection shall develop and implement an effective written respiratory protection program specific to their scope of work.
 15. Filtering face piece type respirators, commonly referred to as dust masks, shall not be utilized for respiratory protection above the Occupational Exposure Limits (OEL's) of any substance.
 16. Where filtering face pieces are being used below the Occupational Exposure Limits (OEL's) their required use, regardless of the filtering face piece's construction, shall constitute the use of a respirator. Thus, employers shall ensure that all personnel required to utilize a filtering face piece is medically qualified, trained, and fit-tested prior to utilizing the filtering face piece.
 17. When the voluntary use of a respirator is being conducted, each employer shall secure written documentation from their employees voluntarily utilizing a respirator acknowledging their receipt of OSHA's "Information for Employees Using Respirators When Not Required Under the Standard".
 18. Loose fitting earrings, necklaces, and/or bracelets shall not be worn. Workers who have the potential to have their hand(s) and/or finger(s) caught in/on material, equipment, tools, or similar shall not wear any form of bracelets and/or rings while exposed to such hazards.
 19. Variances to Oakland's PPE requirements shall only be allowed if it is in compliance with Oakland's [PPE Variance Policy](#).

Subpart F – Fire Protection and Prevention

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Subcontractor shall comply with the fire protection and prevention requirements stipulated throughout this manual. The fire protection and prevention requirements will be evaluated periodically and may be revised to meet the specific needs of the Project. Subcontractor will be advised of any changes during weekly coordination meetings. Subcontractor shall comply with the revisions.
2. Subcontractor shall take all necessary steps to protect all installed life safety systems (e.g., fire suppression sprinkler heads & lines, smoke detection devices, fire stand-pipes, etc.).
3. The fire hydrants located on and around the Project shall remain in service if at all possible.
4. Subcontractor shall not take out of service an existing fire hydrant unless Subcontractor has the explicit written permission from Okland. In addition, Subcontractor shall give verbal notification to, and obtain authorization from the City and/or State Fire Marshal's office, the City Fire Department, and the City Public Utilities office prior to taking any fire hydrant out of service.
5. Subcontractor shall not utilize a fire hydrant as a source of "construction water" without the explicit written permission from Okland's Superintendent and the City Public Utilities office. In addition, Subcontractor shall give verbal notification to, and obtain authorization from, Okland's Superintendent, the City/State Fire Marshal's office, and the City Fire Department prior to the commencement of such work.
6. The fire hydrants and the adjacent occupied buildings' Fire Department Connections (FDCs) shall not be blocked, damaged, or covered for any reason.
7. Okland or its contracted representative shall stage "general location" fire extinguishers in strategic locations throughout the Project as required by federal, state, and local regulations.
 - a. A minimum of a 10-pound ABC type fire extinguisher shall be used for all "general location" fire extinguishers.
8. Subcontractor shall not consider the "general location" fire extinguishers placed by Okland to meet the requirement for Subcontractor to provide its own fire extinguisher during "hot work" (e.g., welding, torch use) or "fire sensitive work" (e.g., the use of flammable liquids).
9. Subcontractor shall not damage, remove, relocate, block, or otherwise render these "general location" fire extinguishers inactive.
10. Subcontractor is required to inspect its fire extinguishers on a daily basis prior to the beginning of work. If any extinguisher is found to be damaged or inactive in any way it shall be taken out of the field, properly "tagged" as "OUT OF SERVICE" and either repaired, recharged, or discarded.
11. Okland will inspect the Okland supplied "general location" fire extinguishers on a monthly basis.

12. At a minimum, a 3' clearance shall be maintained around all fire extinguishers so as to facilitate quick and safe access to the extinguishers.
13. Subcontractor shall not render inactive any portion of an active fire alarm system and/or fire water sprinkler system unless Subcontractor has the explicit written permission from Okland's Superintendent.
14. Fire lanes shall not be blocked for any length of time.
15. Plastic containers shall not be utilized for any flammable or combustible liquid including diesel fuel.
16. All flammable and combustible liquids shall be kept covered when not in immediate use.
17. Subcontractor shall provide a minimum of a ten-pound ABC type fire extinguisher within 25' of the use and/or storage of flammable and/or combustible liquids or other fire sensitive materials or work.
18. Flammable and combustible liquids in containers larger than 5 gallons shall be stored in/on a spill containment system that provides for 110% containment.
19. Flammable materials (e.g., roofing adhesives, coatings, paints) shall not be stored within the building or on the roof.
20. Only the quantity of flammable materials (e.g., roofing adhesives, coatings, paints) that will be used in a single day's shift shall be taken into the building or to the roof.
21. All flammable material and their containers (full, partially used, or empty) (e.g., roofing adhesives, coatings, paints), rollers, rags, and other related tools or materials that have flammable material on them, shall be removed from the building and the roof at the end of each shift.
22. Flammable materials (e.g., roofing adhesives, coatings, paints) shall be stored at least 50' away from the building, in manufacturer approved containers with tightly sealed lids.
23. Liquefied Petroleum Gas (LPG) containers shall be kept upright at all times.
24. LPG containers shall not be stored indoors at any time.
25. LPG containers shall be considered "in storage/stored" if not in immediate use.
26. LPG containers shall not be taken in to nor stored in conexes, hooches, gang-boxes, tented areas, and similar confined areas for any amount of time.
27. Subcontractor shall install the proper signage in LPG storage areas warning personnel of the potential flammability of the area.

28. Where plastic is used for weather protection or temporary barriers/enclosures, and it is anticipated that the plastic will be in close proximity to or be exposed to fire hazard sources, such as welding, cutting, grinding, temporary heaters, etc., fire retardant plastic shall be used.

29. Temporary Heaters:

- a. Prior to the installation/use of a temporary heater, a [Temporary Heater Permit](#) must be completed for the use of the heater and the use of its fuel source delivery system.
- b. Only trained, competent, and authorized workers shall install, adjust, turn off/on, move, and/or maintain a heater and/or its fuel source delivery system.
- c. A detailed inspection of the heater and fuel source delivery system by a competent person shall be conducted prior to placing them into service.
- d. A general daily inspection by a competent person shall be conducted of the heater and the fuel source delivery system while it is in operation. This daily inspection shall be documented on a heater inspection tag. That tag shall be placed on or near each heater.
- e. LOTO shall be implemented anytime maintenance or repairs are being performed on the heaters and/or the fuel source delivery system.
- f. When heaters being used on the project, subcontractor shall instruct their employees:
 - i. That only trained, competent, and authorized workers are allowed to install, adjust, turn off/on, move, and/or maintain a heater and/or its fuel source delivery system.
 - ii. What to do if they smell gas
 - iii. How to handle heater and fuel gas related emergencies
- g. Heaters, hoses, manifolds, and valves shall be protected from damage.
- h. Hoses being placed directly on the ground shall not be ran over by equipment or have material placed on top of them.
- i. Hoses shall never come in contact with sharp edges such as metal studs or be pinched in doorways or windows.
- j. When a mainline fuel source is used to deliver fuel into the building shut-off valves shall be installed at each of these locations:
 - i. At the main, where gas originates from the meter or tank.
 - ii. At each floor of the building.
 - iii. At each "T" in the gas line.
 - iv. At the end of each hose assembly, just prior to the heater connection.
- k. The shut-off valve location at each floor of the building shall be demarcated with a sign for quick identification in the event of an emergency.
- l. Only specific compatible components shall be used for construction of the fuel source delivery system.
- m. Prior to placing the fuel source delivery system into service, each connection shall be tested to ensure leaks are not present.
- n. Manufacturer's clearances for heaters shall be maintained.
- o. Heaters shall only be moved by their designated components, NEVER by the fuel source piping.
- p. Heaters shall not be in operation while being moved.
- q. Safety features on the heaters shall NEVER be overridden.
- r. Absolutely no alterations shall be made to the factory onboard controls of a temporary heating device to bypass factory output discharge air temperature settings

Subpart G – Signs, Signals, and Barricades

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. [Overhead Protection at Building Perimeter Policy:](#)

- a. When any work is taking place on an elevated level, within 10-feet from an elevated edge that is not protected by a solid wall or a floor-to-ceiling net, or when elevated work is taking place on the exterior of building (e.g., from an aerial lift, swing stage scaffold, etc.), the area below the elevated level shall be barricaded with an effective barricade system. At a minimum, RED DANGER TAPE, shall be used as the barricade system.
- b. Signage:
 - i. Signage shall be placed on the barricade system notifying personnel of the overhead hazard.
 - ii. Signage shall state: [“DANGER DO NOT ENTER OVERHEAD WORK”](#)
 - iii. Signage shall be placed so that it is facing all directions that workers could potentially enter the barricaded area.
 1. The number of signs required is dictated by the configuration of the barricaded area and the number of visual obstructions of the signs. Signage shall not exceed more than 80-feet between signs and be placed at the most likely points of entry (doorways, corridors, envelope penetrations, etc.) into the barricaded area. Additional signage shall be placed when a visual obstruction prohibits the other signs from being seen.
 - iv. At least one [Danger Barricade Tape Information Sign](#) shall also be posted on the barricade system so as other workers can be informed of the barricade owner and contact information.
- c. Barricade Distance:
 - i. The barricade system shall be placed from the face of the building and extend outward to the minimum distances listed in the table below and extend 10-feet on both sides of the work area.
 1. If the work is taking place on the exterior of the building from an aerial lift, swing stage or other elevated access system, the barricade system shall be placed from the face of the building and extend outward to the minimum distance listed in the table below, **plus the width of the elevated access system**, and extend 10-feet on both sides of the work area.
 - ii. The barricade system shall also be placed from the face of the building and extend **inward** no less than 2-feet. This is to prevent workers from extending their head or torso into the potential drop zone of an object from the interior of the building.
 1. If a wall is in place that prevents workers from extending their head or torso into the potential drop zone, barricade tape, in the area the wall is in place, is not required.
 - a. All openings in walls must be barricaded with the 2-foot inward extension barricade system.

Working Level Height	Equation	Minimum Barricade Distance
Up to 40'	None - Minimum	10'
Up to 50'	$4\text{mph} \sqrt{\frac{2 \times 50}{15}} = 10.3'$	10' - 4"
Up to 60'	$4\text{mph} \sqrt{\frac{2 \times 60}{15}} = 11.3'$	11' - 4"
Up to 70'	$4\text{mph} \sqrt{\frac{2 \times 70}{15}} = 12.2'$	12' - 3"
Up to 80'	$4\text{mph} \sqrt{\frac{2 \times 80}{15}} = 13'$	13' - 3"
Up to 90'	$4\text{mph} \sqrt{\frac{2 \times 90}{15}} = 13.9'$	13' - 11"
Up to 100'	$4\text{mph} \sqrt{\frac{2 \times 100}{15}} = 14.6'$	14' - 7"
Up to 110'	$4\text{mph} \sqrt{\frac{2 \times 110}{15}} = 15.3'$	15' - 4"
Up to 120'	$4\text{mph} \sqrt{\frac{2 \times 120}{15}} = 16'$	16'

2. Barricade tape, when used, shall be placed around all sides of the hazard area to establish a defined Restricted Access Zone (RAZ).
3. Barricade tape shall be placed in accordance with Okland's [Barricade Tape Guidelines](#).
4. Barricade tape signage links: [DANGER](#), [CAUTION](#), & [CARE](#)
5. "Caution" (yellow) tape shall only be utilized to warn personnel of potential hazards that are not immediately dangerous to life or health. Personnel will be allowed to cross caution tape only after they have identified the hazard/condition, are wearing the required personal protective equipment to protect them from the hazard and the hazard is not immediately dangerous to life or health.
 - a. When "Caution" (yellow) tape is utilized, and specific PPE is required to be worn prior to entering the barricaded area, it shall have a [Caution Barricade Tape Information Sign](#) placed on it identifying the specific PPE required, who placed the barricade tape, their company name, the date it was placed, the contact number for the person who placed the tape and why the barricade tape is in place (i.e., the hazards). The sign shall face away from the exposure
6. "Danger" (red) tape shall be utilized to warn personnel of hazards that are immediately dangerous to life or health. Danger tape shall be placed at an appropriate distance so as to prevent personnel from being exposed to the hazard being barricaded. **Personnel shall not cross** Danger tape until they have identified the hazard(s), have been trained on the proper procedure(s) to control exposure to the hazard(s), are wearing the required personal protective equipment to protect them from the hazards, and have permission to enter the area from the individual(s) who placed the danger tape.
 - a. When "Danger" (red) tape is utilized it shall have a [Danger Barricade Tape Information Sign](#) placed on it identifying who placed the barricade tape, their company name, the da

te it was placed, the contact number for the person who placed the tape and why the barricade tape is in place (i.e., the hazards). The sign shall face away from the exposure.

7. “Special” barricade tape shall be utilized on an as needed basis and be of sufficient type to adequately inform workers of the hazards associated with the work (e.g., RADIATION, ASBESTOS, LEAD)
8. Barricade tape of any type shall NOT be utilized as a means of fall protection/prevention.
9. **“CARE” (blue) Tape:**
 - a. Blue CARE Tape should be used to protect finished surfaces, rooms, stocked building materials, assemblies, equipment, etc. from damage.
 - b. When “CARE” (blue) tape is utilized, it is strongly recommended (i.e., not required) to place a **CARE Tape Information Sign** that identifies the specific finishes that need to be protected and the necessary controls (e.g., boot covers), who placed the barricade tape, their company name, the date it was placed, the contact number for the person who placed the sign. The sign should face away from the finishes being protected.
 - c. Blue CARE tape shall not be used in lieu of Danger or Caution tape when safety hazards are present.
 - d. Danger or Caution tape shall not be used in lieu of Blue CARE tape when no safety hazards are present.
 - e. Reference Okland’s **Barricade Tape Guidelines** document for further guidance.
10. When work must be conducted overhead of workers, sensitive equipment, other sensitive building materials, or within close proximity to other workers below, that are within the barricaded area, tool tethers are required to be used with all tools.
11. Barricading of work areas shall provide for the complete safety of the general public and all construction personnel. Sequencing and barricading shall be performed in such a manner as to create a minimum amount of interference with the normal flow of pedestrian and construction foot, vehicle, and equipment traffic.
12. **Under no circumstance shall barricade tape and signage alone be placed to prevent the general public from entering a hazardous area. Hard/physical barricades and signage shall also be utilized in such areas.**
13. Signs, signals, and barricades shall be maintained in an appropriate manner when the hazard(s) is/are present.
14. Signs, signals, and barricades shall be immediately removed from the work area and properly stored or discarded when the hazard(s) is/are no longer present. In no case, shall signs, signals, and/or barricades be left lying on the ground.
15. Prior to barricading access/egress routes (e.g., stairwells) Subcontractor shall secure the approval from the Okland Supervisor overseeing the specific area being barricaded and coordinate alternate access/egress routes with all affected Subcontractors.

Subpart H – Materials Handling, Storage, Use, and Disposal

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Cranes, Hoisting, and Rigging:
 - a. All crane, hoisting, and/or rigging operations shall be in compliance with [Subpart CC](#) of this document.
 - i. Operations that do not involve a crane **must still** comply with the [Rigging](#) and [Critical Lift](#) sections of [Subpart CC](#) of this document.
 - ii. Operations that do not involve a crane (e.g., rigging to a forklift carriage, rigging to an excavator, chain falls, winches, etc.) that meet the Okland criteria of a Critical Lift, shall complete the [Non-Crane Lift Planning Worksheet](#).
2. Proper lifting techniques:
 - a. Workers shall plan their path prior to lifting items that need to be carried to another location to ensure that the path is clear and safe.
 - b. Workers shall not lift any item that exceeds his/her weightlifting limitations (i.e., workers shall not lift items that are beyond their capability).
 - c. Workers shall squat down so the item being lifted is lifted with the worker's legs and arms rather than their back.
 - d. Workers shall keep the item being lifted close to their torso.
 - e. Workers shall keep their back as straight as possible while the item is being held.
 - f. Workers shall not twist at the waist while carrying a load. Workers shall turn their legs along with their torso.
 - g. Workers shall set loads down by squatting down while continuing to keep the load close to their torso.
 - h. To the extent practical, workers shall utilize equipment, tools and machinery to assist with lifting and moving loads.
3. Dropping Material:
 - a. No material shall be dropped greater than a distance of 20' without the use of chutes. The chute discharge shall be into a dumpster receptacle or similar. The area around the dumpster shall be barricaded with hard barricades. Signage shall be placed instructing personnel not to get on or near the trash chute and/or dumpster. Barricades shall be placed a sufficient distance away from the dumpster to prevent personnel from being struck by material that may ricochet out of the dumpster.
 - i. When trash chutes are installed, a daily documented inspection of the entire trash chute system shall be conducted by a competent person.
 - ii. Prior to maintenance or adjustments being made to a trash chute system, an effective lock-out, tag-out system shall be put into place, on each trash chute accessible floor, that will prevent use of the trash chute system.
 - b. Prior to the construction of any trash chute, a [Trash Chute Permit](#) must be obtained from the Okland Project Management Team.
 - c. Where any material is dropped less than a distance of 20' to an open area below, the area below shall be completely barricaded on all accessible sides. If soft barricades are

utilized, a sufficient number of spotters shall be positioned around the area below so as to ensure personnel do not walk under or near falling material.

- d. Where any material is dropped less than 20' into a dumpster receptacle or similar, the area around the dumpster shall be barricaded with hard barricades. Signage shall be placed instructing personnel not to get on or near the dumpster. Barricades shall be placed a sufficient distance away from the dumpster to prevent personnel from being struck by material that may ricochet out of the dumpster.
- e. Where debris is pushed or lifted over an edge of a structure, the equipment used to push or lift such debris shall not be exposed to an unguarded edge where it may drive or slide over the edge of the structure. Material used to prevent the equipment from driving or sliding over the edge must be sufficiently anchored to stop the movement of the equipment.

Subpart I – Tools, Hand, and Power

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. All manufacturer-provided guards and safety devices on hand and power tools shall be in place and operating properly at all times.
2. The misuse of tools such as the use of a screwdriver as a chisel, a wrench as a hammer, overloading a wrench by using a pipe extension (**cheater bar**) on the handle, or similar action is prohibited.
3. Tools in need of repair shall be tagged out-of-service and be immediately removed from the work area.
4. Grinders with a “lock-on” function are prohibited from Okland Projects.
5. Prior to drilling/penetrating into any enclosed electrical or mechanical unit of any type, the unit shall be opened and inspected to ensure the drilling/penetrating activity will not damage any pre-installed wiring, lines, etc.
 - a. If wiring, lines, etc. could be damaged, adequate controls shall be implemented to ensure damage does not occur.
 - b. If the enclosed unit has already been energized, appropriate de-energization and Lock-Out / Tag-Out controls shall be initiated prior to opening the unit. Controls shall remain in place until the entire operation is complete.
6. When handheld circular saws (e.g., Skill Saws) are used, workers shall keep both hands on the saw while operating the saw.
 - a. The material being cut (e.g., lumber) shall NOT be held in the hand of any worker while it is being cut.
 - i. The material being cut shall be mechanically secured with clamps, nailed down, screwed down, etc.
 - b. Stable cut stations (e.g., sawhorses) shall be utilized to support the material being cut.
 - i. Workers shall NOT utilize their foot, leg, or any other part of the body to support the material being cut.



7. The depth of the blade of all saws shall be appropriately adjusted for the thickness of the lumber being cut so the blade does not extend beyond the lumber more than necessary.
8. Pneumatic power tool hoses with an inside diameter that exceeds $\frac{1}{2}$ in. shall have:
 - a. Pins or wire securing all twist lock connections.
 - i. When pins are used, at least one pin is required at each coupling connection point.
 - ii. When wire is used, wire shall secure both lock locations at each coupling connection point.
 - b. A metal cable spring loaded whip check device installed on both sides of the coupling connection points.
 - c. A safety device at the source of supply or branch line to reduce pressure in case of hose failure.
 - i. When hoses larger than $\frac{1}{2}$ in. are used for cleaning deck forms, this requirement is not applicable.
 - d. Pneumatic power tools shall not be left unattended when they are under pressure (e.g., disconnect the tool or release the hose pressure).
 - i. Unattended means when the tool is not within the line-of-sight of the worker.
 - e. Workers shall ensure all stored energy is released from pneumatic tools prior to leaving the tool unattended (e.g., squeeze the tool trigger to ensure it is in a zero-energy state).
 - i. Unattended means when the tool is not within the line-of-sight of the worker.
9. Whip-check devices shall be installed at all air hose connections from the air compressor to the tool/machine. Whip-check devices shall be utilized in addition to the pin connections in the hose couplings. Whip-check devices shall be spread to the furthest extent possible.
10. Pneumatic and/or hydraulic hoses shall not be dropped from heights, nor shall they be run over with equipment or tools.
11. Pneumatic tools shall not be lowered by the hoses.
12. Workers who utilize powder or gas actuated tools shall be trained and certified in the safe operation of the specific tool being utilized and shall follow the requirements stipulated in the training.
13. Workers shall have certification cards on their person at all times while operating powder or gas actuated tools.

14. Hearing protection shall be worn while utilizing a powder & gas actuated tool.
15. Prior to the use of powder or gas actuated tools, workers shall install signage warning other personnel of the use of powder actuated tools in the area.
- a. The sign shall be a yellow CAUTION sign stating that a Powder or Gas Actuated Tool is in Use, Hearing Protection Required in This Area
16. If a powder or gas actuated tool is used in a horizontal position (e.g., shooting into a steel column), the operator of the tool shall establish a controlled access zone with, at a minimum, red DANGER tape and DANGER signage on the opposite side of the material that the fastener is being fastened into.
17. Powder and gas actuated tools shall be stored unloaded.
18. Subcontractors shall not store powder cartridges on the project. They are to be brought to and from the project each day in a manufacturer's approved storage container.
19. Fired or unfired powder cartridges shall not be disposed of in project trash receptacles or dumpsters. Subcontractors shall remove all fired and unfired powder cartridges from the project on a daily basis and dispose of them off the project in compliance with applicable local, state, and federal regulations.
20. Subcontractor is responsible for the proper offsite disposal of gas actuated tool fuel cells. These fuel cells (full, partially used, or empty) shall not be thrown into project trash receptacles or dumpsters.
- 21. Lithium Batteries (LBs)**
- a. LBs and charges shall be inspected for damage before use.
 - i. LBs and chargers that are cracked, corroded, or melted shall be immediately remove from service.
 - b. LBs and chargers shall be kept clean.
 - c. Only matching Original Equipment Manufacturer (OEM) LBs and chargers shall be used. (i.e., aftermarket LBs shall never be placed to charge in a charger that is NOT of the same manufacturer).
 - d. LBs and chargers shall only be used as instructed by the OEM.
 - e. Workers utilizing LBs shall review the owner's manual and all labels before use.
 - f. Pay attention to LB performance, including runtime and power output. If a significant decline is observed, the LB shall be replaced.
 - g. A minimum of a 10-pound ABC rated fire extinguisher shall be placed within 15' of all active chargers.
 - h. Charging within a Building Under Construction/Improvement/Demolition:
 - i. Daytime: Charging shall be done in well-ventilated areas free of flammable and explosive atmospheres.
 - ii. Overnight/Weekends: Charging shall be done in a 35' diameter area free of all flammable and combustible materials, on a non-combustible surface (e.g., concrete, metal, soil), and in well-ventilated areas free of flammable and explosive atmospheres.

1. LBs SHALL NEVER BE LEFT TO CHARGE OVERNIGHT/WEEKENDS IN ANY TYPE OF STICK FRAMED BUILDING/STRUCTURE (This requirement does not apply to temporary job trailers).
 - i. Charging within a Temporary Job Trailer or Conex:
 - i. Charging shall be done in areas free of flammable and explosive atmospheres.
 - j. Charging should only be conducted between 32°F and 120°F.
 - k. While charging multiple LBs in one location, workers shall ensure the power supply is not overdrawn. (Note: check the amperage demand on each charger's label.)
 - l. If the batteries are frozen (below 32°F), allow them to warm up to room temperature before charging.
 - m. If batteries are overheated due to weather or storage, allow it to cool down below 120°F before charging.
 - n. If batteries are overheating due to use or charging, they shall be immediately removed from service.
 - o. Storage:
 - i. Store batteries in cool (less than 120°F), dry locations.
 - ii. Batteries and chargers should be kept away from direct sunlight or any other heat source.
 - p. When transporting batteries, protect leads to prevent contact with metal (coins, keys, nails, screws, tools, etc.). Batteries should only be transported with the OEMs provided lead covers in place, or in separate plastic bags.
 - q. Dispose of batteries at a local battery recycling location or as directed by the manufacturer and the local authorities. They shall NOT be placed in project trash cans and/or project dumpsters.
 - r. LBs shall immediately be removed from service if they are overheating, emitting any odor, change in color or shape, leaking, emit odd noises, have observed cracks, or holes, etc.
 - s. LBs shall never be modified, altered, or disassembled. Opening battery casings or altering wiring is strictly prohibited. Doing so can expose individuals to chemicals and electrical hazards.
 - t. Field repairs shall not be conducted on LBs or chargers. Only OEM authorized technicians shall conduct repairs.
 - u. All LBs shall be UL listed.
 - v. LB power tools shall not be overloaded. This can strain batteries, decrease performance, and lead to overheating.
 - w. Chargers shall not be exposed to liquids, oils, grease, etc.
 - x. LBs shall not be cleaned with solvents or corrosive chemicals.
 - y. LBs shall not be charged if they are found to be damaged, defective, or overheated.
 - z. Charger vents shall not be covered; Overheating may occur.
 - aa. LBs shall not be stored or charged in wet or damp conditions.
 - bb. Damaged or defective batteries shall not be stored together.

Subpart J – Welding and Cutting

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Compressed gas cylinders shall only be hoisted in an upright and secured position, with the gauges removed, cylinder caps installed and in an approved lifting device. Approved lifting

devices shall only include those devices that are specifically and commercially manufactured for that purpose or those that are constructed per the design of a registered professional engineer and are specifically designed for hoisting compressed gas cylinders.

2. Compressed gas cylinders shall be kept secured and upright at all times.
3. Compressed gas cylinders shall not be stored indoors at any time.
4. Compressed gas cylinders shall be considered “in storage/stored” when not used for greater than a 24-hour period of time.
5. Compressed gas cylinders shall not be taken in to nor stored in conexas, hooches, gang-boxes and similar confined areas for any amount of time.
6. Gauges and hoses shall be removed from compressed gas cylinders at the end of each shift.
7. Valve protection caps shall be installed on compressed gas cylinders when not in use.
8. Western couplers must be used to connect hose to torch and hose to hose. Screw type hose clamps are not acceptable.
9. Flash arrestors **and** back-flow prevention devices are required to be utilized with all oxygen and/or fuel gas operations.
10. Compressed gas cylinders shall be kept within sight and immediately accessible to the individual utilizing them or the fire watch.
11. Only strikers shall be used to ignite a torch, weed burner, and similar fuel gas tools. The use of lighters, matches, etc. for this purpose is prohibited.
12. Hoses shall not be repaired with tape.

13. Hot Work Permits:

- a. Hot work is defined as: a process which, because of its design or function, can cause ignition of a product/material or gaseous/vaporous atmosphere due to direct or indirect contact. Examples of hot work include, but are not limited to, welding, torch cutting, burning, soldering, grinding, use of a chop-saw, use of a demo-saw, etc.
 - i. Hot work also includes activities that generate heat but do not generate a flame or spark, such as, but not limited to, exothermic heat (e.g., two-part foams, two-part epoxies) and hot wires used to cut structural foam blocks. As such, these activities also require a hot work permit.
- b. The supervisor overseeing the hot work operation shall obtain a daily (unless an extended duration is authorized by the Okland Superintendent, if so, it shall not exceed 7-days) written [Hot Work Permit](#) from the Okland Superintendent prior to the execution of any hot work.
- c. **Prior to requesting the [Hot Work Permit](#) the requesting supervisor shall inspect the work area and developed a fire prevention plan for the hot work activity.**

- d. Prior to obtaining a hot work permit for hot work activities that will be conducted on, above, or one level below a roof, Subcontractor shall ensure that their means and methods are explained to the Okland supervisor that is issuing the permit. Subcontractor shall also inspect both the hot work area and the roof with the Okland supervisor to assess the means and methods of how the hot work will be performed and to validate that adequate controls will be in place for those means and methods.
- e. All tools and equipment, including the fire extinguisher(s), shall be inspected prior to beginning the hot work activity.
- f. To the extent practical, hot work operations shall be conducted in areas which pose the least potential exposure for a fire.
- g. Where plastic is used for weather protection or temporary barriers/enclosures, and it is anticipated that the plastic will be in close proximity to or be exposed to fire hazard sources, such as welding, cutting, grinding, temporary heaters, etc., fire retardant plastic shall be used.
- h. Okland's Safety Department shall be notified prior to conducting any hot work on coated metals and prior to any inert gas metal-welding being performed on stainless steel.
 - i. When conducting hot work on surfaces that have a coating, a pre-assessment shall be conducted to determine if the coating is hazardous or will become hazardous once it is heated. If a hazard is identified, the hazard must be effectively controlled.
- i. The [Hot Work Permit](#) process is as follows:
 - i. Subcontractor's supervisor inspects the work area and develops a fire prevention plan for the hot work activity. This inspection shall include an assessment of all potential fire hazards that currently exist or will exist due to the hot work activity.
 - 1. The assessment shall encompass a 35' radial sphere around the proposed hot work activity.
 - 2. All fire hazards shall be removed or adequately protected prior submitting the hot work permit to Okland for signature and prior to the start of any hot work.
 - 3. All openings to adjacent areas should be sealed to prevent sparks, slag, etc. from falling into other areas.
 - ii. Subcontractor's supervisor completes the permit and signs the Supervisor line.
 - iii. The person performing the hot work inspects all hot work tools and equipment, including the fire extinguisher(s).
 - iv. The person performing the hot work reviews the permit and signs the permit.
 - v. The fire watch reviews the permit and signs the permit.
 - vi. The Okland Superintendent reviews the permit for completeness and coordination and signs the permit.
 - vii. The Okland Superintendent is to take a photo of the permit and upload it into Pro-Core system or the project designated system. The Superintendent can designate this to the Subcontractor if the Subcontractor is on the Project's Pro-Core or the project designated system.
 - viii. The physical permit and sleeve are then to be attached to the equipment where the hot work is to be performed.
 - ix. Subcontractor shall monitor the effectiveness of their controls and adjust if improvement is necessary to prevent fire or damage to building materials.

- x. Each time the hot work activity stops (e.g., breaks, lunch, end of day), the fire watch is to remain for the duration stipulated on the permit to ensure fire does not start following the work and then sign the back of the permit.
 - xi. At the end of the duration of the permit, Subcontractor is to return the permit to the Okland Superintendent.
 - xii. Completed permits should remain in project files for at least 90 days.
- j. Additional [Hot Work Permits](#) and/or requirements may be added in cases where hot work is performed within or near occupied structures. In these cases, Okland's Superintendent shall coordinate the securing of such permits from the respective building owner/manager and conveying to Subcontractor the additional requirements.
- k. Hot work permits shall be not be valid for more than one day.
 - i. For hot work that is stationary and only generates sparks (e.g., a stationary chop saw station for cutting metal studs) a hot work permit may be valid for up to 7-days.
 - 1. The Okland supervisor issuing the hot work permit for these fixed stationary activities will make the determination of the duration the hot work permit is valid for, but it shall not extend more than 7-days.
 - 2. A dedicated fire watch is not required for these fixed stationary hot work activities. However, a competent person shall inspect the work fixed station prior to and after each shift for any fire hazards. This inspection shall be documented on the hot work permit.
 - 3. The workstation must be set up to contain all sparks.
 - 4. The fixed station must be marked off or barricaded to section off the area (e.g., use of candle sticks and caution tape).
 - 5. Caution signage must be placed at the fixed station stating: **"CAUTION - HOT WORKSTATION - NO FLAMMABLE OR COMBUSTIBLE STORAGE."**
- l. After the [Hot Work Permit](#) has been issued, but prior to beginning the hot work activity, Subcontractor's supervisor shall review the requirements listed on the [Hot Work Permit](#), as well as the requirements of the Supervisor's fire prevention plan, with the worker(s) conducting the hot work and the fire watch established.
 - i. Subcontractor shall furnish and assign a trained fire watch equipped with at least a 10A 60B:C or greater rated fire extinguisher.
 - 1. Subcontractor's supervisor shall instruct the fire watch on the responsibilities of his/her assignment and specifically on what to do in the event of a fire or fire related injury.
 - 2. The fire watch shall continue to monitor the area in which he/she has been assigned to watch for fire for the duration stipulated on the [Hot Work Permit](#) following the completion of the hot work activity. This duration shall not be less than 30-minutes after the hot work activity has been completed.
 - 3. **A fire watch shall have no other duties other than fire watch.**
 - ii. **Soldering and/or Brazing:** A designated, full time, fire watch is not necessary in a steel/concrete structure where no flammable and/or combustible materials (e.g., drywall, insulation, plywood) are present within a 10' sphere around the employee conducting the work and the worker conducting the work can immediately access the area where soldering/brazing splatter may fall. When a designated, full time, fire watch is not utilized, the employee conducting the soldering and/or brazing shall serve as their own fire watch and is responsible to execute all the fire watch responsibilities noted on the Hot Work Permit.

- m. Sparks and/or slag shall be contained at the source of the hot work operation by use of catch pans/basins, fire blankets or any combination of these or by other effective means.
 - i. Plywood or similar combustible material shall not be utilized for protection against sparks and/or slag.
 - n. In cases where sparks and/or slag cannot be contained at the source of the hot work operation, Subcontractor shall assign additional fire watch personnel to guard against fire on/in all areas and/or floors in which the sparks and/or slag generated from the hot work activity are being dropped.
 - i. The areas where sparks and/or slag can drop to shall be barricaded with red DANGER tape. These areas shall also be protected with adequate fire protection measures such as, but not limited to, removing all flammable materials, removing, or covering all combustible materials with fire blankets, protection of all finishes.
14. Welding screens shall be utilized when conducting welding operations where other trades and/or the general public may be exposed to the hazards of arc-flash.
15. Sufficient mechanical ventilation shall be provided by Subcontractor whenever interior welding or cutting operations are performed to ensure appropriate air exchange.
16. When arc welding or cutting operations are conducted on aluminum, stainless steel, and galvanized materials, workers must also utilize respiratory protection (i.e., respirators) until personal exposure monitoring is conducted and it is determined that respiratory protection is not needed for that specific scope of work.

Subpart K – Electrical

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Pulling wire through conduit, or similar, shall not be conducted from an extension ladder or a standard A-frame ladder (the safe use of a platform ladder, scaffold or a scissor lift would be acceptable).
2. Subcontractor shall conduct monthly documented inspections of all their extension cords, power tool cords, welding leads, temporary lighting, and similar. Those conducting the inspection shall utilize Okland's [Rigging & Electrical Color-Code](#) system to identify that it has been inspected and found safe to use.
3. Where overhead powerlines span above ANY access/egress point to/from the project, where vehicles and/or mobile equipment may travel, large (3' x 3' minimum) prominent, signage shall be posted, that is visible from both directions, warning drivers/operators of the overhead power lines.
4. Where overhead powerlines span above ANY point on the project, where vehicles and/or mobile equipment may travel, large (3' x 3' minimum) prominent, signage shall be posted, that is visible from both directions, warning drivers/operators of the overhead power lines.

5. Cords, Welding Leads & Tools:

- a. All extension cords must be 12 AWG or thicker and be of a 3-wire type (i.e., it must have a ground).
- b. Electrical cords, welding leads, and cables shall be covered, elevated, or otherwise protected from damage and from creating additional hazards to workers (e.g., they must not be run in the **middle of hallways**, walkways, or stairs nor can they be routed in or under water or ice).
- c. Subcontractor shall conduct periodic "site-wide roll-ups" of all cords for inspection. Subcontractor shall also participate in "site-wide roll-ups" that may be periodically mandated by Okland. Cords shall be inspected and properly organized when rolled back out.
- d. Only industrial rated "three-ways" and similar splitters shall be utilized.
- e. Repairs to electrical cords, welding leads and cables with electrical tape is prohibited. Only manufacturer approved repair methods shall be used. Subcontractor shall supply to Okland, if requested, documented proof of the manufacturer's approval of the repair method.
- f. The use of a Ground Fault Circuit Interrupter (GFCI) is required when utilizing extension cords, power tools, and/or electrical machinery that are connected to an electrical receptacle. The GFCI shall be placed at the receptacle. This shall be accomplished with either a GFCI receptacle or a portable GFCI device plugged into the receptacle.
- g. The use of makeshift electrical multi-ways (e.g., knock-out boxes not properly mounted) is prohibited.
- h. All electrical equipment and components shall be utilized in the manner and condition of its Underwriter Laboratories (UL) listing.
- i. Subcontractor shall not utilize Romex electrical cable (or similar) for temporary wiring, temporary lighting, and/or to feed temporary power supply panels, boxes, lighting, outlets or similar.
- j. Subcontractor shall NOT move, alter, repair, disconnect, or similar, any electrical gear of any type that is owned and/or operated by the local power company unless:
 - i. Subcontractor is contracted to do so.
 - ii. Subcontractor has secured authorization from the power company.
 - iii. The power has been validated by Subcontractor to be disconnected.

6. Temporary Lighting:

- a. Subcontractor shall provide all necessary task lighting in their work areas.
- b. Helmet lights and flashlights shall not be the only means of task lighting. Subcontractor shall provide lighting that illuminates the general work area for their worker(s).
- c. As of January 1, 2025, all temporary, interior, lighting, both task lighting and general access/egress lighting, shall be LED type lighting.
- d. All light stringers must be 12 AWG or thicker and be a 3-wire type (i.e., it must have a ground).
- e. If temporary lighting is wired directly into a panel, the respective breaker must be a GFCI type breaker.
- f. All lighting shall be Underwriters Laboratories (UL) listed. The use of "job-made" or "shop-made" lighting is prohibited.
- g. All temporary lighting shall be equipped with a lamp cage to keep other objects from coming into contact with the lamp. The outer lens of a halogen or similar light does not

meet the intent of this requirement. Exception to this requirement is if the temporary lighting is LED and manufactured without a cage.

- h. Night and or low light activities must have sufficient lighting to eliminate shadows that can create visual hazards.
- i. Halogen light bulbs containing mercury shall not be utilized in temporary light stringers unless the contractor responsible for maintenance of the temporary lighting has submitted to Okland a cleanup and remediation process for broken bulbs.

7. Control of Hazardous Energy Sources (Electrical & **Other**) – **Lock-Out Tag-Out:**

- a. The unexpected energizing or start-up of machines, tools or equipment, or the release of stored energy that could cause injury to workers or damage to equipment (e.g., Electrical, Hydraulic, Pneumatic, Magnetic, Heat, Mechanical, Radiation, Gravitational, Chemical, Stored Energy, such as in springs, batteries, and items under tension) shall be controlled through an effective Lock-Out Tag-Out (LOTO) program established by Subcontractor consistent with 29 CFR 1910.147 the control of hazardous energy.
 - i. The LOTO program shall require that all electrical wiring be treated as though it is energized until it is determined that the wiring is not yet connected to an electrical source, or the electrical source is properly locked-out and tagged.
 - ii. The LOTO program shall require that each worker exposed to a potential hazard from the unexpected energizing or startup of machines, tools or equipment, or the release of stored energy, be protected with their own lock and tag.
 - iii. The LOTO program shall require, when it can be done safely, an appropriate test of the LOTO to ensure that it has been effectively implemented.
 - iv. The LOTO program shall require workers to unplug all hand tools prior to servicing the tool (e.g., unplugging a grinder prior to changing the grinding disk).
 - v. The LOTO program shall require all combustion equipment to be turned off prior to fueling the equipment (e.g., turning off a compressor prior to fueling it).

8. Energized Electrical Work

- a. Subcontractor and their employees shall exhaust every effort to perform electrical work with electrical systems de-energized.
- b. Workers conducting live electrical work of any voltage shall only do so when they can be visually seen and heard by other workers within their company.
- c. Energized electrical work includes working on or near any energized electrical system, whether alternating or direct current, including, but not limited to, services entrance sections, distribution switchgear, transformers, distribution panels, UPS systems, and branch circuit wiring where a worker is required to deliberately, or could accidentally, place any part of their body or any type of tool or material in to or around such electrical devices where the voltage has been determined to be in excess of 50 volts. Examples of such work includes, but is not limited to:
 - i. Voltage testing;
 - ii. Circuit testing;
 - iii. Trouble-shooting;
 - iv. Power switching;
 - v. De-energizing and re-energizing procedures;
 - vi. Pushing fish-tapes and/or pushing/pulling wire into an energized enclosure;
 - vii. Work performed on or in energized enclosures; and
 - viii. Excavations near underground energized lines.

- d. Subcontractor and their employees shall comply with the energized electrical work safety requirements stipulated in the most current edition of NFPA 70E, Electrical Safety in the Workplace, including the use of an Energized Electrical Work Permit, when conducting energized electrical work.
 - i. The following shall be submitted to Okland prior to the any live electrical work taking place:
 - 1. An executed copy of Subcontractor's Energized Electrical Work Permit
 - 2. A copy of Subcontractor's Job Hazard Analysis for the specific live electrical work activity that will be taking place
 - 3. A detailed Method of Procedure specific to the live electrical work activity that will be taken place
- e. Access to energized electrical rooms or other areas where electrical work is taking place shall be limited to those workers who are wearing the required PPE and who are engaged in the energized electrical work.
- f. The electrical subcontractor, in coordination with Okland's Superintendent, shall maintain control of all energized electrical rooms and panels.
- g. Physical barriers that are lockable (e.g., doors with locks) and danger signs shall be used to prevent unauthorized entry to areas where energized electrical systems are in place and/or where any energized electrical work is being performed. Signage shall state **"DANGER ENERGIZED ELECTRICAL ROOM AUTHORIZED PERSONNEL ONLY"**
- h. Metal belt buckles, jewelry, key chains, cell phones, pagers, etc. shall be removed when working on any energized system. Hands shall be clean and free of any lotion, sunscreen or similar coating that may reduce the voltage rating of any rated glove liner being worn.

Subpart L – Scaffolds

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

- 1. Capacity
 - a. Except as stipulated in OSHA Regulations 29 CFR 1926.451(a)(1), each scaffold and scaffold component shall be capable of supporting, without failure, its own weight and at least 4 times the maximum intended load applied or transmitted to it.
 - b. Workers must know the capacity of the scaffold they will work from prior to accessing the scaffold.
- 2. Scaffold Construction
 - a. Prior to the construction of a stationary supported scaffold that has a platform height of greater than 24', a [Stationary Supported Scaffold Erection Permit](#) must be obtained from the Okland Project Management Team.
 - i. A copy of the signed permit shall be posted at or near the scaffold.
 - b. Prior to construction of any scaffold stair tower, a [Scaffold Stair Tower Permit](#), must be obtained from the Okland Project Management Team.
 - i. All scaffold stair towers shall be designed by a registered professional engineer and shall be constructed per that design.
 - c. Prior to the construction of any trash chute, a [Trash Chute Permit](#) must be obtained from the Okland Project Management Team.

- d. Scaffold base plates, mud sills, and/or casters shall only be placed on flat surfaces.
- e. Scaffolds shall be erected, used, modified, moved, and dismantled per the manufacturer's recommendations.
- f. Scaffolds shall be fully planked irrespective of the use of a personal fall arrest system.
- g. Scaffold platforms shall be at least 18" wide irrespective of the use of a personal fall arrest system. (e.g., for standard wood planks that are 12" wide, two planks are required).
- h. All scaffold planking shall be secured from movement/displacement (e.g., wired down, cleated, locked in).
- i. Where up lift can occur, which would displace any scaffold end frame or panel, the frames or panels shall be locked together vertically by pins or equivalent means.
- j. Cross braces are required on both sides of scaffolding between each frame section unless the scaffold manufacture has provided written instructions that allow for other cross bracing configurations.
- k. Job built scaffolds shall be constructed with lumber that is construction grade #1 spruce and plywood decking must be construction grade 3/4" at a minimum.
- l. All cantilevered scaffolds and/or cantilevered landing platforms (e.g., Preston Decks) shall be designed by a registered professional engineer and shall be constructed per that design. Design drawings shall be Project and application specific. Documentation of the design shall be submitted to the Okland Superintendent prior to the construction or installation of the scaffold/platform.
- m. All multi-sectional scaffolding must be tied into the building or stable structure, at a minimum, every 30' horizontally and, when the width to height ratio is less than 4:1, every 15' vertically.
- n. All scaffolds wrapped or enclosed with fabric, plastic or similar wrap shall be designed by a qualified person and shall be constructed per that design. Design drawing shall be Project and application specific. Documentation of the design shall be submitted to the Okland Superintendent prior to the construction of the scaffold/platform.
 - i. The scaffold tieback system should be designed by registered professional engineered to address additional loading due to wind.
 - ii. When direct fire heaters are used:
 - 1. The hazards of carbon monoxide buildup must be controlled.
 - 2. Additional housekeeping shall be conducted to ensure that combustibles and debris are not allowed to accumulate around the heat source.
 - 3. Tarps/tenting/wrapping must be of the flame-retardant type.
- o. If mud sills are required, mud sills shall be of adequate dimension to ensure stability of the scaffold and be secured to the scaffold base plate with a minimum of 2 nails per base plate.
 - i. When mud sills are used for scaffolding over 24' in height, the mud sills shall be, at a minimum, 2" thick x 10" wide x 24" in length when not installed on a concrete surface.
 - ii. When scaffolding over 24' in height is installed on a concrete or other similar structurally sound surface, mud sills are still required and shall be consistent with the requirements listed in the [Mud Sill Reference Sheet](#).
 - iii. When scaffolding over 24' in height is installed on soft or loose ground; wet sites; on other earth type surfaces; or will be heavily loaded, the proposed mud sill type shall be reviewed with an Okland Authorized Scaffold Reviewer.
- p. Base plates or casters are required on all fabricated frame scaffolds irrespective of the scaffold being on a solid surface, such as concrete. A [Fabricated Frame Scaffold Quick Reference Sheet](#) is included in this manual for reference.

- q. Oakland's Corporate Risk Management and/or Safety Department, may require, due to the elevated level of risk a scaffold poses, that the scaffold or certain components of the scaffold (e.g., scaffold wrap) be designed/engineered by a registered professional engineer. Such designs may include but not be limited to the scaffolding and scaffold wrap anchoring methods and requirements.

3. Criteria for Suspended Scaffolds

- a. The Oakland Superintendent, in consult with the Project assigned Oakland Safety Manager, shall develop a written plan detailing the safe construction, use and inspection of suspension scaffolds if used.
- b. Oakland's Safety department shall be advised prior to the use of any type of suspension scaffold on a project.
- c. The construction and use of suspension scaffolds shall be per the design of a registered professional engineer. Design drawings shall be Project and application specific. A copy of the design shall be submitted to Oakland's Superintendent prior to the construction of the suspension scaffold. If the system being proposed is a "manufactured system", and the installation and use of the scaffold system is within its intended design, the manufacturer's use/operation/installation manual shall suffice.
- d. Two-point suspension scaffolds (swing stages) shall not be less than 20" or more than 36" wide overall.
- e. A complete guardrail system including toe boards shall be provided on swing stage platforms.
- f. Suspension wires and ropes shall be protected from damage by heat or sharp edges.

4. Access

- a. Scaffolds equipped with guardrail at their access/egress points shall have an access/egress gate that opens onto the platform at each point of access/egress. Climbing over/under/around the guardrail as a means of access/egress is prohibited.
- b. Fixed vertical ladders, greater than 24', shall only be permitted if alternative access systems are infeasible.
 - i. When fixed vertical ladders greater than 24' are used, a PFAS shall be used while working from and/or climbing the ladder (e.g., a retractable lanyard, a fixed ladder climb safety device).
 - 1. Yellow CAUTION tags shall be attached at every access point (all levels) indicating that fall protection is required to utilize the scaffold ladder.
- c. Where direct access is provided from the inside of a building to the scaffold:
 - i. The access must be constructed with a solid platform that fully bridges the gap between the scaffold and the building.
 - 1. Guardrail shall be installed on both sides of the access bridge.
 - 2. Safe access onto, and from, the bridge to the building's floor and to the scaffold platform shall be provided.
 - (a) If the bridge is elevated greater than 19" above the building floor or the scaffold platform, a scaffold ladder, stairs, ramp, or similar safe access shall be provided.

- (b) If the bridge, or the method of access to/from the bridge (e.g., scaffold ladder, stairs, ramp), creates a potential fall hazard, additional fall protection (e.g., adding additional guardrail) shall be provided to control the fall hazard.

5. Use & Inspection

- a. Scaffolds (of all types) shall be inspected by a competent person prior to each shift and following any modifications including new assembly. Each inspection shall be documented on a scaffold tag. Inspection tags shall be attached to the scaffold at the base of the access ladder and any other access point.
 - i. If a scaffold is utilized by different contractors, a competent person of each contractor shall complete a documented inspection of the scaffold unless other written agreements are made. The contractor having care, custody, and control of the scaffold shall establish a process of reporting and correcting deficiencies with the scaffold and a system for complying with the inspection tag requirement.
 - 1. If a scaffold is utilized by different contractors, the scaffold shall not be modified by any other contractor, without the explicit permission from the contractor having care, custody, and control of the scaffold.
 - ii. “Green Tags” shall signify that the scaffold is complete and can be utilized with ordinary precaution.
 - iii. “Yellow Tags” shall signify that there are specific hazards associated with the scaffold that requires specific controls to be executed. Access and use shall be addressed through a JHA.
 - iv. “Red Tags” shall signify that the scaffold is NOT safe to access or use.
- b. If a scaffold does not have an inspection tag posted, employees shall assume that it is “Red Tagged” and not safe for use.
- c. Ladders, buckets, or any other makeshift object shall not be a component of any scaffold.
- d. Employees shall not stand on the guardrail of any scaffold including scissor lifts.
- e. Working from the scaffold ladder is prohibited.
- f. Workers must know the capacity of the scaffold they will work from prior to accessing the scaffold.
- g. Scaffolds, or any component of a scaffold, shall not be placed within 10’ from an electrical hazard. If the voltage is over 50kv, the distance shall be increased by 0.4-inches for every 1kv over 50kv.

6. Fall Protection

- a. Scaffold erectors/dismantlers shall utilize fall protection when exposed to a fall of 6’ or more.
- b. Guardrail, mid-rail and toe-board are required on all scaffolds if at all feasible. Workers do not have the option of utilizing a personal fall arrest system in lieu of completing a scaffold if it is reasonably feasible to install guardrails.
- c. Guardrail is required on all open sides of fixed/stationary scaffolds when the scaffold’s work platform is 6’ or more above a lower level. With the exception of OSHA’s variance under 1926.451(b)(3), (b)(3)(i), & (b)(3)(ii) – (outrigger scaffolds & lathing/plastering operations).
- d. Cross braces on scaffolding do not constitute guardrail. Guardrail systems shall be horizontally installed. The use of diagonal or vertical members as guardrail top-rail or mid-rail is prohibited.

- e. Guardrail is required on all sides of any scaffold which has a work platform of 46" or less in its least dimension when the scaffold's work platform is 4' or more above a lower level (e.g., on "Baker" scaffolds).
- f. Guardrail is required on all open sides of mobile scaffolds (i.e., scaffolds equipped with wheels – locked or unlocked, including Baker type scaffolds) when the scaffold's work platform is 4' or more above a lower level. A Baker Scaffold Quick Reference Sheet is included in this manual for reference.
- g. A complete guardrail system including toe boards shall be provided on swing stage platforms.
- h. When a scaffold, of any type, is positioned within 6' from an edge that is elevated 6' or more above a lower level, guardrail must be installed on the scaffold, regardless of the height of the scaffold.
 - i. If it is infeasible to install a guardrail system on the scaffold, a Personal Fall Arrest System (PFAS) shall be utilized by each worker that works from the scaffold.
 - 1. The PFAS shall be engaged to the anchor point prior to the worker climbing the scaffold ladder to gain access to the scaffold.
- i. Fall protection when working from and/or climbing any scaffold ladder is required where the fall exposure to the employee is greater than 24'.
- j. Fall Arrest wires and ropes shall be protected from damage by heat or sharp edges.
- k. Suspension scaffolds require a rescue/retrieval plan prior to use.
- l. Occupants must be fitted with suspension straps while on the scaffold.
- m. If modifications to guardrail is required (e.g., to facilitate building materials being installed, and/or tools, materials, equipment, or debris, being loading into or out of the building, etc.), the Subcontractor creating the need for the modification is responsible to properly modify the guardrail and/or install equally effective guardrail further back from the fall exposure. Subcontractor shall notify Okland's Project Superintendent and secure a Guardrail Removal/Modification Permit from Okland prior to modifying or removing any guardrail that is under the care, custody and control of Okland and/or provides protection for any other company or the public (i.e., a permit is not needed when Subcontractor is modifying their own guardrail [e.g., ironworkers adjusting cable guardrail on a deck that is not yet turned over to Okland, a mason subcontractor modifying guardrail on their scaffolding that is not controlled by Okland, etc.]).
 - i. A Personal Fall Arrest System must be utilized during the guardrail removal and installation when a fall exposure is present.
 - ii. If a scaffold guardrail system is going to be modified, a Scaffold Competent Person shall be present during the modification.

7. Falling Object Protection

- a. When overhead work must be completed and effective falling object controls cannot be implemented, staggering of work schedule or sequence shall be instituted to eliminate the exposure of the falling objects.
- b. When members of the same crew must work in tandem to execute the assigned task and that work must be conducted with one or more worker working overhead other workers involved in the same task, the overhead worker shall utilize tool/equipment/material tethers/lanyards to ensure a falling object does not strike the lower worker.
- c. When scaffold work takes place at the perimeter of a building, Subcontractor shall comply with Okland's [Overhead Protection at Building Perimeter Policy](#). [See Subpart G.](#)

8. Ladder jack scaffolds
 - a. Ladder jack scaffolds shall not be utilized on Okland Projects.
9. Stair Towers/Stair Scaffolds
 - a. Stair towers shall include landing platforms at each level in each direction of travel.
 - b. Toe-boards shall be installed on each landing platform.
10. Suspension Scaffolds (swing stages)
 - a. Two-point suspension scaffolds (swing stages) shall not be less than 20" or more than 36" wide overall.
 - b. A complete guardrail system including toe boards shall be provided on swing stage platforms.
11. Multi-point adjustable suspension scaffolds, stonemasons' multi-point adjustable suspension scaffolds, and masons' multi-point adjustable suspension scaffolds.
 - a. Meet OSHA Regulation.
12. Mobile & Narrow Frame Scaffolds
 - a. Scaffolds shall be erected, used, modified, moved, and dismantled per the manufacturer's recommendations.
 - b. The surfaces on which mobile scaffolds are erected, moved, used, and/or dismantled shall be:
 - i. Solid and within 3-degrees of level.
 - ii. Free of open pits, holes, unprotected elevation changes, and other similar hazards that could upset the stability of the scaffold as it is moved.
 - iii. Free of debris, hoses, cords, welding lead or any other obstruction that could upset the stability of the scaffold as it is moved.
 - c. Narrow Frame Scaffolds (e.g., Baker's scaffolds) shall NOT be used as a means of support ends to "bridge" scaffold planks/platforms to other "Baker" scaffolds, other scaffolds, other supports, or similar.
 - d. Guardrail is required on all sides of any scaffold which has a work platform of 46" or less in its least dimension when the scaffold's work platform is 4' or more above a lower level (e.g., on "Baker" scaffolds).
 - e. Guardrail is required on all open sides of mobile scaffolds (i.e., scaffolds equipped with wheels – locked or unlocked, including Baker type scaffolds) when the scaffold's work platform is 4' or more above a lower level. A [Baker Scaffold Quick Reference Sheet](#) is included in this manual for reference.
 - f. When freestanding mobile scaffold towers are used in a stationary position, the height shall not exceed 3 times the minimum base dimensions.
 - g. Narrow framed scaffolds shall not exceed a height to base ratio of 3:1.
 - h. Outriggers shall be placed on all narrow frame scaffolds any time the scaffold is stacked (i.e., the scaffold is more than one section in height).
 - i. Outriggers shall be placed on all four legs of the scaffold.
 1. When the scaffold is positioned against a wall that is of sufficient strength to support the scaffold, outriggers shall be installed on the two legs of the scaffold opposite of the wall.
 - i. Employees shall lock all wheels on mobile scaffolds prior to accessing the scaffold.

- j. Employees shall ensure that all wheels stay locked while they are working from a mobile scaffold in a stationary position (i.e., a mobile scaffold that is not being moved uninterruptedly).
- k. Employees shall NOT “ride” on any mobile scaffold which has a work platform of 46” or less in its least dimension (e.g., “Baker” scaffolds). On these scaffolds, wheels shall stay locked at all times an employee is on the scaffold.
- l. Prior to an employee “riding” on any mobile scaffold with a work platform of more than 46” in its least dimension, a JHA shall be completed, the JHA shall be submitted to Okland, and the employee’s supervisor shall ensure that all the requirements under OSHA’s 29 CFR 1926.452(w)(6) are met.
- m. While working from a scaffold, workers shall not stand on the guardrail, ladders, boxes, buckets, or other objects.
- n. Working from the scaffold ladder is prohibited.
- o. Workers must know the capacity of the scaffold they will work from prior to accessing the scaffold.
- p. Base plates or casters are required on all narrow frame scaffolds irrespective of the scaffold being on a solid surface, such as concrete.
- q. Scaffolds, or any component of a scaffold, shall not be placed within 10’ from an electrical hazard. If the voltage is over 50kv, the distance shall be increased by 0.4-inches for every 1kv over 50kv.
- r. When a scaffold, of any type, is positioned within 6’ from an edge that is elevated 6’ or more above a lower level, guardrail must be installed on the scaffold, regardless of the height of the scaffold.
 - i. If it is infeasible to install a guardrail system on the scaffold, a Personal Fall Arrest System (PFAS) shall be utilized by each worker that works from the scaffold.
 - 1. The PFAS shall be engaged to the anchor point prior to the worker climbing the scaffold ladder to gain access to the scaffold.
- s. Positioning Narrow Frame Scaffolds Near an Edge:
 - i. When a narrow frame scaffold is positioned parallel to an edge that is elevated 6’ or more above a lower level, the parallel side of the scaffold shall not be placed within 6’ from the edge.
 - 1. If it is infeasible to reposition the scaffold so that it is not parallel to the edge or use an alternate means to access the work (e.g., a scissor lift, a scaffold that is not a narrow frame scaffold.) a Personal Fall Arrest System (PFAS) shall be utilized by each worker that works from the scaffold.
 - (a) The PFAS shall be engaged to the anchor point prior to the worker climbing the scaffold ladder to gain access to the scaffold.
- t. When a narrow frame scaffold is positioned perpendicular to an edge that is elevated 6’ or more above a lower level, the perpendicular side of the scaffold shall not be placed within 3’ from the edge.
 - i. Workers shall only access the scaffold from the ladder access point furthest from the edge.
 - ii. If it is infeasible to position the scaffold so that it is not closer than 3’ from the edge or use an alternate means to access the work (e.g., a scissor lift, a scaffold that is not a narrow frame scaffold) a Personal Fall Arrest System (PFAS) shall be utilized by each worker that works from the scaffold.

1. The PFAS shall be engaged to the anchor point prior to the worker climbing the scaffold ladder to gain access to the scaffold.

13. Mobile Elevated work Platforms (MEWP)

- a. Each Mobile Elevated Work Platform (MEWP) (e.g., scissor lifts, aerial lift, vertical mast lifts) must also have a documented inspection conducted by a competent person on the [MEWP Operators Inspection Form](#) prior to each shift. The scaffold tagging system is not required to be utilized on these types of equipment unless Subcontractor chooses to use them.
- b. Employees shall not stand on the guardrail of any scaffold including scissor lifts.
- c. Unless explicitly allowed, in writing, within the OEM operators manual, scissor lifts shall be fully lowered prior to moving.
- d. Mobile Elevated Work Platforms (e.g., aerial lifts, scissor lifts) shall only be fitted with attachments that are approved by the Original Equipment Manufacturer (OEM) (i.e., shop made/job made attachments are prohibited).
- e. Spotters:
 - i. During the **Pre-Task Plan (PTP)**, the need for a spotter shall be evaluated, including determining the appropriate number of spotters and how they will be used effectively.
 - ii. When a spotter is used, the spotter shall be on the ground, not in the scissor lift.
 - iii. At a minimum, a spotter is required when the scissor lift **is being operated** (i.e., raised, lowered, or moved horizontally):
 1. In finished areas.
 2. Through any doorway where the finish door frame has already been installed.
 3. In confined or congested areas (e.g., work near mechanical piping, electrical systems, overhead ductwork, within tight areas such as partitioned zones etc.).
 4. Within 3-feet of an elevated edge that is not protected by a solid wall.
 5. On any elevated platform of any type.
 6. Up or down any type of ramp.
 7. In any occupied area.
 8. Where there is any potential for public exposure.
 9. When loading and unloading of MEWPs from man/material hoists or elevators.
 10. During any night or low light work.
 11. In any location where the operator of the lift is in a Lone Worker situation.
- f. MEWPs shall not be operated on elevated decks, roofs, or similar without adequate perimeter protection that can effectively stop the MEWP from traveling beyond the elevated edge, penetration, shaft, skylight, or similar ("edge").
 - i. The elevated edge protection shall be in place around the entire elevated deck, roof or similar that the MEWP can access.
 1. The elevated edges needing protected can be minimized by having adequate area specific protection in place that can effectively stop the MEWP keeping it in a specific area.
- g. Scissor-Lifts shall not be moved horizontally while the operator is walking adjacent to it, commonly referred to as "walking the dog".

Subpart M – Fall Protection

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Irrespective of OSHA regulations governing specific fall protection requirements, a positive means of fall protection (e.g., guardrail system, safety net system, or personal fall arrest system) shall be utilized whenever workers are exposed to a fall which is 6' or more above a lower level. The only three exceptions to this requirement is the fall protection requirements for workers working from mobile ladders which is covered in [Subpart X](#) of this document, the requirement for fall protection when climbing scaffold ladders which is covered in [Subpart L](#) of this document and the requirement for fall protection on "Baker" scaffolds which is required at 4-feet.
2. Workers conducting activities where a Personal Fall Arrest System (PFAS) is required to be used shall only do so when they can be visually seen and heard by other workers within their company.
3. Prior to assigning a worker to conduct work requiring the use of a personal fall arrest system, the supervisor of that worker shall establish a fall rescue plan, ensure the needed rescue equipment is in safe working condition and that equipment is readily available.
4. Personnel utilizing a Personal Fall Arrest System (PFAS) while exposed to a fall hazard shall not work alone. One or more personnel, that are NOT anchored to the same PFAS, must be in sight to initiate the fall rescue plan if a fall occurs.
5. C-clamps or other accessories (e.g., shackles, slings, chain, wire-rope clips) not commercially manufactured by a fall protection specific manufacturer, that are manufactured outside of the United States (e.g., China) shall not be utilized as components of a fall protection system (e.g., guardrail, lifelines).
6. All falls, arrested or not-arrested, shall be immediately reported to the Okland Safety Department so they can support a thorough incident investigation.
7. Workers are expected to clean their assigned harness and lanyard(s), on a regular basis, to ensure they are kept free of contaminants that may affect their safe performance.
8. All components subjected to a fall arrest shall be immediately removed from service.
9. In addition to the user inspecting their fall protection prior to each use, each personal fall arrest systems shall be inspected on an annual basis by a qualified person other than the user.
10. Personal fall arrest systems used for edge work shall be rated by the manufacturer for use in edge work.
11. When a positioning hook device is utilized, and the worker is exposed to a fall of 6' or more above a lower level, a personal fall arrest lanyard shall also be utilized.

12. If a horizontal lifeline system is proposed to be utilized as part of a personal fall arrest system, Subcontractor shall submit written documentation to Okland which demonstrates that the entire personal fall arrest system has been designed by a registered professional engineer and is in compliance with OSHA regulations. If the system being proposed is a manufactured system, the manufacturer's use/operation/installation manual will suffice.
13. The use of a Self-Retracting Lifeline (SRL) shall not expose the user to dangerous swing in the event of a fall.
14. **Prior to** a warning line system being utilized, a written plan and sketch of its configuration shall be submitted to Okland and reviewed by Okland's project assigned Safety Manager.
 - a. **Warning Lines shall only be utilized on ROOFS.**
 - b. Warning Lines when used where **ONLY ROOFERS** are on the roof conducting roofing work, shall be at least **6'** from any edge where a fall exposure greater than 6' exists.
 - c. Warning Lines when used where **OTHER TRADES** (other than roofers) are on the roof shall be at least **15'** from any edge where a fall exposure greater than 6' exists.
 - d. Prior to workers traveling beyond a Warning Line System, 100% tie-off shall be utilized.
 - e. Warning Line Systems should only be utilized when roofing work is taking place. If no roofing work is taking place, a guardrail system or the use of a personal fall arrest system should be utilized.
15. If Subcontractor creates a hole in a walking/working surface, Subcontractor shall ensure that the hole is covered, and the cover is maintained. Hole covers shall be properly constructed, marked, and secured from accidental displacement.
16. Hole covers must meet Okland's [Floor Opening Hole Cover Standard](#).
17. Covers must be readily available in the immediate work area prior to creating a hole.
18. If modifications to a hole cover is required to facilitate building materials being installed, the Subcontractor creating the need for the modification is responsible to properly modify and secure the hole cover from accidental displacement.
19. The tops of all hole covers shall be labeled "HOLE" or "COVER". If the cover is too small to label it, the cover shall be completely coated with orange paint.
20. All hole covers shall be cleated to prevent accidental displacement.
21. Skylights themselves shall not constitute an acceptable hole cover.
22. Skylight covers shall be capable, at a minimum, to support 500-lbs.
23. When skylights will not be covered with an Okland acceptable cover, one of the following shall be implemented:
 - a. A guardrail system shall be installed around the skylight.
 - b. A warning line system shall be installed around the skylight that is set, at a minimum, 15' back from the edge of the skylight.

- i. A personal fall arrest system must be worn and properly utilized by any worker entering the area between the warning line system and the unprotected skylight.
 - 1. The personal fall arrest system shall be rated for leading edge work.
- 24. When floor holes/penetrations that are protected with a guardrail system must be covered for weather/dust protection or similar needs, and the material being used does not meet the requirements of a hole cover (e.g., plastic, Visqueen, corrugated plastic), the top of the material shall be clearly marked with a danger sign(s) stating **“DANGER” “OPEN HOLE” “DO NOT STEP”**. In addition, signage shall be placed on the guardrail system protecting the hole/penetration stating **“DANGER” “100% TIE-OFF BEYOND THIS POINT”**
 - a. When this condition exists and work must take place within the guardrail system, workers shall utilize a personal fall arrest system, shall be tied-off prior to crossing the guardrail system, and shall maintain 100% tie-off at all times.
- 25. Prior to the removal of a guardrail system that is in place around a floor hole/penetration that has been covered with weather/dust protection such as plastic, Visqueen, corrugated plastic, or similar material that does NOT meet the OSHA and Okland requirement for a hole cover, a compliant hole cover shall be installed.
- 26. A personal fall arrest system shall be utilized by all workers exposed to a hole that is greater than 12” in its least dimension when a fall exposure of 6’ or more is present.
- 27. Workers working in incomplete elevator cabs shall maintain 100% tie-off at all times the worker is in the elevator cab.
- 28. Workers working on top of elevator cabs shall maintain 100% tie-off at all times and the elevator shall be properly locked and tagged out.
- 29. Fall restraint systems, at a minimum, shall have the capacity to withstand at least 3,000 pounds of force or twice the maximum expected force that is needed to restrain the worker from exposure to the fall hazard.
- 30. Unless otherwise stipulated in the contract documents, the Subcontractor creating a fall hazard is responsible for correcting the fall hazard. However, this does not relieve the exposing Subcontractor from protecting its employees from all potential fall hazards.
- 31. “Slab-Grabber” guardrail systems shall not be utilized on Okland Projects without Subcontractor first consulting the Okland Project Superintendent and the Okland assigned Safety Manager. Subcontractor shall develop an installation and maintenance plan that is consistent with the manufacturer’s recommendations.
- 32. Guardrails shall be the primary form of fall protection utilized. If guardrail cannot be installed, the use of a personal fall arrest system shall be implemented.
- 33. Guardrail systems shall be horizontally installed. The use of diagonal or vertical members as guardrail is prohibited.

34. When wood, pipe, or structural steel is utilized to construct the guard rail system it shall be constructed to meet or exceed the guidelines in 1926 Subpart M Appendix B.
35. Guardrail shall be installed on the inside or on top of the posts/stanchions.
36. Guardrail shall be mechanically fastened with screws, bolts, nails, etc. Tie-wire, 9-wire, or similar shall not be utilized to secure guardrail in place.
37. If cable is used to construct a guardrail system:
- a. It shall consist of a 3/8" diameter cable top-rail installed 42" (+/- 3") above the height of the finished concrete floor (not the metal deck), a 3/8" diameter cable mid-rail installed 21" above the height of the finished concrete floor (not the metal deck) and a minimum of a 3-1/2" toeboard.
 - b. All cable connections shall be looped (not lapped) and shall be connected with a minimum of three wire rope clips (U.S. only) on each loop regardless of the cable size requiring less.
 - c. Wire rope clips must be installed properly (never saddle a dead horse).
 - d. Wire rope dead tail ends should be shorter than 6". If the dead tail is longer than 6", install an additional wire rope clip or otherwise tie up and secure the tail.
38. 100% tie-off must be utilized if any part of the worker's head or torso is placed outside the guardrail system.
39. If modifications to guardrail is required (e.g., to facilitate building materials being installed, and/or tools, materials, equipment, or debris, being loading into or out of the building, etc.), the Subcontractor creating the need for the modification is responsible to properly modify the guardrail and/or install equally effective guardrail further back from the fall exposure. Subcontractor shall notify Okland's Project Superintendent and secure a Guardrail Removal/Modification Permit from Okland prior to modifying or removing any guardrail that is under the care, custody and control of Okland and/or provides protection for any other company or the public (i.e., a permit is not needed when Subcontractor is modifying their own guardrail [e.g., ironworkers adjusting cable guardrail on a deck that is not yet turned over to Okland, a mason subcontractor modifying guardrail on their scaffolding that is not controlled by Okland, etc.]).
- a. A Personal Fall Arrest System must be utilized during the guardrail removal and installation when a fall exposure is present.
 - b. If a scaffold guardrail system is going to be modified, a Scaffold Competent Person shall be present during the modification.
40. Guardrail, when used to protect leading edge exposures, shall be placed between 6' – 30' back from the leading edge and, when feasible, shall be placed parallel with the leading edge.
41. Danger signs shall be placed on guardrail protecting leading edge work stating "DANGER" "100% TIE OFF REQUIRED BEYOND THIS POINT"
42. When the height of a wall form scaffold, or similar deck, is placed in a configuration that the top of the wall form is less than 39" from the walking/working surface, a guardrail system should be

installed on the backside of the wall form to provide fall protection for the workers on the wall form scaffold. When the installation of guardrail is not feasible, all workers on the wall form scaffold shall utilize a personal fall arrest system and maintain 100% tie-off.

43. Knots of any kind shall not be placed in any type of fall arrest, restraint or prevention system including anywhere along its midpoint or its anchor point.
44. Guardrail or equivalent barriers shall be posted at each excavation and/or trench that is 6' or more in depth. Additional barricading requirements for excavations are listed in [Subpart P](#) of this document.
45. Bike barricades and similar types of non-fall protection barricades, shall not be within 20' of an edge where G-rail or other similar temporary base-weighted guardrail is in place so as the bike barricades are NOT mistaken for a proper guardrail device.
46. Falling Objects:
 - a. Where the potential exists for an object to fall from a person, building, structure, or equipment due to circumstances which require the object to be located at, near, above, or beyond the installed falling object protection (e.g., toe-boards, screens, mesh) additional positive control measures shall be implemented to prevent the object from falling to any lower level. The duty and cost of implementing additional controls shall be the responsibility of the Subcontractor creating the hazard. Examples of additional control measures include, but are not limited to, lanyards, debris nets, and catch basins.
 - b. Where overhead work is taking place, such as steel erection, decking, welding, bolting up, etc. and it is not feasible to implement a positive control measure, the entire area below the operation that could be potentially impacted from a falling object(s) shall be completely barricaded and the necessary number of spotters needed to control the area shall be assigned to stand outside the barricaded area to prohibit access to the area while a falling object hazard exists.
 - c. Screens, panels, or similar must be installed where objects have the potential to fall to from an upper level down to a public area.
 - d. **Designated Covered Access/Egress Points:**
 - i. To protect personnel from overhead hazards, Okland's Project Management Team shall ensure established covered entry and exit points, to buildings under construction, are in place.
 - ii. The covered entry and exit points shall be in compliance with Okland's [Overhead Protection at Building Perimeter Policy](#):
 1. Covers shall extend at least 10-feet out from the face of the building for overhead exposures of 40' or less.
 2. For overhead exposures greater than 40' reference the table below for the required extension from the face of the building:

Building Height	Equation	Minimum Barricade Distance
Up to 50'	$4\text{mph} \sqrt{\frac{2 \times 50}{15}} = 10.3'$	10'- 4"
Up to 60'	$4\text{mph} \sqrt{\frac{2 \times 60}{15}} = 11.3'$	11'- 4"
Up to 70'	$4\text{mph} \sqrt{\frac{2 \times 70}{15}} = 12.2'$	12'- 3"
Up to 80'	$4\text{mph} \sqrt{\frac{2 \times 80}{15}} = 13'$	13'- 3"
Up to 90'	$4\text{mph} \sqrt{\frac{2 \times 90}{15}} = 13.9'$	13'- 11"
Up to 100'	$4\text{mph} \sqrt{\frac{2 \times 100}{15}} = 14.6'$	14'- 7"
Up to 110'	$4\text{mph} \sqrt{\frac{2 \times 110}{15}} = 15.3'$	15'- 4"
Up to 120'	$4\text{mph} \sqrt{\frac{2 \times 120}{15}} = 16'$	16'

3. Covered walkways shall be capable of supporting live loads 150-pounds per square foot (psf).
 - a. If overhead hazards are anticipated to impose live loads greater than 150-psf, the covered walkway shall:
 - i. Be reinforced to support the anticipated live load; or
 - ii. Be relocated to an area where the anticipated live load would not exceed 150-psf; or
 - iii. Be closed and adequately barricaded to prevent its use
4. Be designed and constructed to support all imposed loads
5. Covered walkways shall be a minimum of 4-feet wide and have a minimum clear height of 8-feet from the walking surface to the lowest point of the cover/ceiling.
6. Covered walkways shall be illuminated to a minimum of 5-foot-candles and be free of any trip hazards.
7. SIGNAGE:
 - a. Okland branded SAFE ENTRY/EXIT signage (2'x2') shall be placed adjacent to the exterior entry of the covered walkway.
 - b. Building perimeter signage must be strategically placed at locations around the perimeter of the building, directing workers to the safe entry/exit points.
 - i. If the interior perimeter of the building will be barricaded to guide team members to the safe entry and exit points, use snow fence. Do **not** use caution or danger tape.
8. Covered walkways shall be inspected and approved by the project's assigned Okland Safety Manager prior to putting the walkway into service.

9. While there are multiple methods to construct a covered walkway compliant with Okland's requirements below are 4-recommended methods:
 - a. A [scaffolding system](#)
 - b. A [stick framed](#) system
 - c. A shoring system (see your project's General Superintendent for design recommendations)
 - d. Conexes (See Okland's Equipment/Warehouse Manager for options)
10. If a scaffold system, or similar system, is placed over the covered walkway, extend the covered walkway a distance equal to the width of the scaffold system as measured from the face of the building to the backside of the scaffolding.
11. When there is no space for a covered walkway, or when a covered walkway cannot extend the required distance from the face of the building, a [debris netting system](#) (floor to ceiling) shall be established on each floor (floor-to-ceiling) that is above and in direct line with the entry and exit point and extend 10' on both sides of the entry and exit point.
 - a. Any areas not protected by a floor to ceiling [debris netting system](#) (for example, work taking place on a roof), a [cantilevered debris net system](#) shall be used; the debris net shall not be placed lower than one floor below the uppermost floor and must extend a minimum of 10' horizontally.
 - b. A guardrail system must also be used on both sides of the safe entry/exit point that extends 6' out from the face of the building.
 - c. While there are several debris netting systems in the market, Okland's preferred Net System Vendor is:
www.trionetsinc.com
 Marvyn Ramdeen,
 Telephone: (346) 216-1281
 Email: marvyn@trionetsinc.com
- iii. Workers shall utilize the designated access/egress points established by Okland for access/egress to/from buildings under construction.
- iv. A covered walkway (safe entry/exit point) can be removed when:
 1. The building's envelope system is fully installed, on all levels, directly above the safe entry and exit point AND when all roof work is completed directly above.

47. [Overhead Protection at Building Perimeter Policy:](#)

- a. When any work is taking place on an elevated level, within 10-feet from an elevated edge that is not protected by a solid wall or a floor-to-ceiling net, or when elevated work is taking place on the exterior of building (e.g., from an aerial lift, swing stage scaffold, etc.), the area below the elevated level shall be barricaded with an effective barricade system. At a minimum, RED DANGER TAPE, shall be used as the barricade system.

b. Signage:

- i. Signage shall be placed on the barricade system notifying personnel of the overhead hazard.
- ii. Signage shall state: ["DANGER DO NOT ENTER OVERHEAD WORK"](#)
- iii. Signage shall be placed so that it is facing all directions that workers could potentially enter the barricaded area.
 1. The number of signs required is dictated by the configuration of the barricaded area and the number of visual obstructions of the signs. Signage shall not exceed more than 80-feet between signs and be placed at the most likely points of entry (doorways, corridors, envelope penetrations, etc.) into the barricaded area. Additional signage shall be placed when a visual obstruction prohibits the other signs from being seen.
- iv. At least one [Danger Barricade Tape Information Sign](#) shall also be posted on the barricade system so as other workers can be informed of the barricade owner and contact information.

c. Barricade Distance:

- i. The barricade system shall be placed from the face of the building and extend outward to the minimum distances listed in the table below and extend 10-feet on both sides of the work area.
 1. If the work is taking place on the exterior of the building from an aerial lift, swing stage or other elevated access system, the barricade system shall be placed from the face of the building and extend outward to the minimum distance listed in the table below, **plus the width of the elevated access system**, and extend 10-feet on both sides of the work area.
- ii. The barricade system shall also be placed from the face of the building and extend **inward** no less than 2-feet. This is to prevent workers from extending their head or torso into the potential drop zone of an object from the interior of the building.
 1. If a wall is in place that prevents workers from extending their head or torso into the potential drop zone, barricade tape, in the area the wall is in place, is not required.
 - a. All openings in walls must be barricaded with the 2-foot inward extension barricade system.

Working Level Height	Equation	Minimum Barricade Distance
Up to 40'	None - Minimum	10'
Up to 50'	$4\text{mph} \sqrt{\frac{2 \times 50}{15}} = 10.3'$	10' - 4"
Up to 60'	$4\text{mph} \sqrt{\frac{2 \times 60}{15}} = 11.3'$	11' - 4"
Up to 70'	$4\text{mph} \sqrt{\frac{2 \times 70}{15}} = 12.2'$	12' - 3"
Up to 80'	$4\text{mph} \sqrt{\frac{2 \times 80}{15}} = 13'$	13' - 3"
Up to 90'	$4\text{mph} \sqrt{\frac{2 \times 90}{15}} = 13.9'$	13' - 11"
Up to 100'	$4\text{mph} \sqrt{\frac{2 \times 100}{15}} = 14.6'$	14' - 7"
Up to 110'	$4\text{mph} \sqrt{\frac{2 \times 110}{15}} = 15.3'$	15' - 4"
Up to 120'	$4\text{mph} \sqrt{\frac{2 \times 120}{15}} = 16'$	16'

48. When work must be conducted overhead of workers, sensitive equipment, other sensitive building materials, or within close proximity to other workers below, that are within the barricaded area, tool tethers are required to be used with all tools.

49. When rigging accessories are used as part of a Personal Fall Arrest System (PFAS), the rigging shall:
- Be marked with Green Electrical Tape immediately next to the rigging's identification tag.
 - Be in like new condition.
 - Be used within its manufacture's listing capacity.
 - NEVER have been loaded (e.g., never been used for hoisting, towing).
 - NOT be equipped with any form of mechanical function (e.g., come-alongs, chain-falls).

Subpart N – Helicopters, Hoists, Elevators, and Conveyors

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

- Okland's Safety Department shall be advised and involved in all proposed helicopter operations.
- Personnel & Material Hoists
 - Prior to the assembly and/or disassembly of any personnel or material hoist, [Okland's Personnel/Material Hoist Checklist](#) process shall be executed and a completely signed [Okland Tower Crane & M/M Hoist Assembly/Climb/Disassembly Authorization Form](#) shall be secured.
 - The manufacturer's specifications and limitations applicable to the operation, use and maintenance of personnel and material hoists shall be available on site, complied with, and posted within the operator's station of the specific hoist. A copy of this information shall be submitted to an Okland Superintendent prior to erection of the hoist.

- c. All personnel and/or material hoists shall be operated, inspected and maintained as per the manufacturer's recommendations. All inspections and maintenance shall be documented. A copy of the inspections and maintenance record shall be kept on file at the operator's station.
- d. Construction & operation of personnel hoists shall comply with ANSI A10.4 – current.
- e. Construction & operation of material hoists shall comply with ANSI A10.5 – current.

Subpart O – Motor Vehicles, Mechanized Equipment (*Including Forklifts*), and Marine Operations

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Prior to any mobile equipment being used in the Public Right of Way (i.e., outside our project fence or in any area where a project fence is not established), a [Mobile Equipment in Public Right of Way Permit](#) must be obtained from the Okland Project Management Team.
2. Workers conducting activities before normal working hours or after normal working hours involving the use of earth or material handling equipment (e.g., forklifts, skid-steers, back-hoes, scrapers, graders) shall only do so when they can be visually seen and heard by other workers within their company.
 - a. When this equipment is used during normal working hours and its use is in areas where the operator cannot be visually seen and heard by others within their company (e.g., remote yard, remote areas of the project), the supervisor and the operator shall establish a method to verbally check-in with each other while those operations are taking place (e.g., radio communication, phone communication).
3. Workers conducting activities involving the maintenance or repair of mobile or fixed equipment (e.g., MEWPs, man/material hoists, forklifts, skid-steers, back-hoes, scrapers, graders) where there is a potential for the released of stored energy (e.g., electrical, hydraulic, gravity) shall only do so when they can be visually seen and heard by other workers within their company.
4. Equipment Keys
 - a. Subcontractors shall ensure all keys are removed from their equipment at the end of each shift.
 - b. Subcontractors shall ensure that only authorized individuals are allowed to operate their mobile equipment.
5. Documented daily inspections shall be conducted on all equipment (e.g., forklifts, skid-steers, earthmoving equipment) utilized on the project. Copies of inspection forms must be submitted to the Okland Superintendent.
6. All cab glass shall be safety glass and shall be free from any cracks and/or damage.
7. Operators of **motor vehicles** shall have a valid State issued driver's license issued to them, in their name and in their possession/on their person while operating a motor vehicle in the course of their work while on or off the Project.

8. Operators of **mobile equipment** shall have a valid State issued driver's license, in their name and in their possession/on their person while operating equipment on public streets.
9. Subcontractor shall ensure only trained and qualified operators are assigned to operate mobile equipment. Training shall be conducted by a qualified person and a field evaluation shall be conducted of the operator's skill level prior to the operator being allowed to operate the equipment without continuous monitoring by the trainer.
10. Mobile equipment operators shall have adequately demonstrated to their employer their ability to safely operate the specific type of equipment being assigned prior to operation.
11. Operators shall have an understanding of the owner's/operator's manual for the specific equipment they are operating and ensure a copy of the manual is readily available.
12. Additional Okland and/or project specific training may be required for supervisors and workers prior to operating mobile equipment on the project.
13. Operators shall have no physical, visual, or mental restrictions or impairments that will affect the safe operation of the mobile equipment.
14. Seatbelts shall be installed and utilized in all equipment manufactured with a seat and with a Roll Over Protective Structure (ROPS).
15. Workers shall not ride in the beds, buckets, sides, tops, or similar of motor vehicles and/or equipment.
16. Take 3 Initiative:
 - a. Prior to exiting the cab of any mobile equipment, the operator shall take three seconds to remove their hands and feet from the equipment controls and ensure:
 - i. The equipment has come to a complete stop.
 - ii. The equipment is not on a grade or slope that will allow the equipment to roll if the park break fails.
 - iii. The equipment's transmission is placed in neutral.
 - iv. The equipment is turned off.
 - v. The park brake is set, and it holds.
 - vi. The forks, bucket, or other attachment is lowered to the ground.
 - vii. Their hard hat is on.
 - viii. Their safety glasses are on.
 - ix. Their high visibility shirt, vest, or jacket is on.
 - x. While exiting, use three points of contact (i.e., no jumping out of the equipment).
17. Motor vehicles shall not be left unattended when started (i.e., operator not at the operator's station).
18. No modifications to motor vehicles or equipment shall be made which affects their safe operation. Motor vehicles or equipment with such modifications shall not be allowed on the Project.

19. All mobile equipment shall be equipped with a functional and audible back-up alarm, (audible above the surrounding noise level).
20. All motor vehicles with an obstructed view to the rear and all forklifts shall be equipped with a functional back-up alarm irrespective of the use of signal personnel/spotters.
21. Spotters shall be used when work with mobile equipment is being performed:
- a. Around parked vehicles
 - b. Around sensitive equipment
 - c. Around live utilities
 - d. Tight areas that are within 2' of any structure or equipment
 - e. Around public foot or vehicular traffic
 - f. In areas of high foot, vehicle and/or equipment traffic
 - g. In areas where the operator's view is obstructed
 - h. Specific to MEWPs:
 - i. During the **Pre-Task Plan (PTP)**, the need for a spotter shall be evaluated, including determining the appropriate number of spotters and how they will be used effectively.
 - ii. When a spotter is used, the spotter shall be on the ground, not in the scissor lift.
 - iii. At a minimum, a spotter is required when the scissor lift is being operated (i.e., raised, lowered, or moved horizontally):
 1. In finished areas.
 2. Through any doorway where the finish door frame has already been installed.
 3. In confined or congested areas (e.g., work near mechanical piping, electrical systems, overhead ductwork, within tight areas such as partitioned zones etc.).
 4. Within 3-feet of an elevated edge that is not protected by a solid wall.
 5. On any elevated platform of any type.
 6. Up or down any type of ramp.
 7. In any occupied area.
 8. Where there is any potential for public exposure.
 9. When loading and unloading of MEWPs from man/material hoists or elevators.
 10. During any night or low light work.
 11. In any location where the operator of the lift is in a Lone Worker situation.
22. Motor Vehicles and equipment shall not exceed a speed of five (5) miles per hour while being operated on the Project.
23. Motorized equipment/vehicles with wheels and/or tracks, regardless of it being a ride on or walk behind, shall not be operated on elevated decks, roofs, or similar without adequate perimeter protection that can effectively stop the equipment/vehicle from traveling beyond the elevated edge, penetration, shaft, skylight, or similar ("edge").
- a. The elevated edge protection shall be in place around the entire elevated deck, roof or similar that the motorized equipment/vehicle can access.

- i. The elevated edges needing protected can be minimized by having adequate area specific protection in place that can effectively stop the equipment/vehicle keeping it in a specific area.
- 24. A berm or equivalent of sufficient height and strength shall be constructed on all sides of runways, ramps, and along the perimeter of excavations to keep motor vehicles and/or equipment from driving off the edge. At no time shall a berm be at a height less than the height of the highest axle for the motor vehicle and/or equipment being driven on the runway, ramp or perimeter of the excavation.
- 25. **Stay Clear, Be Seen & Get Acknowledged Initiative:**
When mobile equipment is in use, employees shall **STAY CLEAR** of the mobile equipment. When employees need to approach the operating area of mobile equipment, they shall ensure that they are **SEEN** by the operator. Prior to approaching or crossing in the path of operation of the mobile equipment, employees shall ensure the operator **ACKNOWLEDGES** their presence.
- 26. Subcontractor shall maintain, on the Project, documentation of the training received for the operators of all equipment.
- 27. Workers shall neither walk nor stand on, or near, the sides of dump-trucks/trailers while they are being loaded.
- 28. Subcontractors shall ensure that all combustion engines being operated within a building or a similar structure (e.g., tented area) are equipped with “scrubbers” on their exhaust system. In addition, portable carbon monoxide alarming devices shall be strategically placed within the area the engine is being operated to warn workers of carbon monoxide levels in excess of 35 PPM.
- 29. Irrespective of the use of “scrubbers,” interior carbon monoxide concentrations shall not exceed 35 PPM in any area of a building or structure that a combustion engine is being operated in or adjacent to buildings or structures. In addition, exhaust from combustion engines shall not be in concentrations that result in workers, occupants of adjacent buildings or structures, or the general public becoming ill, irrespective of the carbon monoxide concentration in the specific area.
- 30. All lighting the manufacturer constructed into the design of the equipment shall be fully functional at all times.
- 31. Rigging attachment points that are installed on earthmoving equipment shall not be utilized (e.g., a hook welded onto a bucket of a track-hoe) unless the manufacturer of the equipment specifically designed the equipment in such a manner or the attachment point was installed per the design and specifications of the manufacturer or a registered professional engineer who has provided load charts or equivalent reference data with the design specific to the equipment.
- 32. Drilling Piers:
 - a. Subcontractor shall ensure that the manufacturer’s recommended safe operating procedures for the equipment being utilized are adhered to.
 - b. Equipment which has been modified without the explicit written approval of the manufacturer and/or the explicit written approval of a professional registered engineer licensed in the United States, shall not be utilized on the Project.

- c. Danger tape shall surround the entire pier drilling operation. At a minimum, the tape shall be placed 10' away from any open hole and shall include the swing radius of the drilling rig.
- d. Subcontractor shall ensure that a guardrail system is in place at least 6' away from any open hole creating a restricted access zone.
- e. If a worker must access a restricted access zone, a personal fall arrest system must be utilized.
- f. When personal fall arrest systems are utilized, no part of the system shall cross the hole or get in close proximity to the auger while it is in operation so as the fall arrest system doesn't become entangled into the rotating auger.
- g. Open holes must be attended to or covered at all times. If not immediately attended to, Subcontractor shall immediately place a hole cover over any open hole or maintain a guardrail system around all sides of the open hole.
- h. Each hole cover shall be marked "HOLE", be secured from accidental displacement and be designed to support twice the anticipated load expected to be applied to it.
- i. Holes greater than 3' in diameter must be protected by a guardrail system or an engineered hole cover must be used.
- j. If a backed-out auger is used to protect the hole, the gap between the auger and the edge of the hole shall not be more than 4".
- k. Subcontractor shall provide a barricade system around all test piles. Subcontractor shall minimize worker exposure to test piles. Subcontractor shall ensure that their employees spend only the time necessary near the test piles needed to collect the required data. Barricade systems shall be placed far enough away from the test piles to protect workers from the hazards of a pile or jack failure.

33. Where overhead powerlines span above ANY access/egress point to/from the project, where vehicles and/or mobile equipment may travel, large (3' x 3' minimum) prominent, signage shall be posted, that is visible from both directions, warning drivers/operators of the overhead power lines.

34. Where overhead powerlines span above ANY point on the project, where vehicles and/or mobile equipment may travel, large (3' x 3' minimum) prominent, signage shall be posted, that is visible from both directions, warning drivers/operators of the overhead power lines.

35. All work in public streets and/or sidewalks shall be conducted in compliance with the most current edition of the Manual on Uniform Traffic Control Devices (MUTCD) irrespective of the absence of any legal obligation to do so.

36. All flaggers assigned to control public traffic on public streets shall be certified flaggers and shall have on their person proof of certification while flagging.

37. The use of grey-market equipment is prohibited unless it meets all regulatory (including safety) requirements required in the United States for that specific make and model of equipment.

38. Motorized Carts:

- a. Oakland's Superintendent must approve the use of any motorized cart on the project.
- b. Personally-owned motorized carts and/or trailers are prohibited on the Project.
- c. Operators of motorized carts shall:
 - i. Have a valid State issued driver's license.

- ii. Follow all traffic laws.
 - iii. Not operate motorized carts on any open public sidewalk.
 - iv. Ensure that no unsecured or unstable loads are being transported in the cart.
 - v. Conduct a daily inspection of the cart prior to operation.
 - vi. Ensure all safety devices are operating appropriately on the cart prior to operation.
 - vii. Not operate the cart perpendicular on inclines.
 - viii. Operate the cart in accordance with the manufacturer's recommendations.
 - ix. Give foot traffic and machinery the right-of-way.
 - x. Not travel within 3' of pedestrian foot traffic.
 - xi. Ensure the cart is not loaded beyond its rated capacity.
 - xii. Stop the cart at blind corners and honk the horn before proceeding.
 - xiii. Not operate the cart while distracted (e.g., talking on a mobile telephone).
 - xiv. Switch the engine off, set the parking brake and remove the key when parking the cart.
- d. All operators of and passengers in motorized carts shall:
 - i. Be transported in the seat of the cart only.
 - ii. Be secured to the seat with a seatbelt.
 - iii. Keep their arms, legs, and head within the cart.
- e. All motorized carts shall be equipped with the following:
 - i. A seatbelt for the driver and each passenger
 - ii. A cab.
 - iii. A flag mounted on a flexible 8' pole.
 - iv. A functional beacon light mounted on the top of the cab if the cart is operated on open public streets.
 - v. A slow-moving vehicle marker visible from the rear of the cart.
 - vi. A fire extinguisher.
 - vii. A functional back-up alarm.
- f. Tools or material shall not extend beyond the sides of the bed of the motorized cart.
- g. Tools or material that extend 2' beyond the back end of the bed of a motorized cart shall be secured and shall be properly marked with RED flagging.
- h. If trailers are connected to motorized carts the trailer shall also be equipped with a slow-moving vehicle marker posted on the back of the trailer.
- i. A motorized cart shall not be modified in any way that will affect its safe operation.
- j. All motorized carts shall be adequately secured after hours to prevent unauthorized use and theft.

39. Forklifts:

- a. If Subcontractor or other does not have an established MSA with Okland, a person authorized to commit Subcontractor or other's company must sign the Okland [Release Waiver and Indemnity Agreement for Equipment Use](#)
- b. Prior to operating a forklift for production, the certified operator shall, in a safe and clear area, practice and familiarize themselves with the controls of each forklift BEFORE production operation:
 - i. Example: safety devices (e.g., horn, wipers, lights), forward, neutral, reverse, clutch cut-out foot controls, tilt levers, outrigger controls, etc.
- c. The 29 CFR 1910.178 Powered Industrial Trucks (forklift) standard shall apply to forklift operations on Okland projects (i.e., not just the training requirements).

- d. Prior to operating a forklift on the Project, forklift operators shall submit a copy of their current forklift operator certification to Okland's Superintendent.
- e. Forklift operators shall keep on their person a copy of their current forklift operator certification while operating a forklift on the Project.
- f. If forks/tines are installed on earthmoving or any other type of equipment, the equipment shall be considered a forklift. Thus, Subcontractor shall meet the requirements stipulated in this manual and the OSHA regulations governing forklifts/powered industrial trucks (29 CFR 1910.178).
- g. Forklift operators shall conduct a daily documented inspection of the forklift prior to operating the forklift.
 - i. If deficiencies are found, forklift operators shall document the deficiency on the inspection log and immediately reported it to their supervisor.
 - ii. If the deficiency affects or could potentially affect the safe operation of the forklift, the forklift shall be tagged out of service and shall not be operated until the deficiency is corrected.
- h. Forklifts shall be maintained in accordance with manufacturer's recommendations.
- i. Forklifts shall NOT be used to squash-down debris in a dumpster as this damages the forklift.
- j. Loads shall not be suspended under the tines of a forklift without a manufactured approved attachment which allows for centering of the suspended load and the operation is conducted in compliance with the manufacturer's load chart.
- k. All hoisting attachments installed onto the tines of a forklift shall be fitted with a secondary connection system that will prevent the attachment from sliding off the tines (e.g., if the tines were inadvertently tilted forward).
- l. Forklift rigging shall be marked with white electrical tape and the current monthly inspection color pursuant to Okland's [Rigging & Electrical Color-Code](#) system.
- m. Rigging used for towing and/or with any other material handling equipment (e.g., forklifts, skid-steers, track-hoes, loaders) shall not be allowed to be used on a crane.
- n. Forklift operators shall wear a hardhat and safety glasses while in all open cab-type forklifts.
- o. Forklift Operation.
 - i. When the operator's view is obstructed, a signalman/spotter shall be utilized.
 - ii. Project and public street speed limits shall be adhered to.
 - iii. Forklifts **shall not** travel on public streets **with or without a load** unless:
 - 1. A Job Hazard Analysis is first conducted to assess the risks associated with the operation and the appropriate controls are put into place.
 - 2. Okland's project superintendent is made aware of the operation and given a copy of the completed JHA.
 - 3. The operation is conducted with strict compliance to all municipal and State requirements for operating on public streets.
 - iv. If a forklift is utilized to support a personnel platform, the following shall apply:
 - 1. The operation shall be conducted in accordance with the manufacturer's recommendations (both the forklift's and the personnel platform's).
 - 2. If the personnel platform is a manufactured system, the manufacturer's use/operation/installation manual shall be adhered to and shall be within the platform while in use.
 - 3. If the personnel platform is a non-manufactured system, it shall be designed by a registered professional engineer and it shall be constructed and utilized per that design.

4. The Subcontractor utilizing the personnel platform shall develop a written work plan.
 5. Fall protection shall be utilized; both a guardrail system and a person fall arrest system.
 6. The total lifting capacity shall be reduced by 50%
 7. There shall be a certified operator at the controls of the forklift with the truck engine running at all times when platform is occupied by personnel.
 8. The transfer of loads from one forklift to another while elevated is prohibited.
- p. Forklift Inspections shall, at a minimum, be conducted at the following intervals:
 - i. Daily – Each operator of the unit is responsible for performing a daily inspection that shall include items specified by the manufacturer.
 - ii. Weekly - Weekly inspections are required to be documented and filed in the unit's permanent file. A trained operator or supervisor can perform this inspection.
 - iii. Annual – Conducted by an authorized service center and documented with copies in the permanent equipment file.
 - iv. Post Incident Inspections – Conducted by an authorized service representative
 - q. Prior to an operator exiting the cab, operators must:
 - i. Ensure the forklift has come to a complete stop
 - ii. Ensure the forklift is not on a grade or slope
 - iii. Ensure the forklift's transmission is placed in neutral
 - iv. Ensure the park brake is set
 - v. Ensure the forks or the attachment is lowered to the ground
 - vi. Enter and exit the forklift using 3 points of contact and facing the cab
 - vii. Remove the key from the equipment at the end of the shift
 - r. Fork Adjustment:
 - i. Prior to adjusting the forks, ensure the forklift is in neutral and the park brake is set
 - ii. The operator must not be outside of the forklift cab with the forks elevated. Another person must be used to adjust the forks.
 - iii. If the forks are not sliding, inspect to see if they need to be greased (do not over grease).
 - iv. Do NOT attempt to use the ground to adjust the forks this will damage the forklift.

40. Drones:

- a. Subcontractor shall not operate a drone on or around the project site without the explicit written authorization of Okland's Project Manager and Okland's Superintendent.
- b. Subcontractor shall fully understand all legal obligations with operating a drone and maintain compliance with those obligations.
- c. Subcontractor shall develop and submit to Okland a written JSA for the drone operation prior to the operation taking place.

Subpart P – Excavations & Any Soil Disturbance

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Operations that involve the disturbance of any soil requires Subcontractor to secure and comply with Okland's **Soil Disturbance Permit** process. Such operations include, but are not limited to: driving stakes or posts into the ground, digging in the soil with a shovel, auguring, boring,

trenching (even with a small light duty trenching machine), drilling, directional drilling, vacuum excavating, hydro excavating, potholing, clearing & grubbing, excavating, trenching, etc.

- a. Survey shall comply with Okland's [Survey Layout Standard](#).
- b. Backfilling of utility excavations shall comply with Okland's [Utility Backfill Standard](#).
- c. Supervisors shall reference Okland's [Soil Disturbance Permit Map Process](#) to ensure all aspects of Okland's Soil Disturbance Process are followed.

2. Hand Digging / Non-destructive Excavating (e.g., hydro-vaccing):

- a. All excavations within 2'-0" of each side of existing utilities, regardless of depth, shall be hand dug or a form of non-destructive excavating (e.g., hydro-vaccing) shall be utilized. This is designated as a 'hand dig zone'. Note: Hand digging shall not include any form of tool that can puncture the utility (e.g., pick-axe / pick-mattock)
 - i. For Example:
 1. If you have a 6" pipe, with the 24" hand dig zone, your total hand dig zone would be 54". (24" + 24" + 6" pipe = 54" hand dig zone)
 2. If you have a bank of utilities that are 20" wide. Your total hand dig zone would be 68". (24" + 24" + 20" utility bank = 68" hand dig zone).
- b. The hand dig zone is from the sky, down to the lowest utility in a vertical line, regardless of depth.
- c. The entity conducting the soil disturbance shall mark hand dig locations on the SDP Map and in the field.
- d. The Okland supervisor who issues the SDP is expected to validate that the hand dig zone is marked out on the ground before the SDP is issued.

3. Spotter Zones:

- a. While excavating activities are taking place within 5'-0" of each side of existing utilities, regardless of depth, a spotter shall be present. This area is designated as a 'spotter zone'.
- b. A spotter is an individual that will observe and guide the operator during the excavating activity, looking for any signs of a utility (i.e. warning tape, change in dirt fill, etc.) and **shall ensure the operator does not excavate within the hand dig zone.**
- c. The spotter shall be present at all times while in this zone and shall have no other duties other than spotting for the operator.
- d. The entity conducting the soil disturbance shall mark the spotter zone locations on the SDP Map and in the field.
- e. The spotter zone goes from the sky, down to the lowest utility in a vertical line, regardless of depth.
- f. The entity conducting the soil disturbance shall mark spotter zone locations on the SDP Map and in the field.
- g. The Okland supervisor who issues the SDP is expected to validate that the spotter zone is marked out on the ground before the SDP is issued.

4. Operations that involve excavation and/or trenching activities, and/or work in excavations or trenches, that are equal to or greater than 4' in depth requires Subcontractor to secure and comply with Okland's **Over 4' Excavation & Trench Access Permit** process.
5. Workers conducting activities in a trench 4' or greater in depth shall only do so when they can be visually seen and heard by other workers within their company. Okland's **Over 4' Excavation & Trench Access Permit** process and Okland's [Permit Sleeve / Magnet Tag Tracking Policy](#) shall be followed for all operations that involve excavation and/or trenching activities, and/or work in excavations or trenches, that are equal to or greater than 4' in depth:
 - a. [Soil Type A Over 4' Excavation & Trench Access Permit](#)
 - b. [Soil Type B Over 4' Excavation & Trench Access Permit](#)
 - c. [Soil Type C Over 4' Excavation & Trench Access Permit](#)
6. The supervisor overseeing the soil disturbance operation shall obtain a **Soil Disturbance Permit daily**, and if needed an **Over 4' Excavation & Trench Access Permit**, from the Okland Superintendent **prior to disturbance of any soil**.
7. **Planning:**
 - d. Prior to requesting the written Soil Disturbance Permit and/or an Excavation Permit, the requesting supervisor shall inspect the work area and determine the soil type via a competent person.
 - e. Identification of the location of all underground utilities that may be impacted by the soil disturbance activity shall be conducted prior to the excavating of any soil.
 - i. Subcontractor shall utilize blue stakes or the State's designated locating service to identify all public utilities. The blue stakes reference number needs to be available to add to the soil disturbance permit.
 - ii. Subcontractor shall also identify all nonpublic-ran utilities.
 - f. Subcontractor shall also reference the Okland utility plot plan.
 - g. The location of isolation valves/switches for utilities in close proximity to the soil disturbance operation must be located and identified by all workers involved in the soil disturbance work.
 - h. Subcontractor shall develop an emergency action plan for the operation specific to a utility strike. The plan shall be conveyed to all workers involved in operation.
8. Obtaining a Soil Disturbance Permit and/or an Excavation Permit shall not relieve subcontractor from any responsibility or accountability associated with the soil disturbance activity.
9. Subcontractor shall monitor the effectiveness of their controls and adjust if improvement is necessary.
10. Blue Stake markings and other underground utility markings shall be refreshed as soon as the markings are showing signs of deterioration. Markings shall be inspected at least every two weeks and after each rain.
11. Each Subcontractor shall provide their own competent person to supervise and train their employees engaged in excavation and/or trenching activities or that are working in excavations and/or trenches.

12. At no time shall a worker work alone in a trench that is deeper than 6'. A second person, who is **NOT** in the trench, must be present.
13. The competent person or designee must remain at the excavation/trench until workers exit and are accounted for.
14. Undermining of any structure, including sidewalks and pavement, is not allowed without prior approval from Oakland's Superintendent and Oakland safety in conjunction with an engineered shoring system. The operation must be conducted after the area is closed to any public exposure and not opened up again until it is completed, or the shoring system is appropriately installed.
15. Other than spoil piles that are placed 2' back from the top edge of an excavation, nothing heavier than 1,000-lbs. is allowed to be stored next to the edge of an excavation unless it is placed further than a 1 ½: 1 angle plus 2' as measured from the toe of the excavation. Exceptions to this must be approved by a Registered Professional Engineer in writing.
16. The competent person or designee shall ensure that proper barricading is in place while the excavation is being dug as well as when the excavation is completed and open.
17. All sloped trenches and/or excavations shall at a minimum be demarcated with barricade tape.
18. All vertical trenches and/or excavations less than 6' in depth shall be demarcated with an adequate hazard identification system.
19. All vertical trenches and/or excavations greater than 6' in depth shall be demarcated with a standard guardrail system or an equivalent hard barricade. All workers who must conduct work between the guardrail system and the edge of the excavation shall be protected by a Personal Fall Arrest System.
20. All trenches and/or excavations that the public and/or the building occupants could reasonably be exposed to regardless of depth shall be demarcated with a solid hard barricade or a chain-link fence that is so constructed as to prevent the public and/or building occupants from falling into the trench/excavation. Fences shall be at least 2' away from the edge of the excavation.
21. Where there is pedestrian foot traffic, all temporary pedestrian bridges, trench plates, walkways, hole covers, and similar that are installed over excavations, roadways, site work, etc. must be free of trip hazards, secured from displacement, and have beveled edges; final construction of such shall meet current ADA requirements.
22. If the sloped or vertical trench or excavation is in close proximity to an access/egress pathway for vehicular/equipment, such as a drive lane, additional controls shall be implemented such as berms, jersey barricades/k-rail, or similar protection.
23. All underground utility lines, temporary and permanent, shall be demarcated with underground warning tape.

24. When underground utilities are suspected, they shall be located first by hand digging or by non-destructive excavating (e.g., hydro excavating).
25. Underground systems or lines shall be protected, supported, or removed to protect workers entering excavations.
26. Energized lines or systems shall be protected from physical damage by the excavation, work process, and backfill operations.
27. Drilling Piers:
- a. Prior to drilling any piers, piles, caissons, or similar, a Drilling Piles/Caissons Excavation Permit shall be completed, and Okland's Permit Sleeve / Magnet Tag Tracking Policy shall be followed.
 - b. Subcontractor shall ensure that the manufacturer's recommended safe operating procedures for the equipment being utilized are adhered to.
 - c. Equipment which has been modified without the explicit written approval of the manufacturer and/or the explicit written approval of a professional registered engineer licensed in the United States, shall not be utilized on the Project.
 - d. Danger tape shall surround the entire pier drilling operation. At a minimum, the tape shall be placed 10' away from any open hole and shall include the swing radius of the drilling rig.
 - e. Subcontractor shall ensure that a guardrail system is in place at least 6' away from any open hole creating a restricted access zone.
 - f. If a worker must access a restricted access zone, a personal fall arrest system must be utilized.
 - g. When personal fall arrest systems are utilized, no part of the system shall cross the hole or get in close proximity to the auger while it is in operation so as the fall arrest system doesn't become entangled into the rotating auger.
 - h. Open holes must be attended to or covered at all times. If not immediately attended to, Subcontractor shall immediately place a hole cover over any open hole or maintain a guardrail system around all sides of the open hole.
 - i. Each hole cover shall be marked "HOLE", be secured from accidental displacement, and be designed to support twice the anticipated load expected to be applied to it.
 - j. Holes greater than 3' in diameter must be protected by a guardrail system or an engineered hole cover must be used.
 - k. If a backed-out auger is used to protect the hole, the gap between the auger and the edge of the hole shall not be more than 4".
 - l. Subcontractor shall provide a barricade system around all test piles. Subcontractor shall minimize worker exposure to test piles. Subcontractor shall ensure that their employees spend only the time necessary near the test piles needed to collect the required data. Barricade systems shall be placed far enough away from the test piles to protect workers from the hazards of a pile or jack failure.

Subpart Q – Concrete and Masonry Construction

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Prior to the construction of any masonry wall that will exceed 18-feet in height, Subcontractor shall submit a bracing plan, designed by a registered professional engineer, to Okland's Project Management Team.
2. Supervisors shall plan for the safe set-up of concrete pump-trucks and other concrete conveyance systems.
3. The manufacturer's recommendations for the safe set up of their concrete conveyance systems shall be followed.
4. Workers not involved in concrete placement operations shall stay out from underneath concrete placement booms and shall not stand near surface run slick-lines or placement hoses.
5. Workers shall stay back 20' when the pump is first initiating and when draining the slurry.
6. Deadman Usage:
 - a. When Subcontractor is utilizing Deadman for wall bracing, of any type, Subcontractor shall provide engineered documentation to support their bracing plan.
7. Metal plated caps (e.g., rebar caps) or the equivalent shall be utilized to protect workers from all impalement hazards.
8. The practice of wiring a piece of material (e.g., wood 2x4, rebar, pipe) to the side of the impalement hazard shall not be considered adequate impalement protection.
9. Impalement hazards to which the public and/or the building occupants could reasonably be exposed, shall be capped with metal plated caps and/or the equivalent. In addition, the protective cap or the equivalent shall be secured to the object so it cannot be accidentally knocked off/removed.
10. Only authorized personnel shall erect, alter and/or remove shoring systems.
11. Damaged shoring shall be immediately reported to the Okland Superintendent.
12. Shoring systems shall be inspected daily and upon any event that may have altered the integrity of the shoring system.
13. All forming and shoring system walking/working surfaces shall be, at a minimum, 12" in width. All scaffolds shall be, at a minimum, 18" in width.
14. When the height of a wall form scaffold, or similar deck, is placed in a configuration that the top of the wall form is less than 39" from the walking/working surface, a guardrail system should be installed on the backside of the wall form to provide fall protection for the workers on the wall form scaffold. When the installation of guardrail is not feasible, all workers on the wall form scaffold shall utilize a personal fall arrest system and maintain 100% tie-off.
15. **Gang Form Placement:** Gang forms shall not be released from a crane until adequate bracing is in place to safely support them and all employees on the form.

16. Gang Form Stripping: (*effective 1.5.26. Enforced 2.1.26.*)

- a. Prior to the placement of concrete, a Concrete Superintendent or General Foreman shall coordinate with the Wall Foreman to jointly select the required amount, and location, of ties for each gang form panel that will serve as the designated “Hold Tie(s).”
 - i. The purpose of the “Hold Tie(s)” is to ensure that all of the formwork ties are not prematurely removed from a gang form panel prior to the panel being connected to the rigging and stabilized by the crane.
 - ii. Each gang form panel section shall have at least one designated “Hold Tie”.
 1. Additional “Hold Ties” may be required depending upon the gang form’s size and/or configuration.
- b. The “Hold Tie(s)” shall have a physical tag attached to it to serve as a clear visual indicator identifying it as a “Hold Tie”.
 - i. Only Okland issued “Hold Tie” tags shall be used for this purpose.
 - ii. “Hold Tie” tags shall be placed on BOTH sides of the tie(s).
 - iii. Each “Hold Tie” tag shall have the name of the designated crew member on it that is responsible for confirming that:
 1. The panel is properly connected to the rigging; and
 2. The panel is ready to be stripped and safely removed from the wall.

17. Only the designated crew member can remove the “Hold Tie” tag(s) and disengage the “Hold Tie(s)” allowing for the panel to be safely broken free from the wall.

18. When sliding out table forms, a dedicated spotter is required to be present on the level the table form is being removed from.

- a. The spotter shall have no duties other than visually observing the sliding/hoisting operation and the personnel involved in the operation so as he/she can immediately notify the signal person to stop if a hazard is observed.
- b. The spotter shall be actively spotting from the time the table is connected to the crane until the table and all tag-lines are completely out of, and clear from, the building.

19. Concrete Pumping Safety.

- a. All manufacturer’s safety recommendations for the placement pumping system shall be complied with.
- b. All components of pump trucks shall maintain a 20’ clearance from all energized power lines.
- c. If continuation pipes/hoses are connected to the end hose, they must not impose any load on the boom.
- d. End hoses shall not exceed 13’ unless the manufacturer of the pumping system allows for a longer length.
- e. End hoses, reducers, tremies, and similar conveyance adapters connected to the boom shall be fastened with a secondary safety cable and shall not exceed the weight of the manufacturer supplied end hose.
- f. Pump trucks shall not drive with their placing booms unfolded.
- g. Proper placement of outriggers shall be verified by the pump truck operator prior to the concrete placement operation taking place.
- h. Outriggers shall be placed a safe distance from any excavation/trench so as not to collapse the excavation/trench.

- i. If the pump operator cannot see the discharge end of the placing hose/pipe, a positive means of communication shall be established between the pump operator and a designated spotter positioned at the discharge end of the placing hose/pipe.
- j. Placement booms shall not be unfolded when lightning is present.
- k. Blockages in the pump or delivery pipelines/hoses shall not be removed by applying high pressure to it. If “rocking” the concrete back and forth with the forward/reverse function doesn’t break the blockage loose, the blockage shall be removed manually.
- l. Pressurized sections of pipelines/hoses shall not be opened. Prior to opening any section of pipeline/hose, the operator shall ensure the pressure is removed by first putting the pump in the “reverse” mode for several strokes.
- m. Unless required for the placement of concrete, workers shall not stand directly under placement booms.
- n. Workers shall neither attempt to hold down a hose or pipe nor shall they be in close vicinity to a hose or pipe that is being cleaned with pressurized air.
- o. Workers shall neither attempt to hold down a hose or pipe nor shall they be in close vicinity to a hose or pipe that is being “rocked” to remove a blockage.
- p. Workers shall not sit on, stand on or straddle pipelines/hoses while pressurized.
- q. Workers shall not place any part of their body between the end of the delivery system and a fixed object.
- r. Workers shall not look into a delivery hose or pipe that is connected to a pumping system.
- s. Workers shall neither stand in front of a delivery hose or pipe that is connected to a pumping system nor shall they point/aim the end of a delivery hose or pipe toward another worker.

Subpart R – Steel Erection

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

- 1. Okland’s [Site Specific Steel Erection Plan and Checklist](#) shall be used as a planning tool prior to steel erection taking place.
- 2. The fall prevention/protection requirements listed in [Subpart M](#) of this document shall apply.
- 3. All rigging shall be conducted per the requirements stipulated in [Subpart H](#) and in [Subpart CC](#) of this document.
- 4. Workers working from steel shall be tied off in a manner that minimizes their free-fall. Thus, the use of “loose” rigging/beam-straps wrapped around the steel is prohibited.
- 5. “Climbing Columns” is prohibited irrespective of the use of a personal fall arrest system.
- 6. When workers other than those engaged in decking operations are on an incomplete deck, barriers around openings in decking and where leading-edge work is taking place, has been stopped, or has been suspended shall be protected with hard barriers and those barriers shall be equivalent to the criteria of a guardrail system.

7. Multiple lift rigging (Christmas Treeing/Treeing) shall be in compliance with [Subpart CC](#) of this document.

Subpart S – Underground Construction, Caissons, Cofferdams, and Compressed Air

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Where a check-in/check-out procedure is required per Subpart S of OSHA regulations, Subcontractor shall be responsible to maintain a check-in/check-out log for its employees.
2. Subcontractor is responsible to provide the necessary equipment and is responsible for all costs associated with air monitoring where required.
3. The Subcontractor creating a suspected atmospheric hazard shall have the burden of proof to show that levels of atmospheric hazards are below the OSHA PEL for the suspected hazard. Where mechanical ventilation is required due to the Subcontractor's activities, the subcontractor shall install a mechanical ventilation system that is sufficient in keeping the atmospheric hazards below the OSHA PEL for all affected workers.

Subpart T – Demolition

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Prior to beginning any selective or total demolition activity, Subcontractor shall secure a [Selective Demolition Permit](#) or a [Total Demolition Permit](#) from Okland's Project Management Team.
2. Subcontractor shall have a registered professional structural engineer evaluate Subcontractor's planned means and methods in relation to the engineering survey required by OSHA regulations for demolition to ensure that the proposed means and methods are safe. This evaluation shall be conducted prior to the start of demolition. Written documentation of the review shall be kept on file in Subcontractors on site office.
3. Where a registered professional engineer has issued, instructions, directives and or plans pertaining to the demolition operation, Subcontractor shall make arrangements to have the registered professional engineer on site to adequately review and inspect the Subcontractor's means and methods used to comply with the registered professional engineer's instructions, directives and/or plans.
4. Prior to the construction of any trash chute, a [Trash Chute Permit](#) must be obtained from the Okland Project Management Team.
5. Subcontractor shall be responsible for exploratory activities needed to evaluate the structural integrity and stability of a structure and or for demolition engineering purposes.

6. Workers conducting demolition work shall only do so when they can be visually seen and heard by other workers within their company.
7. Fall hazards created during the course of demolition shall be immediately protected by hole covers and/or guardrail systems by the Subcontractor creating the hazard.
8. Subcontractor performing demolition operations shall be responsible for evaluating and controlling, worker and public exposure to dusts, silica, and other hazardous emissions generated from the demolition activity.
9. All costs incurred for evaluating exposures to the environment, workers, the general public, etc. associated with the demolition activity and costs incurred for complying with the associated regulations for the demolition activity, shall be the responsibility of the Subcontractor conducting the demolition activity.
10. Equipment used for demolition shall have steel cages installed sufficient to protect the operator from flying debris.
11. Equipment shall not be placed, operated, and/or driven on elevated structures that cannot sufficiently support the weight of the working equipment. Thus, maximum capacities of elevated structures shall be known prior to equipment being placed, operated, and/or driven on the elevated structures.
12. Debris and material shall not be permitted to accumulated on or against or fall onto structures unless they are of sufficient strength to hold such loads.
13. Subcontractor performing demolition shall be responsible for all costs associated with the protection of the public from demolition hazards it creates.
14. Prior to starting demolition or renovation of any existing building or portion of a building, Subcontractor shall:
 - a. Conduct a Hazardous Materials Survey via a third-party Industrial Hygienist. The survey must be a full hazardous materials survey and NOT just a survey for asbestos. Hazardous materials that must be inspected for include but are not limited to:
 - i. Asbestos
 - ii. Lead paint via XRF analyzer or paint chip samples
 - iii. Mercury-containing thermostats
 - iv. Mercury-containing fluorescent light fixture lamps
 - v. PCB-containing fluorescent light fixture ballasts and/or transformers
 - vi. Refrigeration units containing CFCs
 - vii. Misc. containers of liquid or hazardous waste
 - viii. Underground storage tanks
 - ix. Waste Batteries (e.g., vehicle batteries)
 - x. Grease/oil separators
 - b. Remove any/all hazardous materials
 - c. Submit a pre-demolition/renovation building inspection form/notice to the city/county health department as required by local law.

- d. Submit a pre-demolition/renovation notice to the State's Department of Environmental Quality as required by state law.

Subpart U – Blasting and the Use of Explosives

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Subcontractors who propose to use explosives on the Project shall develop a site-specific written safety plan detailing their proposed use. The plan shall be submitted to Oakland's Superintendent in advance of the explosive Subcontractor's mobilization on the Project to allow for adequate time for review of the plan by Oakland and the Owner. Subcontractor shall be responsible for all direct and indirect costs associated with the use of explosives on the Project.

Subpart V – Power Transmission and Distribution

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. The safety requirements in NFPA 70E shall be adhered to.

Subpart W – Rollover Protective Structures; Overhead Protection

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Roll Over Protective Structures (ROPS) shall not be altered or removed from any equipment in which the manufacturer has installed and/or recommended them.

Subpart X – Stairways and Ladders

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Oakland expects ladders to be a choice of last resort. First, all work that can feasibly be conducted on the ground should be. If work has to be performed at heights, alternate means of access to perform work should be considered before ladders; These include, scaffolding, scissor lifts, aerial lifts, vertical mast lifts, etc. If a ladder must be used a podium or platform ladder is preferred.
2. Pulling wire through conduit, or similar, shall not be conducted from an extension ladder or a standard A-frame ladder (the safe use of a platform ladder, scaffold or a scissor lift would be acceptable).
3. The use of wood A-frame type ladders is prohibited.

4. Portable ladders shall be fiberglass and at a minimum a Type IA (300lbs), heavy-duty industrial ladder and be suitable for its intended use.
5. Ladders shall be inspected prior to use. Ladders found to be damaged shall not be used and shall be tagged (i.e., with a danger tag) and removed from service.
6. In addition to the pre-use inspection, all ladders shall have a thorough annual inspection. The date of the last annual inspection shall be posted on the ladder.
7. Field and/or shop repairs to ladders are prohibited (e.g., adding splints, braces).
8. Ladders shall not be spliced together.
9. Extension ladders shall be secured at the top of the ladder to prevent displacement during use and when feasible, also at the bottom. If an extension ladder cannot be secured at the top, the base of the ladder shall be stabilized by an additional person to prevent displacement of the ladder while it is being used.
10. Extension ladders shall not be separated, and the sections utilized separately for any reason whatsoever (e.g., use of a section of the ladder to access/egress trenches/excavations is prohibited).
11. The top three rungs of an extension ladder shall not be used to step on.
12. Job made ladders shall comply with the most current ANSI 14.4 Job-Made Wooden Ladders standard.
13. Double cleated job made ladders shall not exceed 24' in length and single cleat ladders shall not exceed 30' in length.
14. Extension and A-frame ladders shall be located at least 1 ½ times the ladder height from an exposed edge.
 - a. If this is not feasible, the worker shall utilize a fall restraint or a personal fall arrest system while working from and/or climbing the ladder.
 - b. Exception: When podium or platform ladders are being used, these ladders may be placed as close as 3' to the exposed edge if the ladder is orientated so that the access to podium or platform is away from and perpendicular to the exposed edge.
15. Fall protection during ladder use is also required when working from and/or climbing any type of ladder, including scaffold ladders, where the fall exposure to the worker is greater than 24'.
16. Workers shall not use step stools or stand on any materials, tools, or equipment that is near an exposed edge that would create a fall exposure to the employee over that edge if the fall exposure is greater than 6'.
17. Three points of contact shall be maintained while climbing a ladder.
18. Nothing shall be carried in the hands while climbing a ladder.

19. Workers shall not lean out beyond the side rails of the ladder (i.e., their center of mass shall not extend outside the side-rails of the ladder).
20. A-frame ladders must not be used as straight ladders unless designed and manufactured for that purpose.
21. The top two steps of an A-frame ladder shall not be utilized to sit or step on.
22. Straddling the top of an A-frame ladder is prohibited.
23. Workers shall face the ladder at all times.
24. Ladders shall not be placed on scaffolds, in scissor-lifts, or in aerial lifts.
25. Stepstools shall not be greater than 32”.
26. Metal style stepstools shall not be used where an electrical hazard could potentially exist (e.g., corded electrical tools cannot be used while working from metal stepstools).

Subpart Y – Diving

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. There are no additional requirements other than those stipulated elsewhere in this manual.

Subpart Z – Toxic and Hazardous Substances

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. If Subcontractor’s operations will involve exposure to silica, Subcontractor shall have previously developed a Silica Exposure Control Plan. This plan shall be submitted to Okland upon request.
2. Prior to the demolition or renovation of any facility a hazardous materials survey shall be conducted. The Owner is responsible to conduct the hazardous materials survey that will identify areas where hazardous materials are on the Project. This survey shall be kept on file and available for review at the request of any worker. Workers may contact Okland’s Superintendent to obtain access to the survey.
3. Prior to starting demolition or renovation of any existing building or portion of a building, Subcontractor shall:
 - a. Conduct a Hazardous Materials Survey via a third-party Industrial Hygienist. The survey must be a full hazardous materials survey and NOT just a survey for asbestos. Hazardous materials that must be inspected for include but are not limited to:

- ii. Asbestos
 - iii. Lead paint via XRF analyzer or paint chip samples
 - iv. Mercury-containing thermostats
 - v. Mercury-containing fluorescent light fixture lamps
 - vi. PCB-containing fluorescent light fixture ballasts and/or transformers
 - vii. Refrigeration units containing CFCs
 - viii. Misc. containers of liquid or hazardous waste
 - ix. Underground storage tanks
 - x. Waste Batteries (e.g., vehicle batteries)
 - xi. Grease/oil separators
 - b. Remove any/all hazardous materials
 - c. Submit a pre-demolition/renovation building inspection form/notice to the city/county health department as required by local law.
 - d. Submit a pre-demolition/renovation notice to the State's Department of Environmental Quality as required by state law.
4. All areas in which the Subcontractor is required to work should be free of asbestos and other hazardous materials. However, in the unlikely event the Subcontractor encounters material reasonably believed to be asbestos, or any other hazardous waste or substance, the Subcontractor shall immediately stop work that may disturb the suspected material. Subcontractor shall immediately report the condition to Okland's Superintendent verbally followed by a written report to the Okland Project Manager within 24-hours of the encounter. The work in the affected area shall only resume in the absence of asbestos or reported hazardous material or when it has been rendered harmless according to federal, state, and local standards.
5. Unless otherwise stipulated in other portions of the contract documents, asbestos shall not be disturbed. However, if disturbance of asbestos is necessary to execute Subcontractor's scope of work, the following conditions shall be met:
- a. All supervisors and workers who are working with asbestos regulated or not regulated by the EPA shall receive training in accordance with OSHA's asbestos regulations. A copy of the documentation of such training shall be submitted to Okland's Superintendent prior to beginning work.
 - b. At no time shall Subcontractor render non-friable asbestos friable or regulated (e.g., via the use of power tools, sanding).
 - c. At no time shall Subcontractor disturb asbestos that is regulated by the Environmental Protection Agency (EPA) and/or classified as friable as defined by the EPA unless explicitly contracted to do so.
 - d. Friable and/or EPA regulated asbestos shall only be disturbed by Subcontractors and its employees licensed and certified by the State that the Project is in. Copies of such licenses and certificates shall be submitted to Okland's Superintendent prior to beginning work.
6. Subcontractor shall provide for all required training to conduct asbestos work.
7. Subcontractor shall be responsible for all costs related to the disposal of all asbestos that is rendered friable and/or EPA regulated due to the intentional act, unintentional act or negligence of the Subcontractor or anyone for whom the Subcontractor may be liable.

8. Subcontractor shall submit to Okland's Superintendent a copy of the manifest which documents the proper disposal of all regulated asbestos immediately after the asbestos is disposed of.
9. Subcontractor shall be responsible for the costs associated with all controls necessary to safely conduct asbestos work (e.g., administrative & engineering controls) unless otherwise stipulated in the contract documents.

Subpart AA – Confined Spaces in Construction

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. Subcontractors who must enter a confined space shall develop and implement an effective written confined space entry procedure specific to their scope of work.
2. Subcontractors shall utilize Okland's confined space entry permit unless Subcontractor's permit is equal or better in detail than Okland's.
3. Workers entering a permit or a non-permit required confined space shall only do so when they can be visually seen and heard by other workers within their company.
4. Subcontractors shall provide for all necessary equipment needed to safely enter a confined space (e.g., emergency rescue, ventilation, atmospheric testing).
5. Subcontractors shall conduct their own atmospheric testing. Okland will NOT conduct this for subcontractors.
6. Subcontractor shall, at a minimum, assign **one** attendant (i.e., hole watch) **for each** confined space.
7. Attendants shall have NO other duties other than being an attendant.
 - a. The attendant(s) shall be equipped with an operating telephone to summon rescue and emergency assistance if they are needed.
 - b. The attendant(s) shall be capable of understanding and relaying emergency information in English and the language of those workers entering the confined space.
8. The supervisors of those entering the confined space, the entering workers and the attendant(s) shall be trained on the emergency action plan by their employer prior to the entry.

Subpart CC – Cranes

In addition to the regulations set forth in OSHA 29 CFR 1926 (construction regulations), and other Local, State, and Federal regulations, and those requirements established in the [EXPECTATIONS](#) section of this manual, the following requirements shall apply:

1. General Requirement:
 - a. Prior to the mobilization of any Subcontractor crane that will have any part of the crane, or any crane load, extend beyond the project fence line, Subcontractor shall work with

[Crane Request Form](#), validate that the necessary permits and agreements can be secured, and obtain the required signatures.

- b. Prior to the selection, assembly, and disassembly of any type of tower crane (hammer head, luffing, self-erector, cast in place foundation, ballast foundation, etc.), regardless of the crane being an Okland owned/leased crane or a subcontractor owned/leased crane, [Okland's Tower Crane Assembly Checklist](#) or [Okland's Tower Crane Disassembly Checklist](#) process shall be executed and a completely signed [Okland Tower Crane & M/M Hoist Assembly/Climb/Disassembly Authorization Form](#) shall be secured.
- c. Subcontractor shall submit a lift plan to the Okland Superintendent at least 14-days prior to the crane lift taking place and prior to the crane being allowed on site. This is to allow adequate review of the operation by Okland's Crane Superintendent or his designee.

The lift plan shall include:

- i. Okland's **Lift Planning Worksheet** filled out completely for each of the crane's locations and for its heaviest lift at those locations ([Hydraulic/Lattice Crane – Tower Crane](#))
- ii. A copy of the load chart(s) for the exact configuration(s) the crane will be used in
- iii. A copy of the crane's annual inspection report (i.e., the actual report document NOT a photo of the inspection sticker)
 - 1. NOTE: Annual inspections of all cranes shall be conducted by a qualified **third-party** inspector. Annual inspections conducted in-house will **NOT** be accepted.
- iv. A copy of the crane operator's current NCCCO card
- v. A copy of the crane operator's current medical certification card
- vi. A copy of the certifications/qualifications for all riggers and signal personnel (**this is typically a card or a certificate**).

2. Qualifications:

- a. Qualified Rigger (QR): All riggers shall meet the qualifications stipulated by OSHA for a qualified rigger and the qualifications listed in this document.
 - i. Unless otherwise agreed to in contract documents, Qualified Riggers who are actively rigging and/or signaling hoisted loads, of any type, shall be wearing a green hardhat or a high visibility hard hat sock/cover.
 - 1. This includes rigging/signaling activities that do not involve a crane (e.g., rigging with chain falls, come-alongs, electric hoists, coffering hoists, forklifts, earthmoving equipment).
 - 2. Green hard hats are not allowed on any Okland Project unless the worker wearing it is a 'Qualified Rigger,' as defined by OSHA, and is actively engaged in rigging activities for the project.
 - ii. QR shall understand spoken and written English.
 - iii. QR shall have previously demonstrated to their employer, or their employer's representative, their skill and competency in using rigging principles.
 - iv. The QR's employer shall field verify the QR's capabilities to adequately and safely conduct rigging operations.
 - v. QR shall be certified by their employer, or their employer's representative, to conduct their assigned rigging tasks.
 - vi. QR shall be capable of calculating the capacities of the rigging being utilized for the configuration and conditions the rigging is placed in.
 - vii. QR shall understand spoken and written English.

- b. Qualified Signal Person (QSP): All signal personnel shall meet the qualifications stipulated by OSHA for a signal person and the qualifications listed in this document.
 - i. QSP shall have previously demonstrated to their employer, or their employer's representative, their skill and competency in signaling cranes.
 - ii. The QSP's employer shall field verify the QSP's capabilities to adequately and safely conduct signaling operations.
 - iii. QSP shall be certified by their employer, or their employer's representative, to conduct their assigned signaling tasks.
 - iv. QSP shall understand spoken and written English.
- c. Operator: All crane operators shall meet the qualifications stipulated by OSHA and the qualifications listed in this document.
 - i. Operators of all cranes shall be certified by The National Commission for the Certification of Crane Operators (NCCCO). The certification shall be for the specific type of crane they are operating (e.g., Telescopic Boom Cranes—Swing Cab, Lattice Boom Crawler Cranes, tower crane). Certification from other organizations may be permitted so long as they are considered to be equivalent to NCCCO certification, meet the requirements of the ASME B 30.5 mobile crane standard, and are accredited by the US National Commission for Certifying Agencies (NCCA).
 - (1) As of the date of this manual, the only two organizations that meet the Oakland crane operator criteria are:
 - (a) NCCO
 - (b) Operating Engineers Certification Program (OECF)
 - (2) No variance from the Oakland stipulated Crane Operator qualification requirements, or acceptance of certifications from any organization except for NCCO or the OECF, shall be allowed without the explicit written approval from:
 - (a) Oakland's Crane Superintendent; and
 - (b) Oakland's Crane Safety Manager; and
 - (c) Oakland's Regional Safety Director; and
 - (d) Oakland's Field Operations Director
 - ii. Operators shall meet ASME B30.5 physical requirements while operating crane(s) on the Project.
 - iii. Operators shall have adequately demonstrated to their employer their ability to safely operate the specific type of crane being assigned prior to the operator conducting hoisting operations on the project.
 - iv. Operators shall have a complete understanding of the owner's/operator's manual for the specific crane they are operating and ensure a copy of the manual and hard copy load charts are in the crane's cab.
 - v. Operators shall understand standard hand and voice signals.
 - vi. Operators shall have no physical, visual, or mental restrictions or impairments that will affect the safe operation of the assigned crane.
 - vii. Operators shall be able to understand spoken and written English.

3. Crane Set-up:

- a. Prior to the start of the erection and dismantle of a crane, Subcontractor and/or Supplier of the crane, including those involved in supporting the erection and dismantle process of another crane, must identify to Oakland a competent and qualified person as the

Assembly/Disassembly Director. The Assembly/Disassembly Director shall meet the criteria of the designation pursuant to OSHA regulations and execute the responsibilities of the position.

- b. Tubular crane booms shall only be handled with nylon slings.
- c. Boom lacings shall never be used to lift or handle the boom.
- d. The entire swing radius of the rotating superstructure of a mobile crane shall be barricaded to prevent workers and/or equipment from entering the area. Barricades shall be, at a minimum, red DANGER tape.
- e. Cribbing shall be utilized under all outrigger pads.
- f. Cribbing shall be of the appropriate size and dimension to adequately support the crane.
- g. Wire rope spool ends shall not be used for outrigger pads.
- h. All cranes shall be equipped with a functional anti-two-block device.
- i. Personnel erecting, climbing, and/or dismantling a tower crane, when exposed to a fall greater than 6', shall be equipped with and shall utilize an appropriate personal fall arrest system. Lanyards shall be a "dual/Y" type and the end hooks shall be compatible with the anchorage points on the tower crane.
- j. The Federal Aviation Administration (FAA) requires a permit be issued for a crane any time they will exceed a specific height (e.g., typically 200' but this may vary in different states and/or when in close proximity to any type of airport). Subcontractor shall ensure all FAA required permits are secured prior to crane erection.
- k. The base of each tower crane shall be secured with, at a minimum, an 8' plywood wall structure with an access door that is capable of being locked and unlocked from both sides.
 - i. As the structure is built adjacent to or around the crane's tower, all accessible points to the cranes tower shall be similarly barricaded with, at a minimum, an 8' plywood wall structure.

4. Crane Inspections:

- a. Annual and post incident inspections of all cranes shall be conducted by a qualified **third-party** inspector. Annual and post incident inspections conducted in-house will **NOT** be accepted.
- b. Operators shall conduct **documented** inspections of the crane they will be operating prior to its operation as stipulated by OSHA regulations. Documentation of these inspections shall be kept in the crane's cab.
- c. Technicians, and/or their company, overseeing the erection, climbing, and/or disassembly of a tower crane shall not be used for the third-party inspection of the tower crane post erection, climbing, and/or disassembly. Nor shall they be used for the annual inspection of the tower crane.
- d. All Tower Cranes, of any type, including self-erectors, shall have a third-party inspection of the crane at the following intervals:
 - i. Pre-erection – on the ground. Unless otherwise approved by Okland, the inspection must occur **after transportation to the work site**.
 - ii. Post-erection - in the air
 - iii. Annually – no later than the post-erection annual anniversary date
- e. Documentation of crane inspections shall be provided to Okland upon request.

5. Operation:

- a. All crane operations shall be conducted with a certified operator, a qualified rigger, and a qualified signal person. It is acceptable to have the qualified rigger and the qualified signal person be the same person if they are qualified to do both.
- b. All safety devices and operational aids installed by the manufacturer of the crane shall be operating effectively.
- c. The crane's operator's manual and hard copy load charts shall be available to the operator in the cab of the crane.
- d. Loads shall not be flown over or immediately adjacent to the public right-of-way without adequate controls in place as stipulated in the General Liability section of this manual. This may require temporarily closing the affected public right-of-way.
- e. Loads shall not be flown over any building occupied by the public.
- f. Operators, QRs and QSPs shall ensure all needed public safety measures are in place prior to the hoisting operation (e.g., closing of streets, closing of pedestrian walkways, barricading of areas).
- g. Operators, QRs and QSPs shall ensure any required LOTO procedures have been properly executed prior to the hoisting operation (e.g., de-energization and LOTO of power lines).
- h. Operators, QRs and QSPs shall understand the maximum loading capacity of the floor/slab/scaffold/platform/deck or similar surface prior to setting any load on it. Loads set on such surfaces shall not exceed 90% of the surface's rated capacity (taking into account the weight of existing loads already on the surface).
- i. Operators shall ensure that the crane is not operated in winds that exceed the manufacturer's recommended wind speed.
- j. Operators of lattice boom cranes, tower cranes, and any crane with a luffing jib in place, shall have immediate access to and utilize an operating anemometer to determine wind speeds.
- k. Operators shall immediately suspend all hoisting operations that are in progress that if continued, would be classified as a critical lift (i.e., a critical lift was not previously anticipated). Operators shall then ensure that the critical lift process is implemented.
- l. Operators shall respond to signals from only one person at a time.
- m. Operators shall not follow any signal that is not understood.
- n. Operators shall always follow the stop signal.
- o. Operators shall not leave the control station while loads are suspended or connected to the crane.
- p. When conflict is possible between other crane booms, lines, or loads, it is the responsibility of the operators of each crane and their supervisor to establish an effective method of communication and placement of an adequate number of spotters and signal personnel to prevent contact of the crane booms, lines, or loads.
- q. The QSP shall ensure that only one QSP is giving signals to the crane operator at a time.
- r. The QSP shall ensure that loads are routed in a manner that presents the least exposure to personnel injury and property damage.
- s. All Qualified Signal Persons and Qualified Riggers shall have in their immediate possession a safety whistle while conducting crane signaling operations.
- t. As needed, the Qualified Signal Person or Qualified Rigger, shall sound a safety whistle to notify personnel in the vicinity of a rigged load, that a load is being hoisted or set

down near them. The intent of this requirement is to gain the immediate attention of personnel in the vicinity of the load, on an as needed basis.

- u. Mobile cranes with rubber tires shall not be used for hoisting unless outriggers are deployed, and all tires are off of the ground (i.e., cannot pick “on rubber”).
- v. When left unattended all cranes shall be secured in a manner that prevents unauthorized startup, operation, or movement of the machine.
- w. Where a crane is operated from a remote control the operator must be in the position where the entire crane and load are in full view of the operator. Where this not possible, a sufficient number of persons, having radio communication with the operator, must serve as spotter(s) for the obstructed portion(s) of the crane.
- x. Operators of self-erector tower cranes shall not rig and signal their own loads.
- y. Cranes shall have an accessible fire extinguisher readily available.
- z. Suspended loads shall not be “anchored” to any object/structure.
- aa. “Shock Loading” & “Side Loading” the crane is prohibited.
- bb. Hoisting of loads that are secured to the surface they are being hoisted from is prohibited (e.g., loads frozen to the ground).
- cc. Increasing the load weight after it has been hoisted is prohibited.

6. Rigging:

- a. Operations that do not involve a crane must still comply with this section as stipulated in [Subpart H](#) of this document.
- b. Operations that do not involve a crane (e.g., rigging to a forklift carriage, rigging to an excavator, chain falls, winches, etc.) that meet the Okland criteria of a Critical Lift, shall complete the [Non-Crane Lift Planning Worksheet](#).
- c. All riggers shall meet the qualifications stipulated by OSHA for a qualified rigger and the qualifications listed in this document ([Qualified Rigger Qualifications](#)).
- d. The QR shall determine the approximate weight of each load prior to the hoisting operation.
- e. The QR shall read the identification tags on the rigging prior to use and shall determine what the “weakest link” is in their rigging assembly to ensure the load is within the rigging’s capacities.
- f. The QR shall conduct a pre-use inspection of the rigging, attachment points, and any specialty type of rigging equipment, such as pallet-forks, concrete buckets, skips, approved Oxygen Acetylene carts, bins, bags, cages, or similar containers, prior to each use. Damaged components shall not be used and shall be immediately taken out of service, properly tagged, or destroyed.
- g. Subcontractors shall conduct a monthly documented inspection of all of Subcontractor’s rigging and specialty type of rigging equipment. Subcontractor shall utilize Okland’s [Rigging & Electrical Color-Code](#) system to mark their rigging once inspected each month.
- h. Rigging attachment points that are installed on earthmoving equipment shall not be utilized (e.g., a hook welded onto a bucket of a track-hoe) unless the manufacturer of the equipment specifically designed the equipment in such a manner or the attachment point was installed per the design and specifications of the manufacturer or a registered professional engineer who has provided load charts or equivalent reference data with the design specific to the equipment.
- i. Loads shall be placed on dunnage or similar blocking to prevent damage to the rigging.

- j. A crane or other mechanical means shall not be used to pull rigging out from under a load.
- k. A crane shall not be used to drag or pull a load or any other object.
- l. Hosting of equipment shall only be executed utilizing the manufacturer's specified rigging attachment points.
- m. The QR shall determine the load's center of gravity prior to rigging the load.
- n. The load shall be rigged in a manner to ensure that the center of gravity is maintained in alignment with the hoist line.
- o. Personnel shall not ride on a suspended load, hooks, ball/blocks, or rigging.
- p. Loads with plastic or similar weak banding holding dunnage under the load shall not be hoisted by a crane without first removing the dunnage.
- q. All loose material shall be removed from the load prior to it being lifted.
- r. All rigging shall have its capacity identified on the rigging.
- s. The use of job-made rigging is prohibited (e.g., wire rope with wire rope clips, makeshift hooks).
- t. Lifting inserts which are embedded or otherwise attached to loads shall be capable of supporting at least four times the maximum intended load applied or transmitted to them.
- u. **All custom designed hooks; grabs; rigging; "flyable" carts, boxes, bins, drums, barrels containers, spreader bars, and similar** shall be:
 - i. Designed by a registered professional engineer.
 - ii. Have a safety factor of at least two times its maximum rated capacity.
 - iii. Have its rated capacity clearly marked/tagged on it for each configuration it can be placed in.
 - iv. Proof tested prior to use to **125%** of its maximum rated load by a third-party testing/rigging company.
- v. When rigging equipment is not in use it shall be removed from the work area and properly stored to protect it from damage and exposure to degradation from environmental elements (i.e., weather).
- w. Rigging shall be kept free of mud, ice, and chemicals.
- x. All hooks shall have an operable self-closing throat latch.
- y. Rope, knots, and non-commercially manufactured splices shall not be utilized in rigging applications.
- z. Rigging accessories (e.g., C-clamps, shackles, slings, chain, wire-rope clips) manufactured outside of the United States (e.g., China) shall not be utilized.
- aa. Shackles shall be utilized when nylon straps are placed in a chocked configuration (pin in eye).
- bb. Identification tags on all rigging shall be placed in the up position – meaning, towards the hook.
- cc. All rigging, including crane load blocks and headache balls, with missing or illegible manufacturer's name, size, trademark and/or rated load capacity identification shall be not be utilized and be immediately taken out of service.
- dd. Basket hitches shall not be utilized unless there is a positive means to prevent the rigging from sliding.
- ee. Adequate and appropriate softeners shall be utilized on all sharp edges – (Sharp is relative to the weight of the material being lifted and the type of rigging assembly being utilized).
- ff. Workers shall keep their hands away from rigging pinch points at all times.
- gg. Workers shall wear leather or equivalent gloves when handling wire rope.

- hh. **Pallets** shall not be rigged/picked with a crane unless a proper set of crane pallet forks (e.g., Jeffery Forks) are used. Loads must be positively anchored to the forks. Loose items must be positively secured and shrink wrapped (e.g., CMU, stone, brick)
- ii. **Flexible Intermediate Bulk Containers (FIBC) and similar rigging containers (e.g., Bulk Bags)** shall not be hoisted **overhead or above 5'** by any crane, for any purpose other than to place them in an **immediately adjacent** staging area or in an **immediately adjacent** approved and rated hoisting skip or similar approved and rated metal hoisting container. The secondary hosting skip or similar metal hoisting container can then be utilized as the overhead hoisting apparatus to carry the FIBC or similar rigging container during the overhead hoisting activity. All FIBCs shall be inspected prior to hoisting as stipulated in **Subpart CC – Cranes 6.e. – pre-use inspection** of this document.
- jj. **Compressed gas cylinders** shall only be hoisted in an upright and secured position, with the gauges removed, cylinder caps installed and in an approved lifting device. Approved lifting devices shall only include those devices that are specifically and commercially manufactured for that purpose or those that are constructed per the design of a registered professional engineer and are specifically designed for hoisting compressed gas cylinders.

kk. **Portable Chemical Toilets**

- i. Portable toilets shall not be connected to any crane hook without first verifying that the toilet is not occupied. Once a portable toilet is connected to the crane, the rigger/signal person shall stay immediately adjacent to the front of the toilet to ensure that no one enters the toilet prior to it being hoisted.
- ii. Signage shall be placed on all four sides of each portable toilet that is equipped with a fixed crane hoisting assembly advising the rigger: CAUTION – Prior to Connecting the Rigging to a Crane, verify that the Toilet is NOT Occupied
- iii. Signage, in English & in Spanish, shall be placed on each portable toilet that is equipped with a fixed crane hoisting assembly advising users: WARNING – This Chemical Toilet is Pre-Rigged to be Hoisted by a Crane. Ensure that it is NOT Connected to a Crane Prior to Use. This signage shall be placed in on the exterior of the door.

ll. **Tag line**

- i. A tag line(s) shall be attached to all loads.
- ii. The use of more than one tag line is permitted so long as it does not interfere with the safety of the operation.
- iii. Tag-lines shall be the appropriate length to provide adequate control of the load.
- iv. Tag-lines shall be adequately secured to the load.
- v. Tag-lines shall be secured in a manner as not to upset the load's center of gravity when the tag-line is pulled on.
- vi. Tag-lines shall be free of knots and similar obstructions that could cause the tag line to get caught on another object.
- vii. Tag-lines shall not be utilized to "anchor" the load.

mm. **Multiple Lift Rigging** (Christmas Treeing/Treeing) shall only be allowed for **steel erection** activities.

- i. Prior to a multiple lift rigging operation, Subcontractor shall complete and submit to the Okland Superintendent a detailed written lift plan.

- ii. If, at the planned boom angle and lift height, any load will be able to contact the crane boom or if the headache ball/load line wedge socket will be hoisted closer than 5' from the boom sheaves, the number of loads shall be reduced accordingly.
- iii. Wind limits shall be established, and wind speed shall be monitored.
- iv. During multiple lift rigging, each piece of steel must be equipped with an individual tag-line.

7. Critical Lifts:

- a. Operations that do not involve a crane must still comply with this section as stipulated in [Subpart H](#) of this document.
- b. **A critical lift is any one or more of the following:**
 - i. Any lift with a **mobile crane** in excess of **90%** of the crane's rated capacity for the configuration it is in.
 - ii. Any lift with a **tower crane** that is in excess of **95%** of the crane's rated capacity for the configuration it is in.
 - iii. Any lift involving **steel erection** with any crane that is in excess of **75%** of the crane's rated capacity for the configuration it is in.
 - iv. Any lift where any component of the **rigging** will be loaded in excess of **90%** of the rigging's rated capacity for the configuration it is in.
 - v. Any lift that requires the concurrent use of two or more cranes or two or more of any other type of hoisting equipment.
 - vi. Any lift that involves hoisting with a crane and lifting or movement of the load with any other type of equipment or vehicle.
 - vii. Any lift involving the pick and carry of a load with a track/crawler crane(s).
 - viii. Any lift of **100,000-pounds** or more.
 - ix. Any **lifting of personnel** (e.g., hoisting personnel in a man-basket).
 - x. Using a crane where fully extended outriggers can **NOT** be used.
 - xi. Using a crane where the **360°** load chart can **NOT** be used.
 - xii. Crane operations in which any part of the equipment, load line, or load (including rigging and lifting accessories) **could become closer, intentionally or unintentionally**, than the minimum clearance distance to overhead power lines as listed in OSHA regulation 29 CFR 1926.1408 Table A.
 - xiii. Lifting a load with **unknown weight** and/or **center of gravity**.
 - xiv. A lift that Okland has determined requires exceptional care in handling. This includes, but is not limited to: the size of the load, the weight of the load, close-tolerance installation, a load that is highly susceptible to damage, a load that is of high monetary value, a load that has a long lead time for replacement, etc.
- c. Critical Lift Procedure:
 - i. Subcontractor shall complete an Okland [Critical Lift Permit & Lift Planning Worksheet](#) that is specific to the Critical Lift operation. ([Hydraulic/Lattice Crane – Tower Crane](#)).
 - ii. Subcontractor shall complete a **Written Critical Lift Plan** for the critical lift operation. The **Written Critical Lift Plan** shall include all pertinent information necessary to adequately evaluate the safety of the lift. This may include, but is not limited to the following:
 - 1. Details of the crane's specifications and/or other hoisting equipment specifications.

2. Details of the load.
 3. Details of the rigging components being used.
 4. Details of the necessary ground conditions.
 5. Weather limitations.
 6. Operational sequence of the lifts.
 7. Detailed signaling procedures.
 8. Duties of personnel involved in the lift.
 9. Considerations of all obstructions (e.g., buildings, boom clearances, other equipment, power lines).
 10. Drawings and diagrams.
 11. Job Hazard Analysis.
- iii. When planning **multiple crane or crane in combination with other equipment lifts**, no crane shall be loaded to more than 75% of the crane's capacity at the given radius as posted in the load chart for the specific crane and its configuration, unless the lift is engineered by a qualified registered professional engineer.
 - iv. Prior to hoisting personnel Subcontractor shall utilize a checklist to validate compliance with OSHA regulations for hoisting personnel.
 - v. The completed Lift Planning Worksheet and the Written Lift Plan shall be submitted to the Okland Superintendent at least **14-days prior to the lift**. This is to allow adequate review of the operation by Okland's Crane Superintendent or his designee.
 - vi. Okland, at its sole discretion, may require, Subcontractor, at Subcontractors cost, to have their Critical Lift Plan or certain portions of the Critical Lift Plan developed/designed/engineered by a registered professional engineer due to the complexity or high risk of the planned lift.
 - vii. Subcontractor shall hold a **pre-operation meeting** at least two days prior to the critical lift with the supervision of all involved individuals so as the Critical Lift Plan can be reviewed, and expectations clearly conveyed to each. This pre-meeting will give an opportunity for the plan to be adjusted, if necessary, before the operation takes place.
 - viii. Subcontractor shall hold a **pre-lift meeting** each day prior to the critical lift(s) with all involved individuals so as the Critical Lift Plan can be reviewed, and expectations clearly conveyed to each. Subcontractor shall ensure all questions are answered and all information is effectively communicated prior to the lift taking place.
 - ix. A copy of the Written Critical Lift Plan shall remain on site, in the immediate work area so as it can be referenced as needed.

8. Landing Platforms:

- a. All landing platforms shall be capable of supporting without failure four times the maximum intended load applied or transmitted to it.
- b. The maximum allowable load for each platform shall be clearly marked on the platform.
- c. All loads being applied or transmitted to the platform shall be equally distributed across the surface of the platform.
- d. All landing platforms shall be designed by a registered professional engineer.

9. Tower Crane Climbing

- a. During climbing activities, Okland, at Okland's expense, may elect to utilize a qualified third-party consultant to be on site to actively monitor the climbing activity and provide

expert opinion to the Okland project management team on the quality and safety performance of the climbing activity. This third-party consultant would be in addition to the crane technician that is directly overseeing the climbing operation for the subcontractor. If Okland exercises this option, Subcontractor shall cooperate with the third-party consultant.

Pollution Liability

1. Subcontractor shall comply with Okland's [Water Crash Cart Policy](#).
2. Subcontractor shall comply with Okland's project specific Water Mitigation Plan.
3. Storm Water Pollution Prevention:
 - a. Subcontractor and its employees shall comply with the requirements stipulated in the Project's Storm Water Pollution Prevention Plan (SWPPP).
 - b. All modifications or removal of any SWPPP controls that are already in place shall only be done with prior authorization from Okland's project Superintendent.
 - c. Subcontractor is responsible for all costs associated with maintaining individual compliance with the project's SWPPP; this includes costs associated with preventing and/or cleaning up of mud subcontractor, its employees, and/or its tiered subcontractors tracks off of the project site.
4. Fugitive Dust:
 - a. Subcontractor and its employees shall comply with the requirements stipulated in the Project's Fugitive Dust Control Plan.
5. Noise:
 - a. Subcontractor and its employees shall comply with all local noise ordinances and project specific noise restrictions.
6. Mold:
 - a. Subcontractor shall immediately report any water damage, leaks, or water intrusion to the Okland Superintendent.
 - b. Subcontractor shall immediately report the presence of mold that Subcontractor observes on any portion of the project to the Okland Superintendent.
 - c. Unless otherwise directed by Okland, Subcontractor shall not remove or disturb visible mold contaminated materials. If such direction is given it shall be conducted in compliance with industry standard safety requirements and under the guidance of Certified Industrial Hygienist.
 - d. Subcontractor shall be responsible for any mold remediation that is necessary as the result of subcontractor's actions or inactions.
7. Hazardous Substances:
 - a. Subcontractor shall comply with the hazardous substances' requirements stipulated in [Subpart D](#) of this document, Occupational Health and Environmental Controls.

Substance Abuse Screening (i.e., Drug Testing)

1. Subcontractor shall conform to Oakland's [Subcontractor Substance Abuse Program Compliance Requirements](#).

NOTE: Quick/Rapid Screenings **MUST** be sent to a lab for final confirmation regardless of initial result. The results of a quick/rapid test alone shall not be accepted.

SECTION 3

RESERVATION OF RIGHTS AND SEVERABILITY

RESERVATION OF RIGHTS

Okland reserves the right to administer the requirements of this manual and the attached programs and policies and to interpret the requirements set forth at its sole discretion. In addition, Okland reserves the right to change and/or rescind the requirements in whole or in part at its sole discretion.

SEVERABILITY

If any portion of this manual and/or the attached programs and policies or the application of any portion to any person or circumstance is held invalid, the remainder of the manual and/or the attached programs and policies shall remain in effect without the invalid portion or application.

ANNUAL REVIEW

The requirements in this manual will be reviewed for effectiveness of safety, health, and environmental protection at least annually.

This manual was last reviewed on:

[December 2025](#)

This manual is due for its next review in:

[December 2026](#)

END